General Mathematics Solutions

©Ezy Math Tutoring | All Rights Reserved

Copyright © 2012 by Ezy Math Tutoring Pty Ltd. All rights reserved. No part of this book shall be reproduced, stored in a retrieval system, or transmitted by any means, electronic, mechanical, photocopying, recording, or otherwise, without written permission from the publisher. Although every precaution has been taken in the preparation of this book, the publishers and authors assume no responsibility for errors or omissions. Neither is any liability assumed for damages resulting from the use of the information contained herein.

Learning Strategies

Mathematics is often the most challenging subject for students. Much of the trouble comes from the fact that mathematics is about logical thinking, not memorizing rules or remembering formulas. It requires a different style of thinking than other subjects. The students who seem to be "naturally" good at math just happen to adopt the correct strategies of thinking that math requires – often they don't even realise it. We have isolated several key learning strategies used by successful maths students and have made icons to represent them. These icons are distributed throughout the book in order to remind students to adopt these necessary learning strategies:



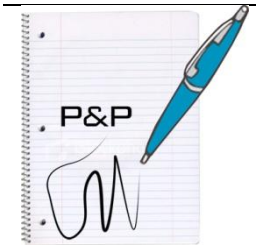
Talk Aloud Many students sit and try to do a problem in complete silence inside their heads. They think that solutions just pop into the heads of 'smart' people. You absolutely must learn to talk aloud and listen to yourself, literally to talk yourself through a problem. Successful students do this without realising. It helps to structure your thoughts while helping your tutor understand the way you think.



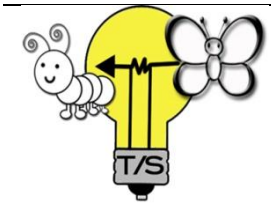
BackChecking This means that you will be doing every step of the question twice, as you work your way through the question to ensure no silly mistakes. For example with this question: $3 \times 2 - 5 \times 7$ you would do "3 times 2 is 5 ... let me check – no 3×2 is 6 ... minus 5 times 7 is minus 35 ... let me check ... minus 5×7 is minus 35. Initially, this may seem time-consuming, but once it is automatic, a great deal of time and marks will be saved.



Avoid Cosmetic Surgery Do not write over old answers since this often results in repeated mistakes or actually erasing the correct answer. When you make mistakes just put one line through the mistake rather than scribbling it out. This helps reduce silly mistakes and makes your work look cleaner and easier to backcheck.



Pen to Paper It is always wise to write things down as you work your way through a problem, in order to keep track of good ideas and to see concepts on paper instead of in your head. This makes it easier to work out the next step in the problem. Harder maths problems cannot be solved in your head alone – put your ideas on paper as soon as you have them – always!



Transfer Skills This strategy is more advanced. It is the skill of making up a simpler question and then transferring those ideas to a more complex question with which you are having difficulty.

For example if you can't remember how to do long addition because you can't recall exactly how to carry the one: $\begin{array}{r} 5889 \\ +4587 \\ \hline \end{array}$ then you may want to try adding numbers which you do know how to calculate that also involve carrying the one: $\begin{array}{r} 5 \\ +9 \\ \hline \end{array}$

This skill is particularly useful when you can't remember a basic arithmetic or algebraic rule, most of the time you should be able to work it out by creating a simpler version of the question.

©Ezy Math Tutoring | All Rights Reserved



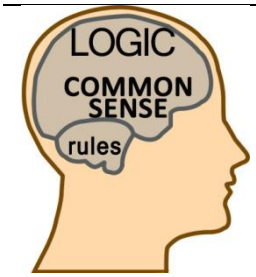
Format Skills These are the skills that keep a question together as an organized whole in terms of your working out on paper. An example of this is using the "=" sign correctly to keep a question lined up properly. In numerical calculations format skills help you to align the numbers correctly.

This skill is important because the correct working out will help you avoid careless mistakes. When your work is jumbled up all over the page it is hard for you to make sense of what belongs with what. Your "silly" mistakes would increase. Format skills also make it a lot easier for you to check over your work and to notice/correct any mistakes.

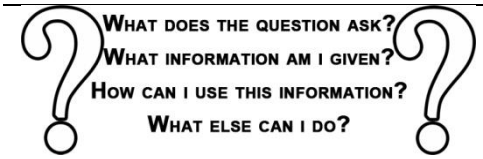
Every topic in math has a way of being written with correct formatting. You will be surprised how much smoother mathematics will be once you learn this skill. Whenever you are unsure you should always ask your tutor or teacher.



Its Ok To Be Wrong Mathematics is in many ways more of a skill than just knowledge. The main skill is problem solving and the only way this can be learned is by thinking hard and making mistakes on the way. As you gain confidence you will naturally worry less about making the mistakes and more about learning from them. Risk trying to solve problems that you are unsure of, this will improve your skill more than anything else. It's ok to be wrong – it is NOT ok to not try.



Avoid Rule Dependency Rules are secondary tools; common sense and logic are primary tools for problem solving and mathematics in general. Ultimately you must understand Why rules work the way they do. Without this you are likely to struggle with tricky problem solving and worded questions. Always rely on your logic and common sense first and on rules second, always ask Why?



Self Questioning This is what strong problem solvers do naturally when they get stuck on a problem or don't know what to do. Ask yourself these questions. They will help to jolt your thinking process; consider just one question at a time and Talk Aloud while putting Pen To Paper.

©Ezy Math Tutoring | All Rights Reserved

Table of Contents

CHAPTER 1: Financial Mathematics **5**

Exercise 1: Earning Money 6

Exercise 2: Taxation 11

Exercise 3: Credit & Borrowing 19

Exercise 4: Annuities & Loan Repayments 26

Exercise 5: Depreciation 30

CHAPTER 2: Data Analysis **34**

Exercise 1: Data Collection & Sampling 35

Exercise 2: Mean, Median & Spread of Data 42

Exercise 3: Representing Data (I) 49

Exercise 4: Representing Data (II) 56

Exercise 5: Normal Distribution 66

Exercise 6: Correlation 71

CHAPTER 3: Measurement **78**

Exercise 1: Units of Measurement 79

Exercise 2: Applications of Area & Volume 88

Exercise 3: Similarity 99

Exercise 4: Right Angled Triangles 105

Exercise 5: Further Applications of Trigonometry 120

Exercise 6: Spherical Geometry 135

CHAPTER 4: Probability **140**

Exercise 1: Simple Probability 141

Exercise 2: Multi-stage Events 150

Exercise 3: Applications of Probability 157

CHAPTER 5: Algebraic Modeling **163**

Exercise 1: Algebraic Skills & Techniques 164

Exercise 2: Modelling Linear Relationships 172

©Ezy Math Tutoring | All Rights Reserved

©Ezy Math Tutoring | All Rights Reserved

General Mathematics

Financial Mathematics

©Ezy Math Tutoring | All Rights Reserved

Exercise 1

Earning Money

©Ezy Math Tutoring | All Rights Reserved

1) Mark earns a gross salary of $78000 per annum. To the nearest cent how much does Mark earn:

a) Per month

$6500

b) Per fortnight

$3000

c) Per week

$1500

d) Per day

$213.70

e) Per hour

$37.50

Assume Mark works a 40 hour week, does not work weekends, and ignore public holidays

2) Tom earns a gross salary of $900 per 37 hour week. Matt earns $22 per hour, but is required to work for 42 hours per week.

a) Who earns more per hour?

Tom ($24.32)

b) Who earns more per week?

Matt $924

c) What is the difference in their annual earnings? (Assume they each work for all 52 weeks of the year, not on weekends, and ignore public holidays)

Matt earns $1248 more

3) Peter's pay rates for a week's work are as follows

- $25 per hour for the first 35 hours
- $40 per hour for each hour worked over 35 hours
- An extra $2.50 per hour for each hour worked over 40 hours

Calculate Peter's earnings before tax for the following scenarios

a) Worked 32 hours

25 x 32 = $800

b) Worked 35 hours

25 x 35 = $875

c) Worked 43 hours

875 + (8 x 40) + (3 x 2.50)
= $1202.50

©Ezy Math Tutoring | All Rights Reserved

d) Worked 60 hours

875 + (25 x 40) + (2.5 x 20) = $1937.50

4) When James takes holidays he is allowed a 7.5% extra on top of his holiday pay. James' salary is currently $82500. If he takes two weeks holiday, how much will he be paid for this period?

82500/26 = 3173.08

3173.08 x 1.075 = $3411.06

5) Ronald works as a car salesman. He gets paid a base wage of $900 per week. He also gets paid commission for every car he sells, according to the sale price. If the car is valued below $20000 he gets 1% of the sale price. For cars sold in the $20000 to $39999 price range, he receives 1.5% commission. If the value of the car sold is $40000 or more he receives 2%.

What does Ronald earn per week under the following scenarios?

a) He sells no cars

$900

b) He sells one car valued at $32000

900 + (32000 x 0.015) = $1380

c) He sells a car for $35000 and one for $41950

900 + (35000 x 0.015) + (41950 x 0.02) = $2264

d) He sells 4 cars all for $37500

If Ronald wanted to earn $2000 for a week's work, what must he sell a luxury car (valued at over $40000) for?

2000 – 900 = $1100

1100 is commission on a car valued at 1100/0.02 = $55000

6) Petra dyes flowers and gets paid 1.5 cents for every stem she dyes.

a) If she dyes 3000 stems how much does she earn?

3000 x 0.015 = $45

b) If she dyes 15000 stems, how much does she earn?

15000 x 0.015 = $225

c) How many stems must she dye in order to earn $750?

750/0.015 = 50000

©Ezy Math Tutoring | All Rights Reserved

7) New start allowance is paid to unemployed job seekers. A single person receives $492.60 per fortnight, whilst a couple receives $444.70 each per fortnight. A job seeker with a dependent child receives $533 per fortnight.

A carers pension is paid to anybody caring for a disabled child and pays $115.40 per fortnight

The aged pension is $712 per fortnight for a single pensioner and $536.70 each per fortnight for a married couple

Calculate how much each household brings in under the following conditions

a) Bill and Doris are both old aged pensioners, and their son Malcolm is currently seeking work

b) Jill is seeking work and also cares for her 10 year old son who is not disabled

c) Bob is a single pensioner who shares a house with his grandson John who is seeking work and also cares for his own son who has a disability

8) Bernard worked 37 hours last week. His hourly rate is $31.50, and he pays tax at a flat rate of 15% of his earnings. In addition he pays 1.5% of his gross pay toward the Medicare levy, and he also has to pay 4.5% of his gross pay in HECS repayments. Union fees of $8 and social club fees of $2.50 per week are also deducted.

Bernard makes voluntary superannuation contributions of 3% of his gross pay.

How much money did Bernard actually take home last week?

9) Max works a 37 hour week and is paid for all public holidays also. He has the following weekly financial commitments

- Rent $350
- Electricity $35
- Petrol $50
- Gas $25
- Entertainment $75
- Food etc. $125
- Credit card $18
- Car costs $30

Max also wishes to put money away for such things as clothing, furniture, household items etc. so that he can pay cash for them when he needs them. He estimates he will need $1500 for the year.

Max also wishes to save $40 per week.

What must Max's hourly pay rate be to be able to meet his

©Ezy Math Tutoring | All Rights Reserved

commitments and savings needs? (Assume Max does not pay taxation nor has any other deductions from his wages)

©Ezy Math Tutoring | All Rights Reserved

Exercise 2

Taxation

©Ezy Math Tutoring | All Rights Reserved

1) Martin works for a salary of $52000 per annum before tax. The weekly tax on this income is $162.44. How much does Martin take home per fortnight?

$52000 \div 52 = \$1000\ per\ week$

$1000 - 162.44 = \$837.56$

2) Income between $18201 and $37000 per annum is currently taxed at the rate of 19 cents per dollar for amounts over $18200. How much tax is payable for the following incomes?

a) $19200

$19200 - 18200 = 1000$

$1000 \times 0.19 = \$190$

b) $26000

$26000 - 18200 = 7800$

$7800 \times 0.19 = \$1482$

c) $36999

$36999 - 18200 = 18799$

$18799 \times 0.19 = \$3571.81$

d) $50000

$50000 - 18200 = 31800$

$31800 \times 0.19 = \$6042$

e) $15000

Zero

©Ezy Math Tutoring | All Rights Reserved

3) People earning over $180000 per annum pay tax according to the following formula. $54547 plus 45 cents per dollar for each dollar over $180000. How much tax is payable for the following incomes?

a) $190000

$190000 - 180000 = 10000$

$10000 \times 0.45 = 4500$

$54547 + 4500 = \$59047$

b) $225000

$225000 - 180000 = 45000$

$45000 \times 0.45 = 20250$

$54547 + 20250 = \$74797$

c) $500000

$500000 - 180000 = 320000$

$320000 \times 0.45 = 144000$

$54547 + 144000 = \$198547$

d) $100000

$1000000 - 180000 = 820000$

$820000 \times 0.45 = 369000$

$54547 + 369000 = \$423547$

The rates mentioned in questions 2 and 3 are taken from the following table which shows the formula to calculate tax payable on all incomes. Use the table to answer the following questions

©Ezy Math Tutoring | All Rights Reserved

Taxable income	Tax on this income
0 - $18,200	Nil
$18,201 - $37,000	19c for each $1 over $18,200
$37,001 - $80,000	$3,572 plus 32.5c for each $1 over $37,000
$80,001 - $180,000	$17,547 plus 37c for each $1 over $80,000
$180,001 and over	$54,547 plus 45c for each $1 over $180,000

4) What is the annual tax payable for the following incomes?

a) $39125

$39125 - 37000 = 2125$

$2125 \times 0.325 = 690.625$

$3572 + 690.625 = \$4262.62$

b) $125432

$125432 - 80000 = 45432$

$45432 \times 0.37 = 16809.84$

$17547 + 16809.84 = \$34356.84$

c) $12000

Zero

d) $37000

$\$3572$

©Ezy Math Tutoring | All Rights Reserved

e) $180002

$180002 - 180000 = 2$

$2 \times 0.45 = 0.9$

$54547 + 0.9 = \$54547.90$

f) $1,000,000

$423547 as per q 3d

5) Jim earns $42 per hour for a 38 hour week. How much tax should be deducted from his wages each week to meet his taxation commitment?

Yearly wage $= 42 \times 38 \times 52 = \$82992$

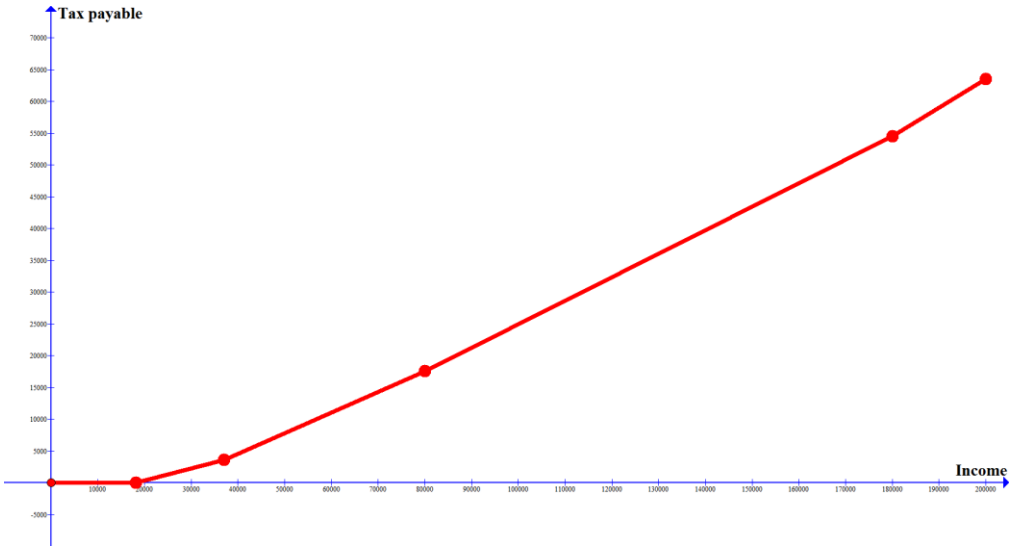
$82992 - 80000 = 2992$

$2992 \times 0.37 = 1107.04$

$17547 + 1107.04 = \$18654.04$ tax per year

$18654.04 \div 52 = \$358.73$ tax per week

6) Graph tax payable per annum versus taxable income for incomes from $0 to $200000



©Ezy Math Tutoring | All Rights Reserved

7) The Medicare levy is payable by all taxpayers who earn more than $20542 per annum, and is charged at the rate of 1.5% of taxable income. How much Medicare levy is payable for the following incomes?

a) $42222

$42222 \times 0.015 = \$633.33$

b) $17000

Zero

c) $82000

$82000 \times 0.015 = \$1230$

d) $53149

$53149 \times 0.015 = \$797.23$

8) If an unmarried taxpayer is not covered by private health cover and they earn more than $84000 per annum, they are liable for the Medicare levy surcharge, which is a further 1% of taxable income

What is the total levy (including surcharge if applicable) payable for the following incomes?

a) $2000

Zero

b) $73250

$73250 \times 0.015 = \$1098.75$

c) $83999

$83999 \times 0.015 = \$1259.98$

©Ezy Math Tutoring | All Rights Reserved

d) $92000

$92000 \times 0.025 = \$2300$

e) $113000

$113000 \times 0.025 = \$2825$

9) Alan is single, and earned $93450 in the past financial year. His employer deducted $500 per week to cover his tax and Medicare commitments. At the end of the financial year is Alan due a refund from the government, or is he liable for additional tax?

Tax on 93450$= 17547 + (13450 \times 0.37) = \$22523.50$

Total Medicare levy (including surcharge) $= 93450 \times 0.025 = \$2336.25$

Total liability $= \$24859.75$

Tax deducted $= 500 \times 52 = \$26000$

Refund$= \$1140.25$

10) GST is a tax placed on many items by the government; it is added to the base price of the item and is included in the total cost of the item. The current rate of GST is 10%. What is the total cost of the following items with base prices of:

a) $1.50

$1.50 \times 1.1 = \$1.65$

b) $12.50

$12.50 \times 1.1 = \$13.75$

c) $105.00

$105 \times 1.1 = \$115.50$

©Ezy Math Tutoring | All Rights Reserved

d) $32000

$32000 \times 1.1 = \$35200$

e) $12243.56

$12243.56 \times 1.1 = \$13467.92$

11) Use guess check and improve, or develop a method to calculate the base price of the following items that have a total cost of:

a) $11

$10

b) $44

$40

c) $36.19

$32.90

d) $111.32

$101.20

e) $8938.05

$8125.50

Develop a formula that enables you to calculate the base price of an item given its total cost

Base price = Total cost ÷(1.1)

©Ezy Math Tutoring | All Rights Reserved

Exercise 3

Credit & Borrowing

©Ezy Math Tutoring | All Rights Reserved

1) Calculate the total simple interest paid under the following conditions

a) Principal of \$10,000 at a rate of 10% p.a. for 10 years

$I = PRT$

$I = 10000 \times \frac{10}{100} \times 10$

$I = \$10000$

b) Principal of \$2000 at a rate of 5% p.a. for 5 years

$I = PRT$

$I = 2000 \times \frac{5}{100} \times 5$

$I = \$500$

c) Principal of \$4000 at a rate of 7.5% p.a. for 2 years

$I = PRT$

$I = 4000 \times \frac{7.5}{100} \times 2$

$I = \$600$

d) Principal of \$25,000 at a rate of 12.5% p.a. for 3 years

$I = PRT$

$I = 25000 \times \frac{12.5}{100} \times 3$

$I = \$9375$

e) An interest rate of 8% p.a. for 5 years on a principal of \$6,000

$I = PRT$

$I = 6000 \times \frac{8}{100} \times 5$

$I = \$2400$

2) Calculate the amount of time it would take to repay a loan under the following conditions (assume simple interest)

a) Principal of \$5,000 at 10% p.a. interest with total interest payable of \$2000

$T = \frac{I}{PR}$

$T = \frac{2000}{5000 \times 0.1}$

$T = 4\ years$

b) Principal of \$12,000 at 12% p.a. interest with total interest payable of \$6000

$T = \frac{I}{PR}$

$T = \frac{6000}{12000 \times 0.12}$

©Ezy Math Tutoring | All Rights Reserved

$$T = 6\ years$$

c) Principal of \$2,000 at 20% p.a. interest with total interest payable of \$6400

$$T = \frac{I}{PR}$$

$$T = \frac{6400}{2000 \times 0.2}$$

$$T = 16\ years$$

d) Principal of \$800 at 11% p.a. interest with total interest payable of \$440

$$T = \frac{I}{PR}$$

$$T = \frac{440}{800 \times 0.11}$$

$$T = 5\ years$$

3) A man borrows \$11500 to buy a car. He agrees to a simple interest rate of 6% per annum and agrees to pay the loan off in 5 years. How much will he repay in total?

$$I = PRT$$

$$I = 11500 \times 0.06 \times 5$$

$$I = \$3450$$

He must also repay the principal, so total repaid is

$$11500 + 3450 = \$14950$$

4) Kerry borrows \$4000 and is required to repay the loan with equal monthly instalments. If the simple interest rate is 9% p.a. how much will she have to repay each month to finalise the loan in 3 years?

$$I = PRT$$

$$I = 4000 \times 0.09 \times 3$$

$$I = \$1080$$

Total amount payable
$= 1080 + 4000$

$$= 5080$$

$$5080 \div 36 = \$141.11$$

5) A man takes out a loan of \$10000 at 6.5% p.a. simple interest rate for 4 years. After 2 years the interest rate was increased to 8%. How much did his repayments have to increase by to still have the loan repaid in the same time?

If Interest rate had not changed

$$= 10000 \times 0.065 \times 4 = \$2600$$

Initial repayments were $\frac{2600+10000}{48}$

$$= \$262.50$$

For first two years paid
$262.50 \times 24 = \$6300$

©Ezy Math Tutoring | All Rights Reserved

Total interest to be repaid is

$10000 \times 0.065 \times 2 = \$1300$

Plus $10000 \times 0.08 \times 2 = \$1600$

Total amount to be repaid is now

$1300 + 1600 + 10000 = \$12900$

Amount left to pay after two years is $12900 - 6300 = \$6600$

$6600 \div 24 = \$275$ per month for last two years

6) Complete the following table

Home loan table				
	Amount = $100,0000	Assume the same number of days per month		
	Interest Rate = 15% p.a.			
	Monthly repayment = $3000			
N	**Principal**	**Interest**	**P+I**	**P+I-R**
1	100000	1250	101250	98250
2	98250	1228.12	99478.12	96478.12
3	96478.12	1205.98	97684.10	94684.02
4	94684.02	1183.55	95867.57	92867.57
5	92867.57	1160.84	94028.41	91028.41

©Ezy Math Tutoring | All Rights Reserved

7) From the table above, what would the amount owing be after 5 months if the monthly repayment was doubled? Why is this amount not equal to half the amount owing after 5 months in question 6?

By completing the above table using the new repayment, the amount owing after 5 months is $75678.78

It is not half the original amount because interest is still being charged on the amount owing which does not halve. The amount the loan is reducing by has doubled, but the amount owing has not

8) Tom buys a new lounge suite for $2400 using the store's credit facility. The store offers a two year non-interest period. After that time the interest charged on the outstanding balance is 18% p.a. simple interest payable monthly.

a) If Tom wishes to avoid any interest charges, what is the minimum amount per month he should pay?

Must pay 2400 per month in 24 months

$2400 \div 24 = \$100\ per\ month$

b) If Tom repays the loan after 3 years with equal instalments, how much did he repay each month?

Let m be the monthly repayment

The total amount paid, A equals the monthly repayment times 36

$A = 36m$

The total amount paid also equals 2400 plus the interest paid

$A = 2400 + I$

Interest is paid for one year only

Let B be the balance owing when interest is charged (that is after the two year non interest period). That is $B = 2400 - 24m$

©Ezy Math Tutoring | All Rights Reserved

$I = PRT = B \times 0.18 \times 1$

So $A = 2400 + (0.18B)$

From above, $A = 36m$

So $36m = 2400 + (0.18B)$

Substituting for B from above gives:

$36m = 2400 + [2400 + 0.18(2400 - 24m)]$

Solving for m gives

$m = \$70.24$

c) The store has a policy that if no repayments have been made in the first 30 months, the debt is referred to a collection agency. How much gets referred to the agency?

Amount owing after 30 months = 2400 + 6 months interest charge

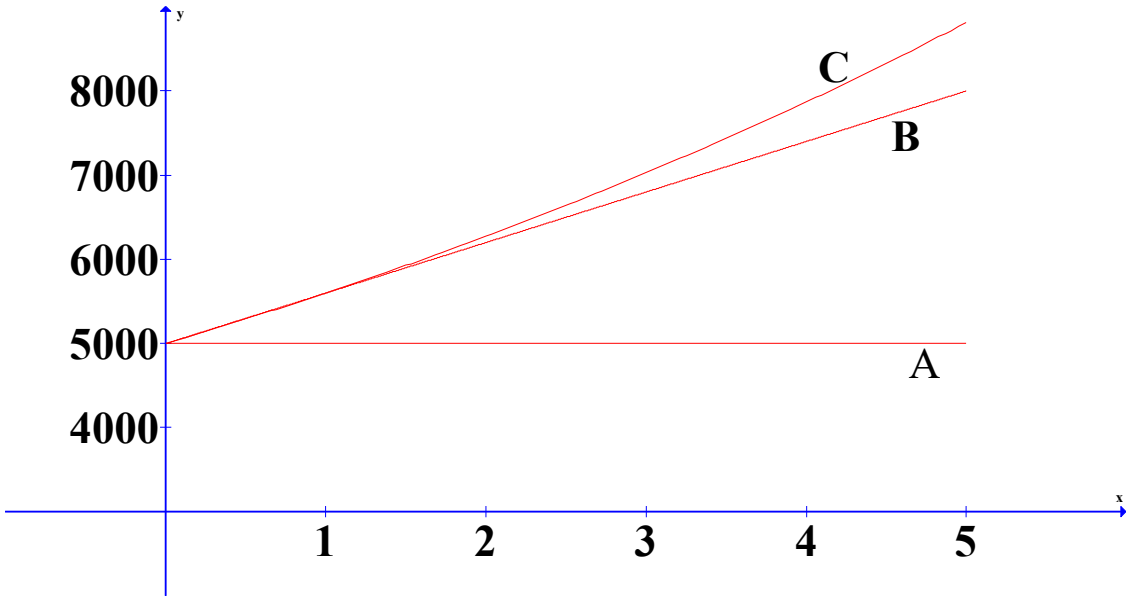
$I = PRT = 2400 \times 0.18 \times 0.5 = \$216$

Amount owing $= 2616$

9) Which of the following curves represents

- The amount paid on a $5000 loan that is repaid with a simple interest rate (B)
- The amount paid on a $5000 loan with a compound interest rate (C)
- The amount paid on a $5000 loan repaid with no interest rate (A)

©Ezy Math Tutoring | All Rights Reserved



10) Calculate the effective interest rate on a loan of $8000 at 15% p.a. interest paid monthly for 3 years

$$r = \left(1 + \frac{i}{n}\right)^n - 1$$

$$r = \left(1 + \frac{0.15}{12}\right)^{12} - 1 = 0.1607 = 16.07\%$$

©Ezy Math Tutoring | All Rights Reserved

Exercise 4

Annuities & Loan Repayments

©Ezy Math Tutoring | All Rights Reserved

1) What is the future value of an annuity with a contribution of \$100 per year for 15 years, if the interest rate is 10% p.a.?

$$\text{FV} = 100\left[\frac{(1+0.1)^{15}-1}{0.1}\right]$$

$$= 100 \times 31.77 = \$3177$$

2) What is the future of an annuity with a contribution of \$2000 per 6 months for 20 years if the interest rate is 8% p.a.?

$$\text{FV} = 2000\left[\frac{(1+0.04)^{40}-1}{0.04}\right]$$

$$= 2000 \times 120.03 = \$190051$$

3) The future value of an annuity after 15 years is \$80,000. If the interest rate was 20% p.a. what were the yearly contributions?

$$80000 = C\left[\frac{(1+0.2)^{15}-1}{0.2}\right]$$

$$C = \frac{80000}{72} \cong \$1111.11$$

4) The future value of an annuity after 30 years is \$250,000. If the interest rate was 9% p.a. and the contributions were made monthly, how much were these contributions?

$$250000 = C\left[\frac{(1+0.0075)^{360}-1}{0.0075}\right]$$

$$C = \frac{250000}{1830.74} \cong \$136.56$$

5) Which has a greater future value; an annuity of \$100 per month at 6% p.a. interest, or an annuity of \$300 per quarter at the same interest rate? Assume the period of investment is 20 years, and explain why the two are not equal even though \$100 per month is equal to \$300 per quarter

$$\text{FV}_1 = 100\left[\frac{(1+0.005)^{240}-1}{0.005}\right]$$

$$= 100 \times 462 \cong \$46200$$

$$\text{FV}_2 = 300\left[\frac{(1+0.015)^{80}-1}{0.015}\right]$$

$$= 300 \times 152.71 \cong \$45813$$

The amounts are not the same due to the compounding nature of the investments. Interest is calculated each month versus monthly, so slightly more interest is earned in the same period

6) Colin is saving for a place in a retirement village. If he needs \$200,000 by the time he retires in 10 years, how much should he pay into an account each year if the rate of interest paid is 8% per annum?

$$200000 = C\left[\frac{(1+0.08)^{10}-1}{0.08}\right]$$

©Ezy Math Tutoring | All Rights Reserved

$$C = \frac{200000}{14.487} \cong \$13805.50$$

7) John is planning to take the trip of a lifetime in ten years' time and estimates that the amount of money he will need at that time is $50 000. He is advised to contribute $4000 each year into an account that pays 5% pa, compounded annually. Will John have enough money in ten years time to make his dream come true? By how much will he fall short of or overshoot his goal?

$$\text{FV} = 4000\left[\frac{(1+0.05)^{10}-1}{0.05}\right]$$

$$= 4000 \times 12.58 \cong \$50311$$

John will just make the money he needs

8) What is the present value of an annuity of $150 per month @ 18% p.a. compounded monthly for 5 years?

$$\text{PV} = 150\left(\frac{1-(1+0.015)^{-60}}{0.015}\right)$$

$$= 150\left(\frac{1-0.409}{0.015}\right) \cong \$5910$$

9) Peter has two options when saving for his retirement. Either invest $50000 today at 7% p.a. interest compounded annually for 10 years or pay $400 per month commencing immediately at 9% p.a. interest compounded monthly. Which option gives Peter more money to retire with?

Option 1:

Investing lump sum

$$A = 50000(1+0.07)^{10} = \$98357.57$$

Option 2:

Payment per month

$$\text{FV} = 400\left[\frac{(1+0.0075)^{120}-1}{0.0075}\right]$$

$$= 400 \times \$77405.71$$

Peter should invest the lump sum

10) In 8 years time a business plans to replace its fitting and fixtures. It is estimated that the replacement will cost $15000. How much does the business need to save per year if it receives 6% p.a. compounded annually on their savings?

$$15000 = x\left[\frac{(1+0.06)^{8}-1}{0.06}\right]$$

$$x = \frac{15000}{9.9} \cong \$1515$$

©Ezy Math Tutoring | All Rights Reserved

11) Arnold deposits \$200 per month into his account. How much does he have in his account at the end of 5 years if the bank pays 8% p.a. interest compounded every 2 months?

$$\text{FV} = 400\left[\frac{(1+0.0133333)^{30}-1}{0.0133333}\right]$$

$$= 400 \times 36.59 \cong \$14636.60$$

12) A couple take a home loan of \$250000 over 30 years at 12% p.a. compounded monthly. What are the monthly repayments, total amount paid, and total interest paid over the course of the loan?

$$P = 250000\left[\frac{0.01(1.01)^{360}}{(1.01)^{360}-1}\right]$$

$$250000\left[\frac{0.3595}{34.95}\right] \cong \$2571.53$$

Total amount paid $= 360 \times 2571.53$

$\cong \$925751$

Interest paid $\cong \$675751$

13) Use the table below to calculate the value of an ordinary annuity of \$200 per month which is invested at 4% per month for 4 months

Future values of \$1

	Interest rate				
Period	**1%**	**2%**	**3%**	**4%**	**5%**
1	1.0000	1.0000	1.0000	1.0000	1.0000
2	2.0100	2.0200	2.0300	2.0400	2.0500
3	3.0301	3.0604	3.0909	3.1216	3.1525
4	4.0604	4.1216	4.1836	4.2465	4.3101
5	5.1010	5.2040	5.3091	5.4163	5.5256
6	6.1520	6.3081	6.4684	6.6330	6.8019
7	7.2135	7.4343	7.6625	7.8983	8.1420
8	8.2857	8.5830	8.8923	9.2142	9.5491

From the table, the future value of an annuity of \$1 over 4 time periods at 4% per time period $= \$4.2465$

\$200 per month for 4 months @4 % is therefore $= 200 \times 4.2465 \cong \$849.30$

©Ezy Math Tutoring | All Rights Reserved

Exercise 5

Depreciation

©Ezy Math Tutoring | All Rights Reserved

1) Assuming straight line depreciation, what is the financial life of the assets having a depreciation rate of?

a) 10%

$100\% \div 10\% = 10\ years$

b) 8.5%

$100\% \div 8.5\% = 12\ years$

c) 20%

$100\% \div 20\% = 5\ years$

d) 12.5%

$100\% \div 12.5\% = 8\ years$

e) 5%

$100\% \div 5\% = 20\ years$

2) What is the depreciation rate of an asset that has the following financial life? (Assume straight line depreciation)

a) 5 years

$100\% \div 5 = 20\%$

b) 20 years

$100\% \div 20 = 5\%$

c) 12 years

$100\% \div 12 = 8.5\%$

d) 25 years

$100\% \div 25 = 4\%$

e) 10 years

$100\% \div 10 = 10\%$

3) A car with a book value of $50,000 is bought by a business in July 2006. If its value is depreciated by 20% using the straight line method, what is its book value in July 2010?

$2010 - 2006 = 4\ years$

$4\ years \times 20\% = 80\%$

$\$50000 \times 80\% = \$40000$

Value $= \$50000 - \$40000 = \$10000$

4) In July 2003 a computer system was valued at $8000. In July 2006 its value was $5000. Assuming straight line depreciation what was the depreciation rate?

Time passed $= 2006 - 2003$
$= 3\ years$

Change in value is $3000

Change per year $= \$1000$

$1000 \div 8000 = \frac{1}{8} = 12.5\%$

©Ezy Math Tutoring | All Rights Reserved

5) A car originally bought for $40,000 was depreciated using the reducing balance method at a rate of 12%. What was its value after 1, 2 and 3 years?

Value after each year is 88% of its previous value

After 1 year,
Value $= 40000 \times 88\% = \$35200$

After 2 years,
Value $= 35200 \times 88\% = \$30976$

After 3 years
Value $= \$27258.88$

6) In July 2006 office furniture was bought for $18000. It was depreciated using the reducing balance method, and in July 2009 its value was $13122. What rate of depreciation was used?

$18000 \times r^3 = 13122$

$r^3 = 0.729$

$r = 0.9$

Therefore depreciation rate $= 10\%$

7) In July 2001 a car having a value of $35000 was purchased. It was depreciated at a rate of 10% using the straight line method. When did the value of the car equal zero?

Depreciation amount per year $= \$3500$

Financial life of car = 10 years

8) In July 2001 a car having a value of $35000 was purchased. It was depreciated at a rate of 10% using the reducing balance method. When did the value of the car equal zero?

The value of the car will never reach zero, since the reducing balance method is used. The value of the car is 35000×0.9^n

By solving the equation for value zero, or graphing the equation it can be seen that the value will approach but never be exactly zero

9) A boat having a value of $75000 was purchased and it was depreciated at a rate of 15% using the reducing balance method

a) Write a formula that calculates the value of the boat after one year

Value $= 75000 \times 0.85$

b) Write a formula that calculates the value of the boat after 2 years

Value $= 75000 \times 0.85^2$

©Ezy Math Tutoring | All Rights Reserved

c) Write a formula that calculates the value of the boat after 5 years

Value $= 75000 \times 0.85^5$

d) Write a formula that calculates the value of the boat after n years

Value $= 75000 \times 0.85^n$

10) A car having a value of V dollars was purchased and then depreciated at a rate of 10% using the reducing balance method.

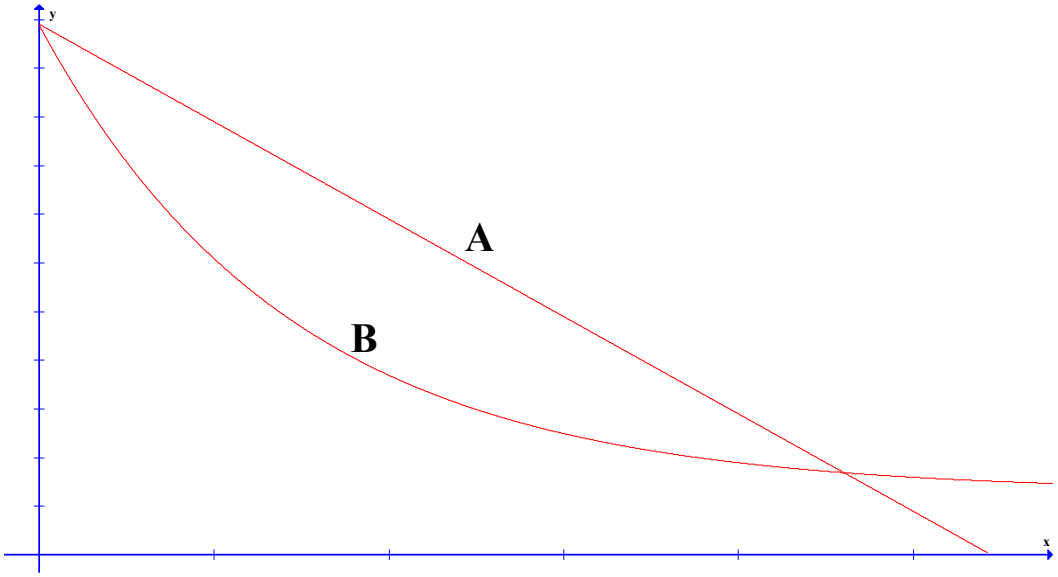
Write a formula that could be used to calculate the value of the car after n years

Value $= V \times 0.9^n$

11) A car having a value of V dollars was purchased and then depreciated at a rate of r%. Write a formula that could be used to calculate the value of the car after n years

Value $= V\left(1 - \frac{r}{100}\right)^n$

12) Which of the graphs below represents the value of an asset depreciated using the reducing balance method of depreciation? Explain your answer



Graph B since the value of the asset never reaches zero under the reducing balance method

©Ezy Math Tutoring | All Rights Reserved

General Mathematics

Data Analysis

©Ezy Math Tutoring | All Rights Reserved

Exercise 1

Data Collection & Sampling

©Ezy Math Tutoring | All Rights Reserved

1) For which of the following would all data be available for analysis, and which would require a sample to be taken?

a) Score distribution in a basketball competition

All data

b) Voting intentions of the Australian people

Sample

c) Favourite colour of your class

All data

d) Favourite car of the people of Sydney

Sample

e) Types of dogs owned by the people of Victoria

Sample

2) Classify the following data as either quantitative or categorical. If the data is quantitative, indicate if it is discrete or continuous

a) Heights of your class members

Quantitative continuous

b) Attendance at football games

Quantitative discrete

c) Car colours

Categorical

©Ezy Math Tutoring | All Rights Reserved

d) Dog breeds

Categorical

e) Courses offered at a university

Categorical

f) Number of people enrolled in each course at a university

Quantitative discrete

3) Describe the differences and similarities between the random, stratified and systematic methods of sampling

Random sampling

- Least biased of all sampling techniques, there is no subjectivity - each member of the total population has an equal chance of being selected
- Can be obtained using random number tables

Advantages:

- Can be used with large sample populations
- Avoids bias

Disadvantages:

- Can lead to poor representation of the overall parent population or area if large areas are not hit by the random numbers generated. This is made worse if the study area is very large
- There may be practical constraints in terms of time available and access to certain parts of the study area

Systematic sampling

Samples are chosen in a systematic or regular way.

- They are evenly/regularly distributed in a spatial context, for example every two metres along a transect line
- They can be at equal/regular intervals in a temporal context, for example every half hour or at set times of the day
- They can be regularly numbered, for example every 10th house or person

©Ezy Math Tutoring | All Rights Reserved

Advantages:

- It is more straight-forward than random sampling
- A grid doesn't necessarily have to be used, sampling just has to be at uniform intervals
- A good coverage of the study area can be more easily achieved than using random sampling

Disadvantages:

- It is more biased, as not all members or points have an equal chance of being selected
- It may therefore lead to over or under representation of a particular pattern

Stratified sampling

This method is used when the parent population or sampling frame is made up of sub-sets of known size. These sub-sets make up different proportions of the total, and therefore sampling should be stratified to ensure that results are proportional and representative of the whole.

Advantages:

- It can be used with random or systematic sampling, and with point, line or area techniques
- If the proportions of the sub-sets are known, it can generate results which are more representative of the whole population
- It is very flexible and applicable to many geographical enquiries
- Correlations and comparisons can be made between sub-sets

Disadvantages:

- The proportions of the sub-sets must be known and accurate if it is to work properly
- It can be hard to stratify questionnaire data collection, accurate up to date population data may not be available and it may be hard to identify people's age or social background effectively

4) A company employs workers under various conditions

- 50 workers are males who work full time
- 25 are males who work part time
- 75 are females who work full time
- 100 are females who work part time

©Ezy Math Tutoring | All Rights Reserved

If stratified sampling is to be used, how many of each group should be sampled under the following conditions?

a) 50 people are to be surveyed in total

$$\frac{50}{250} \times 50 = 10\ full\ time\ males$$

$$\frac{25}{250} \times 50 = 5\ part\ time\ males$$

$$\frac{75}{250} \times 50 = 15\ full\ time\ females$$

$$\frac{100}{250} \times 50 = 20\ part\ time\ females$$

b) 25 females are to be surveyed

$$\frac{75}{175} \times 25 \cong 11\ part\ time\ females$$

$$\frac{100}{175} \times 25 \cong 14\ full\ time\ females$$

c) 75 part time workers are to be surveyed

$$\frac{25}{125} \times 75 = 15\ part\ time\ males$$

$$\frac{100}{125} \times 75 = \ 60\ part\ time\ females$$

d) 10 male part time workers are to be surveyed

10

5) The population of Australia is approximately 23 million. Of that number approximately 1,955,000 are over 65 years old. To gain an accurate representation of a sample set of 5000, how many of them should be over 65 years old?

$$\frac{1955000}{23000000} \times 5000 = 425$$

©Ezy Math Tutoring | All Rights Reserved

6) A sample of 5000 people included 100 in the age range 20 to 40. Comment on the appropriateness of the sample distribution, given that the survey conducted related to services for parents of school aged children.

The sampled group represents only 2% of the age range 20 to 40. Given that the survey would be most appropriate for this age group, the sample of them is too low

7) Tom made a table of the numbers of boys and girls in each year group in his school

YEAR	BOYS	GIRLS
1	12	15
2	9	14
3	13	12
4	9	10
5	16	15
6	11	14
7	12	17
8	14	17
9	13	15
10	9	11
11	8	10
12	6	8

Based on his data, approximately how many of the students in Tom's state are female? (The total number of students in Tom's state is 1,120,000)

$$\frac{158}{290} \times 1120000 = 610207$$

8) Peter also made a table of the number of boys and girls in each year group in his school

YEAR	BOYS	GIRLS
1	15	0
2	19	0
3	23	0
4	29	0
5	26	0
6	31	0
7	22	0
8	14	0
9	13	0
10	9	0

©Ezy Math Tutoring | All Rights Reserved

11	8	0
12	6	0

Comment on the suitability of using Peter's data for the same purpose as Tom's, the probable reason for its unsuitability, and what the data could possibly be used to estimate

Since there are no girls at Peter's school, the data could not be used to estimate general number of girls. It could be used to estimate the age breakup of boys in larger areas

9) 100 animals are caught, tagged and released. Later 250 animals are caught, of which 50 have tags. Based on this data what is the approximate population of these animals?

$$\frac{100 \times 250}{5} = 5000$$

10) Based on tagging data, the population of fish in a lake is estimated to be 10000. Of the sample of 300 taken, 45 had tags already placed by a previous catch and release. How many fish were originally tagged and released?

$$\frac{300}{300/45} \times original\ catch = 10000$$

$$Original\ catch = 222$$

©Ezy Math Tutoring | All Rights Reserved

Exercise 2

Mean, Median & Spread of Data

©Ezy Math Tutoring | All Rights Reserved

1) Calculate the mean of the following data sets

a) 2, 4, 6, 8, 10

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{30}{5} = 6$$

b) 0, 2, 4, 6, 8

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{20}{5} = 4$$

c) 1, 3, 5, 7, 9

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{25}{5} = 5$$

d) 2, 2, 2, 2, 2

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{10}{5} = 2$$

e) 10, 30, 40, 50

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{130}{4} = 32.5$$

f) 7, 11, 15, 17, 25, 52, 55

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{182}{7} = 26$$

2) Calculate the mean of the following data sets

a) 2, 4, 5, 7, 8

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{26}{5} = 5.2$$

b) 2, 4, 5, 7, 8, 500

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{526}{6} = 87.666$$

c) 950, 970, 990, 1000, 1100

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{5010}{5} = 1002$$

d) 2, 950, 970, 990, 1000, 1100

©Ezy Math Tutoring | All Rights Reserved

$$\text{Mean} = \frac{\Sigma}{n}$$

$$= \frac{5012}{6} = 835.33$$

3) From your answers to question 2, what effect does an outlier have on the mean of a set of data?

It pushes the mean score toward the outlier (that is the mean becomes higher if the outlier is higher than the rest of the data set and vice versa)

4) The mean of a set of data is 15. The scores in the data set are

18, 3, 15, x, 30, 12, and 20

What is the value of x?

$$\text{Mean} = \frac{\Sigma}{n}$$

$$\frac{\Sigma}{7} = 15$$

$$\Sigma = 15 \times 7 = 105$$

Therefore $x = 7$

5) Fifteen students sat a maths test and their mean mark was 60%. Alan was sick for the test and sat it later. When his score was added to the data set, the mean mark had increased to 62%. What score did Alan get on the test?

$$\text{Mean} = \frac{\Sigma}{n}$$

$$\frac{\Sigma}{15} = 60$$

$$\Sigma = 900$$

$$\frac{900 + Alan}{16} = 62$$

$$900 + Alan = 62 \times 16 = 992$$

$$Alan = 92\%$$

6) There are 15 girls and 15 boys in a class. On a test the girls mean mark was 80%, while the mean mark of the boys was 70%. What was the mean mark for the class?

Sum of girls' scores
$= 15 \times 80 = 1200$

Sum of boys' scores
$= 15 \times 70 = 1050$

$$\Sigma = 1200 + 1050 = 2250$$

$$\text{Mean} = 2250 \div 30 = 75\%$$

7) There are 20 girls and 10 boys in a class. On a test the girls mean mark was 80% while the mean mark of the boys was 70%. What was the mean mark for the class?

Sum of girls' scores
$= 20 \times 80 = 1600$

Sum of boys' scores
$= 10 \times 70 = 700$

$$\Sigma = 1600 + 700 = 2300$$

©Ezy Math Tutoring | All Rights Reserved

Mean $= 2300 \div 30 = 76.666\%$

8) Why are the answers to questions 6 and 7 different, given that the mean marks of the boys and girls in both classes were the same?

There are more girls than boys in the second class. The mean of the set will therefore be pushed toward the mean of the larger group

9) What is the median of the following data sets?

a) 1, 2, 3, 4, 5

3

b) 2, 4, 6, 8, 10

5

c) 9, 12, 15, 22, 30, 40, 60

22

d) 2, 4, 6, 12, 14, 21, 22, 22

Even number of scores, so median is between two middle scores; that is between $12\ and\ 14$

Median $= 13$

10) What is the median of the following data sets?

a) 2, 4, 5, 7, 10

5

b) 2, 4, 5, 7, 10, 1000

Even number of scores, so median is between two middle scores; that is between 5 and 7

Median $= 6$

c) 1000, 982, 979, 977, 960

979

d) 1000, 982, 979, 977, 960, 2

Even number of scores, so median is between two middle scores; that is between 979 and 977

Median $= 978$

11) From your answers to questions 10 and 11, what effect does an outlier have on the median of a set of data?

Little; the median is pushed very slightly at most in the direction of the outlier

©Ezy Math Tutoring | All Rights Reserved

12) The following set of data is in order. Its mean is 30 and its median is 14. What are the values of x and y?

5, 8, x, 12, y, 40, 50, 100

Even number of scores, so median is between two middle scores; that is between 12 and y

Median $= 14$

Therefore $y = 16$

Mean $= \frac{\Sigma}{n}$

$\frac{\Sigma}{8} = 30$

$\Sigma = 30 \times 8 = 240$

$x = 9$

13) Find the range of the following sets of data

a) 1, 2, 5, 7, 10

Range $= 10 - 1 = 9$

b) 3, 6, 18, 19, 100

Range $= 100 - 3 = 97$

c) 1, 1, 1, 1, 1

Range $= 1 - 1 = 0$

d) 17, 3, 18, 22, 30, 4, 10

Range $= 30 - 3 = 27$

e) 40, 30, 20, 10, 0

Range $= 40 - 0 = 40$

f) -5, 7, 15, 22, 40, 51

Range $= 51 - 5 = 46$

14) Find the inter-quartile range of the following data sets

a) 7, 15, 20, 22, 25, 32, 40

Median $= 22$

Lower quartile $= 15$

Upper quartile $= 32$

IQR $= 32 - 15 = 17$

b) 1, 5, 6, 12, 20, 30, 50

Median $= 12$

Lower quartile $= 5$

Upper quartile $= 30$

IQR $= 30 - 5 = 25$

c) 2, 10, 18, 24, 32, 80, 82, 90

Median is score between 24 and 32 $= 28$

©Ezy Math Tutoring | All Rights Reserved

Lower quartile is score between 10 and 18 $= 14$

Upper quartile is score between 80 and 82 $= 81$

IQR $= 81 - 14 = 67$

d) 23, 25, 4, 12, 21, 50, 32, 43, 5, 60, 45

Firstly put scores in order

4, 5, 12, 21, 23, 25, 32, 43, 45, 50, 60

Median $= 25$

Lower quartile $= 12$

Upper quartile $= 45$

IQR $= 45 - 12 = 33$

15) Can the inter-quartile range be less than the range for a set of data? Explain

No, since the range covers the whole data set, and the IQR a subset of this

16) Can the inter-quartile range be equal to the range for a set of data? Explain

Yes, if all scores in the data set are the same, or if the upper quartile is equal to the maximum score, and the lower quartile is equal to the minimum score

17) What is the standard deviation of the following sets of data?

a) 2, 2, 2, 2, 2, 2

0

b) 1, 2, 3, 4, 5

1.414

c) 3, 6, 9, 12, 15

4.24

d) 4, 20, 40, 60, 100

33.41

18) Calculate the mean and standard deviation of the following

a) 2, 4, 6, 8, 10

Mean $= 6$

S.D. $= 2.83$

b) 4, 6, 8, 10, 12

Mean $= 8$

S.D. $= 2.83$

©Ezy Math Tutoring | All Rights Reserved

c) What effect does adding two to every score have on the mean and standard deviation of a set of data?

The mean increases by 2, the standard deviation is unchanged

19) Calculate the mean and standard deviation of the data set

4, 8, 12, 16, 20

Mean $= 12$

S.D. $= 5.66$

What effect does doubling every score have on the mean and standard deviation of a set of data?

The mean and standard deviation are both doubled

©Ezy Math Tutoring | All Rights Reserved

Exercise 3

Representing Data (I)

©Ezy Math Tutoring | All Rights Reserved

1) Create a tally chart and frequency table to represent the following data set more effectively

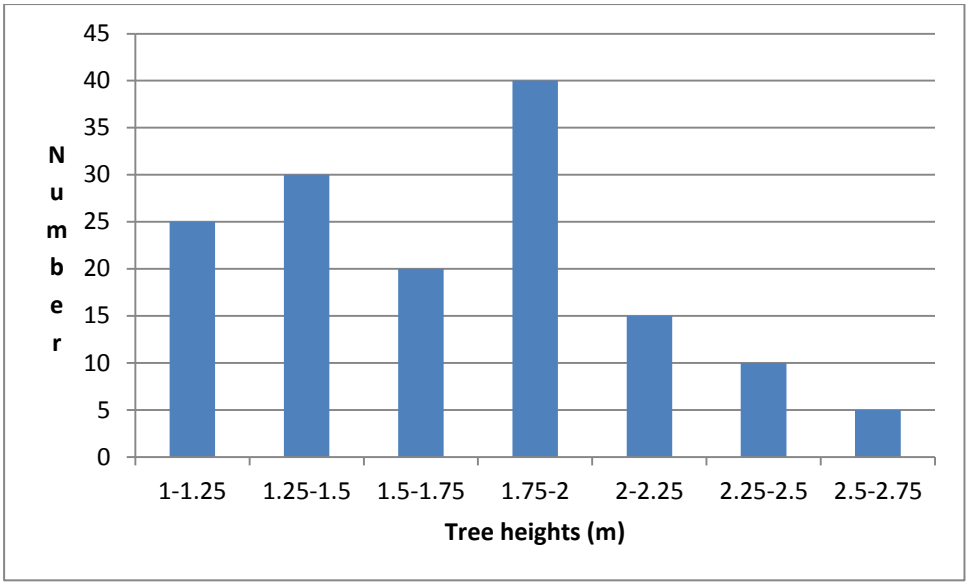
5, 7, 10, 16, 20, 6, 17, 9, 14, 4, 11, 12, 1, 2, 19, 14, 19, 10, 2, 15, 12, 17, 5, 1, 11, 13, 9, 7, 4, 8, 7, 3, 6, 16, 4, 1, 8, 5, 18, 13, 19, 9, 2, 11, 17, 17, 14, 10, 16, 4, 13, 1, 11, 15, 6, 3, 2, 7, 20, 8, 15, 6, 8, 5, 3, 11, 4, 10, 9, 13, 12, 18, 2, 17, 1

Number	Frequency
1	4
2	4
3	2
4	4
5	2
6	3
7	3
8	3
9	3
10	3
11	4
12	2
13	3
14	2
15	2
16	2
17	4
18	
19	2
20	1

2) Construct a frequency histogram for the following grouped frequency table

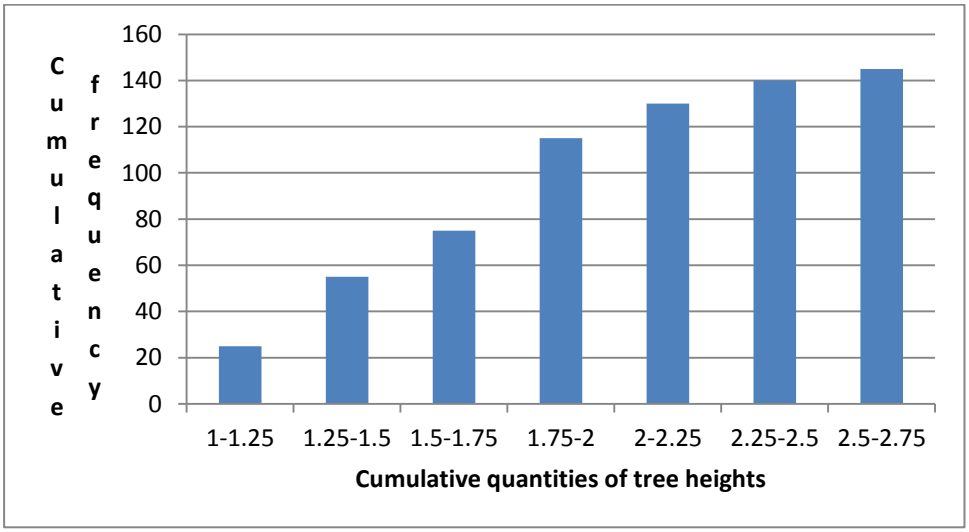
Height of trees (metres)	**Frequency**
1 – 1.25	25
1.25 - 1.5	30
1.5 – 1.75	20
1.75 – 2	40
2 – 2.25	15
2.25 – 2.5	10
2.5 – 2.75	5

©Ezy Math Tutoring | All Rights Reserved



3) Construct a cumulative frequency table and graph for the data from question 2

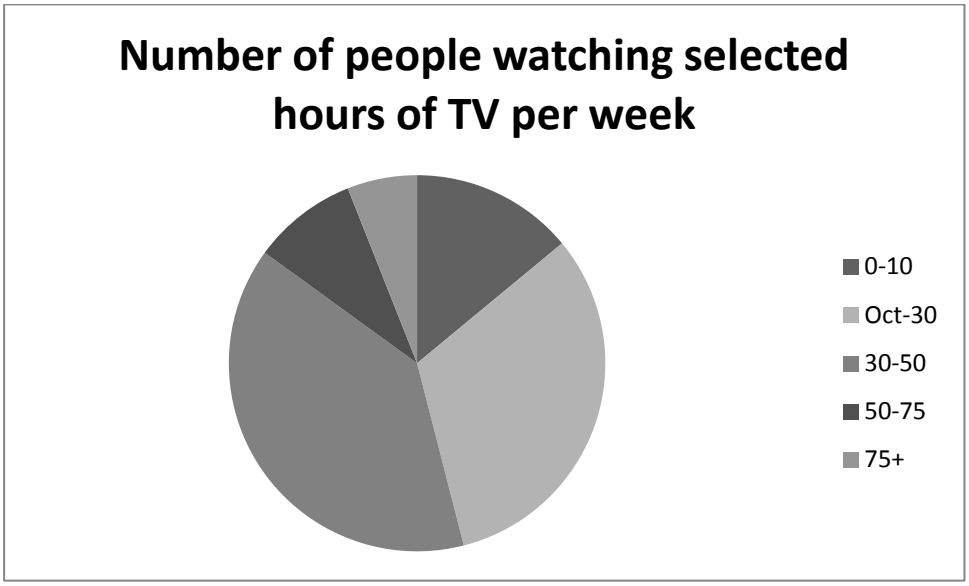
Height of trees (metres)	Frequency	Cumulative frequency
1 – 1.25	25	25
1.25 - 1.5	30	55
1.5 – 1.75	20	75
1.75 – 2	40	115
2 – 2.25	15	130
2.25 – 2.5	10	140
2.5 – 2.75	5	145



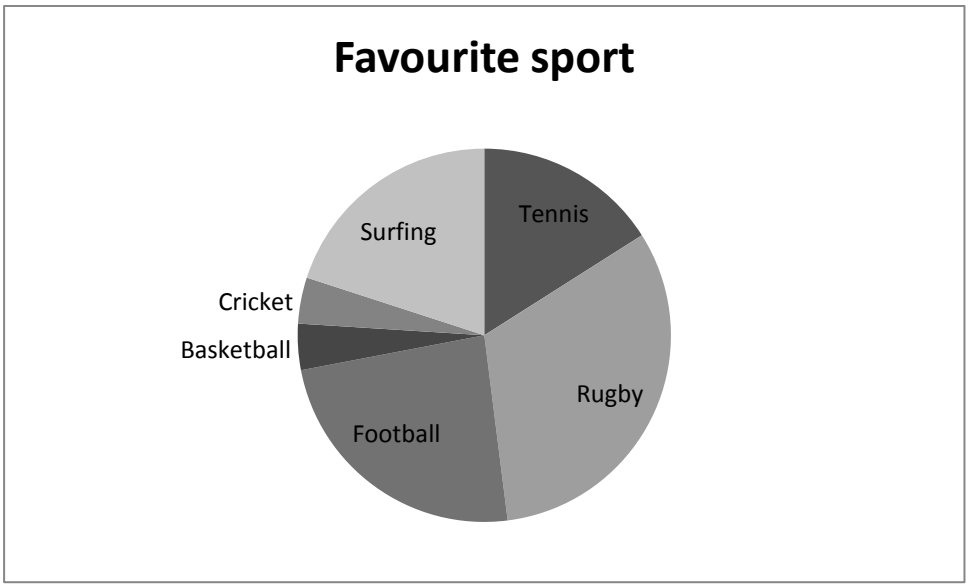
©Ezy Math Tutoring | All Rights Reserved

4) Construct a pie graph to represent the following data

Hours of TV watched per week	Number of people
0-10	14
10-30	32
30-50	39
50-75	9
75+	6



5) Using the following pie graph



©Ezy Math Tutoring | All Rights Reserved

a) Which sport was most popular of those surveyed?

Rugby

b) Which two sports were equally popular?

Cricket and baseball

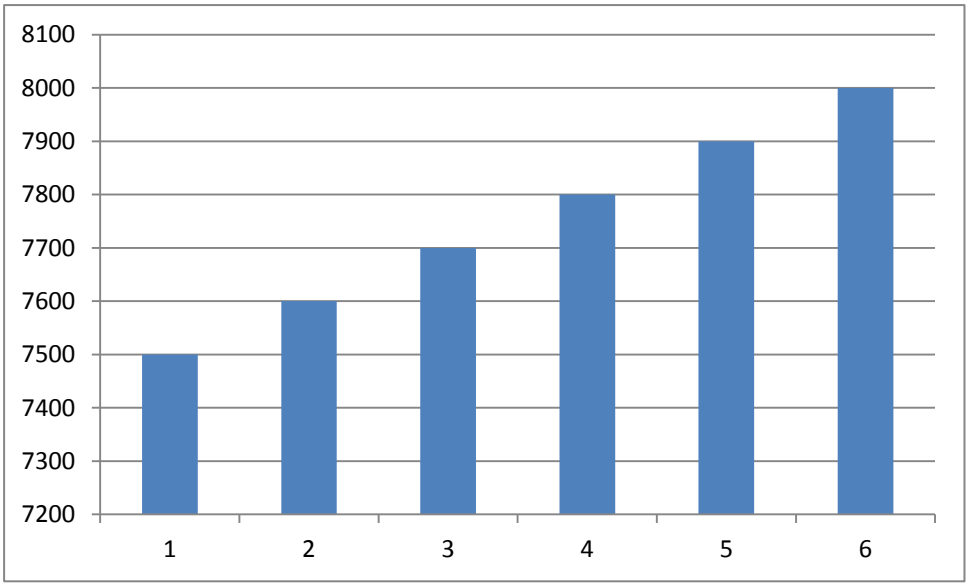
c) Which sport was the favourite of half the number of people who voted for rugby?

Tennis

d) If 50 people chose surfing, approximately how many people were surveyed?

300

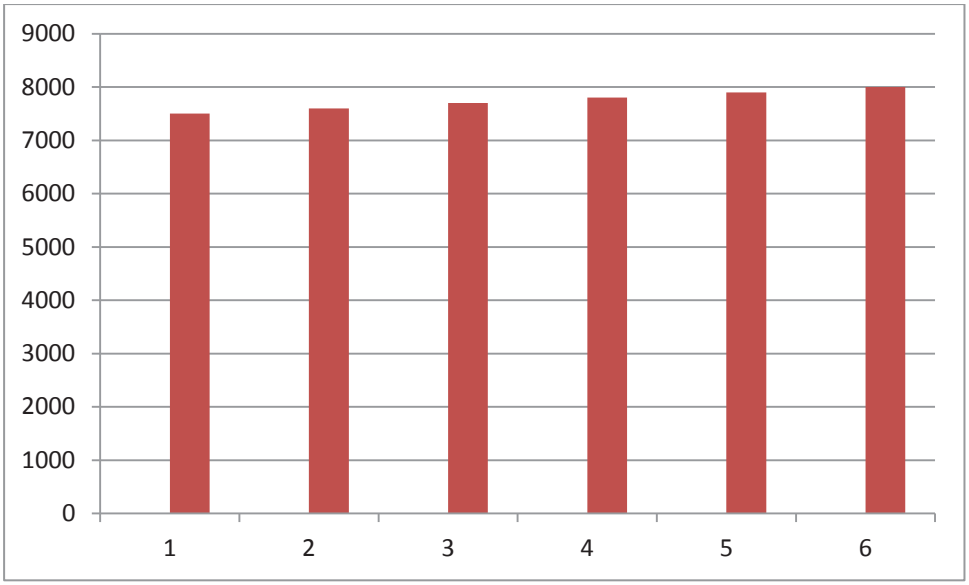
6) Explain why the following graph is misleading, and redraw it so as to make it realistic



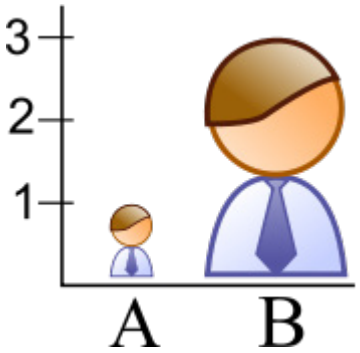
The scale of the y axis begins at 7200, therefore the scale is misrepresented. A difference in quantities between each category looks larger than they actually are

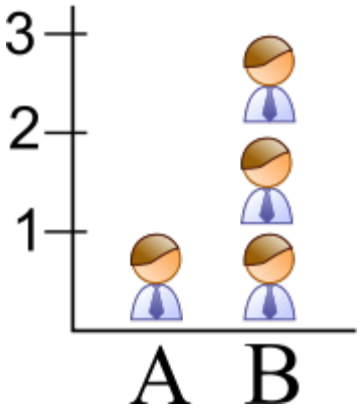
Redraw the graph starting the vertical scale at zero

©Ezy Math Tutoring | All Rights Reserved



7) Which of the picture graphs shown below is less misleading and why?

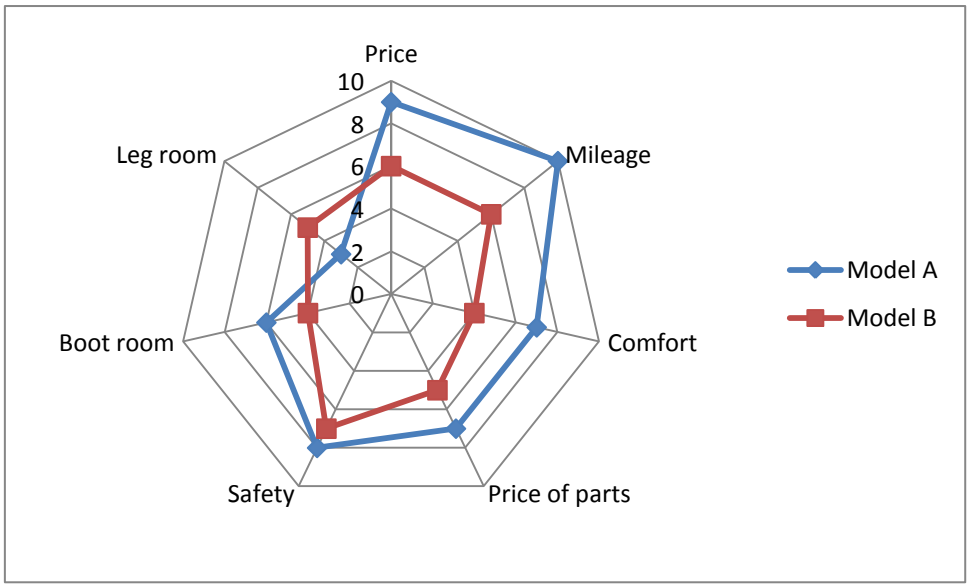




The first graph is more misleading; it shows the same category as two different sizes. It indicates that one person was in each category instead of three in category B

8) A magazine compared two cars named A and B in 7 criteria. The higher the score, the better the value. For example a high price score indicates that a car is cheaper, whilst a high safety score indicates that a car is safer

©Ezy Math Tutoring | All Rights Reserved



a) Which car is cheaper and by what fraction?

B is 33.3% cheaper than A

b) Which car has more leg room?

B

c) Which feature scored almost the same for both cars?

Safety

d) What was the only category in which car B performed better than car A?

Leg room

©Ezy Math Tutoring | All Rights Reserved

Exercise 4

Representing Data (II)

©Ezy Math Tutoring | All Rights Reserved

1) Represent the following data set in a stem and leaf plot and determine the median score using the plot

14, 15, 16, 16, 22, 23, 23, 23, 23, 24, 26, 31, 32, 38, 39, 44, 44, 45, 46, 47, 47, 47, 48

1	4,5,6,6,
2	2,3,3,3,3,4,6
3	1,2,8,9
4	4,4,5,6,7,7,7,8

Median $= 31$

2) The daily maxima for Perth during the month of June 2012 were

19, 20, 22, 24, 23, 23, 17, 20, 21, 21, 19, 21, 20, 17, 18, 19, 18, 21, 24, 21, 16, 16, 17, 18, 19, 15, 21, 20, 19, 17,

Represent this data in a stem and leaf plot.

1	5,6,6,7,7,7,7,8,8,8,9,9,9,9,9
2	0,0,0,0,1,1,1,1,1,1,2,3,3,4,4

What was the median maximum temperature in Perth for June?

19.5

3) The following data set is the set of scores of football team A during its season

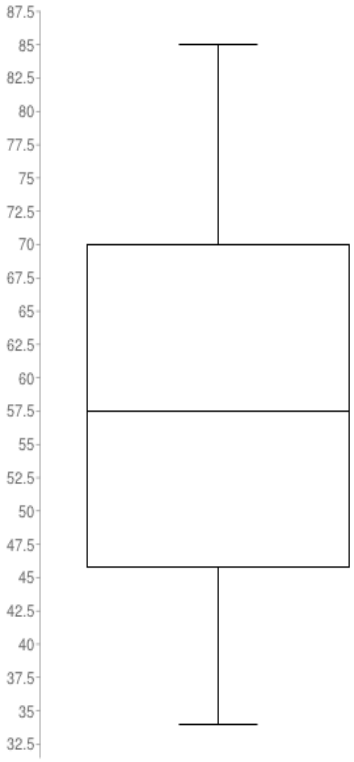
34, 38, 42, 43, 45, 48, 49, 51, 53, 57, 58, 60, 61, 63, 67, 71, 74, 77, 79, 85

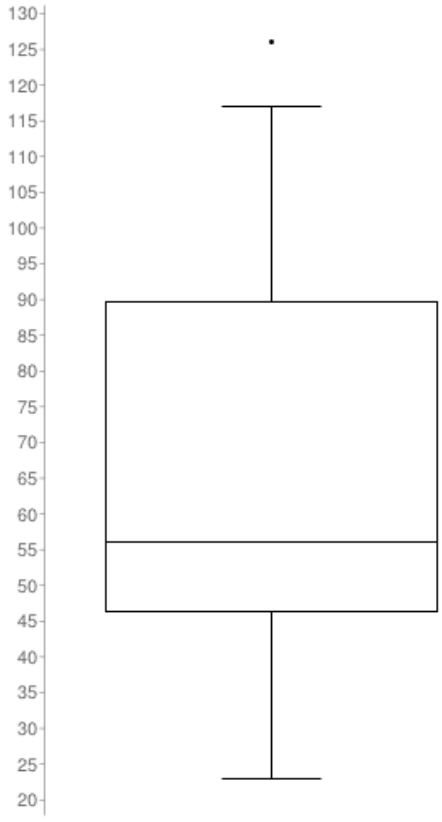
The following data set is the set of scores of football team B during its season

23, 29, 35, 39, 46, 47, 49, 52, 53, 53, 59, 67, 73, 79, 86, 91, 97, 101, 117, 126

Display the data in a back to back stem and leaf plot

©Ezy Math Tutoring | All Rights Reserved





What were the respective median scores, and which team was more consistent during the season?

©Ezy Math Tutoring | All Rights Reserved

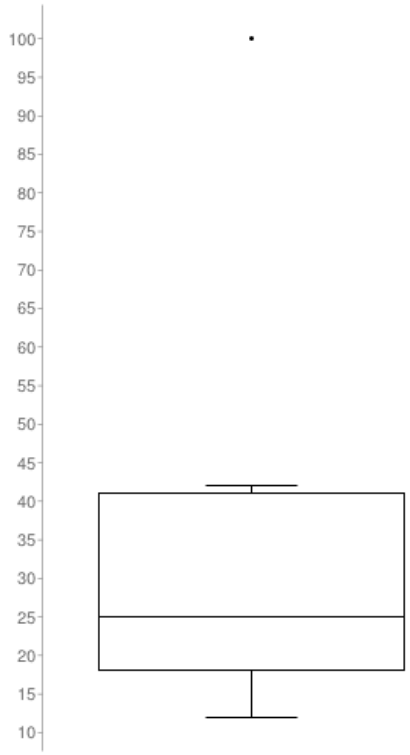
The median for team A was 57.5, and the median for team B was 56

Team a was more consistent; it had a smaller range and interquartile range

4) Represent the following data set in a box and whisker plot

12, 16, 20, 24, 25, 30, 40, 42, 100

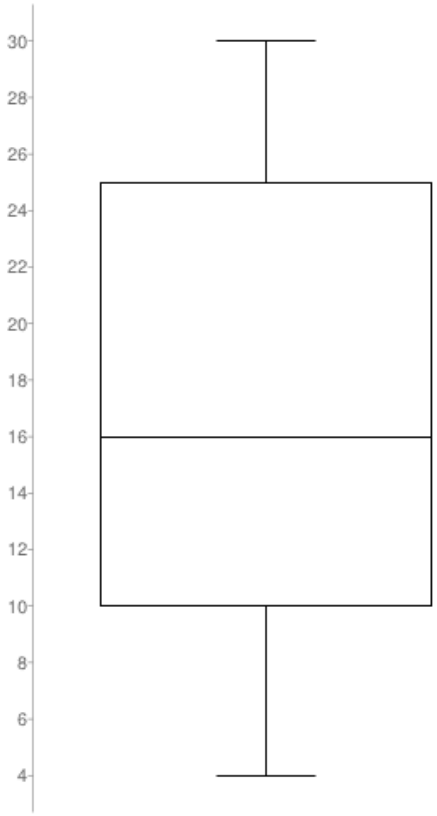
Show and evaluate the range and the inter-quartile range



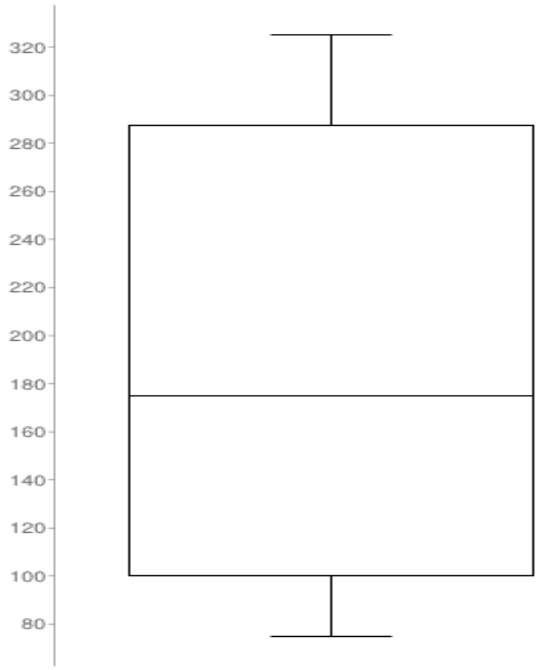
Note that 100 is an outlier

5) A set of data has a minimum of 4, an inter-quartile range of 15; range of 26 and a third quartile of 25. Draw a possible box and whisker plot for this data

©Ezy Math Tutoring | All Rights Reserved

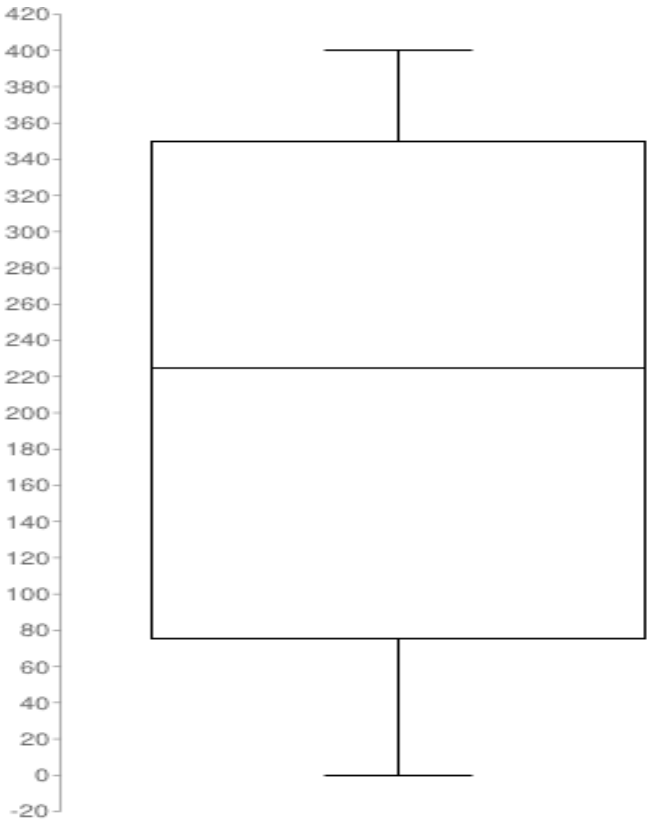


6) The following box plot shows the distribution of the average rainfall for Great Lake for the past 40 years



©Ezy Math Tutoring | All Rights Reserved

The following box plot shows the same data set for Water World



a) Which site has the greater median average rainfall?

Water World (approximately 225)

b) Which site has the record lowest annual rainfall and record highest annual rainfall?

Water World has both, 0 lowest and 400 highest

c) Which site has the greater variation in average rainfall?

Water World

d) Which site has a greater chance of receiving 300 inches or more of rain?

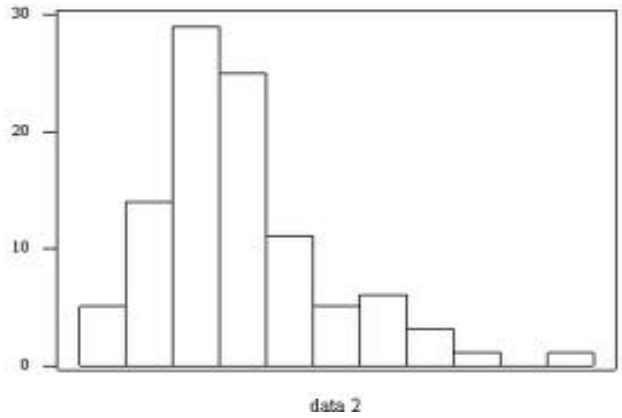
Water World (greater than 25% of its rainfall is more than 300 inches)

©Ezy Math Tutoring | All Rights Reserved

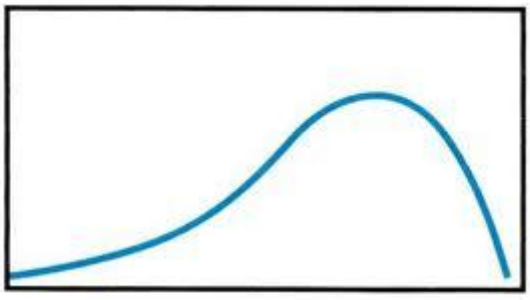
e) Too much or too little rain affects the water levels in the dam to the point where water skiing is too dangerous. Which site would give a person a better chance of being able to water ski?

Although Water World has the best chance of receiving more than 300 inches of rain, its rainfall is also more variable, and hence has a greater chance of receiving too much or too little rain. Therefore Great Lake would offer a person a better chance of water skiing

7) Describe the following graphs in terms of skewness

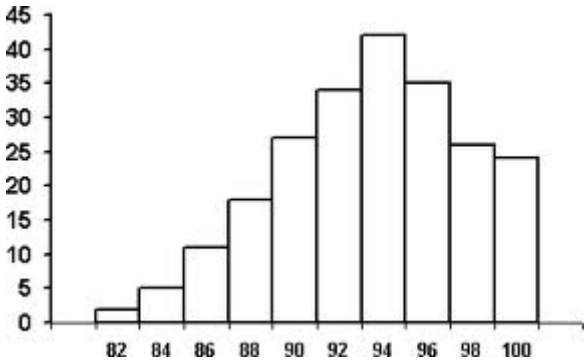


Skewed toward the left (lower values)

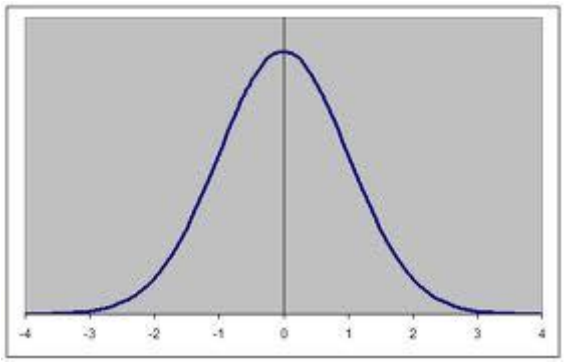


Skewed toward the right (higher values)

©Ezy Math Tutoring | All Rights Reserved

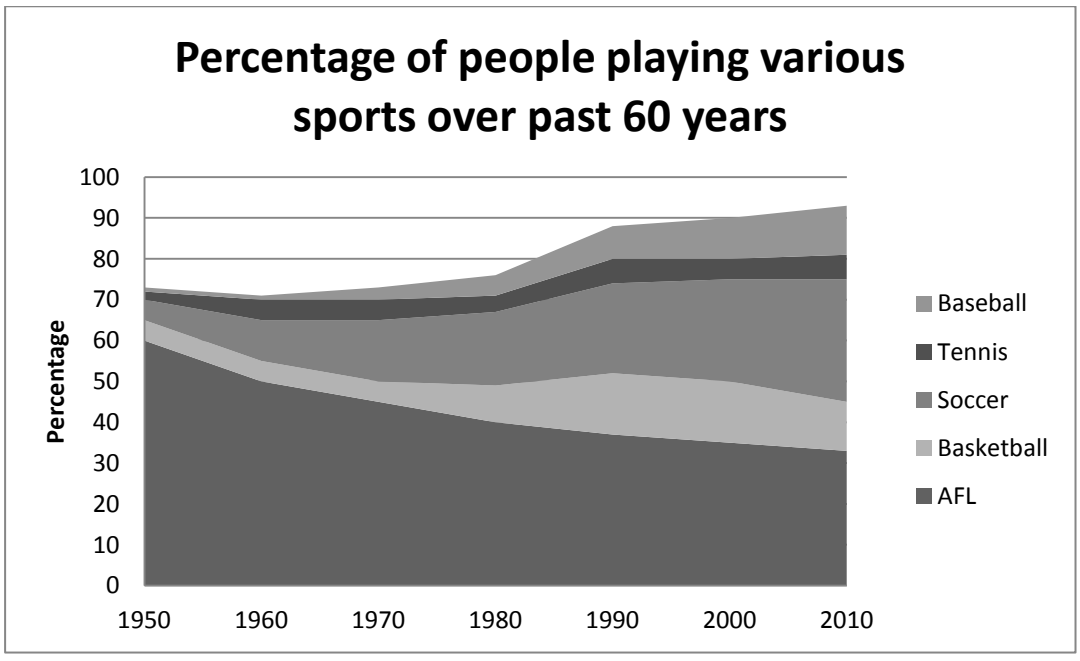


Slightly skewed to the right



No skew (normal distribution)

8) Answer the questions below by using the following area graph



a) Which sport has had a steady decline in percentage participation rates?

©Ezy Math Tutoring | All Rights Reserved

AFL

b) To which sport has most of this percentage gone to?

Soccer

c) Which sport had the most rapid increase in participation percentage in the 1980s?

Basketball

d) During which year was the total participation in these sports combined the highest?

2010

e) Has the number of people playing AFL fallen over the past 60 years? Explain your answer.

Not necessarily. The percentage of people playing AFL has declined, but the population has also increased. Without actual population figures one cannot say for certain, however there is a strong chance that participation numbers have still increased

f) The participation rate for which sport has remained relatively constant?

Tennis

9) Answer the questions based on the following table

	Studied for test	**Did not study for test**	
Passed test	80	20	100
Failed test	10	90	100
	90	110	

©Ezy Math Tutoring | All Rights Reserved

a) What percentage of students passed the test?

$$\frac{100}{200} = 50\%$$

b) What percentage of students who studied for the test passed it?

$$\frac{80}{90} = 89\%$$

c) What percentage of students who did not study for the test failed?

$$\frac{90}{110} = 82\%$$

d) If you failed the test what is the chance that you did not study?

$$\frac{90}{100} = 90\%$$

10) 500 people were asked their preferred colour from red and blue. There were 150 women, 100 of whom liked blue. 200 men preferred red. What percentage of men preferred blue?

	Red	**Blue**	
Men	200	150	350
Women	50	100	150
	250	250	500

$$\frac{150}{350} = 43\%$$

©Ezy Math Tutoring | All Rights Reserved

Exercise 5

Normal Distribution

©Ezy Math Tutoring | All Rights Reserved

1) Describe what the following z values tell us about the data point in relation to the mean

a) $z = 0$

Equals the mean

b) $z = 1$

One standard deviation above the mean

c) $z = -2$

Two standard deviations below the mean

d) $z > 2$

More than two standard deviations above the mean

2) Calculate the z score of a score of 8 in a data set that has a mean of 6 and a standard deviation of 2. Describe the position of the data point in relation to the mean

$$z = \frac{x - \mu}{\sigma} = \frac{8 - 6}{2} = 1$$

Score is one standard deviation above the mean

3) A data point has a z score of 1.5. The data set has a mean of 5 and a standard deviation of 3. What is the data point?

$$z = \frac{x - \mu}{\sigma}$$

$$1.5 = \frac{x - 5}{3}$$

$$x = 4.5 + 5 = 9.5$$

4) A data set has a mean of 17.5. The data point 33.5 is 1.6 standard deviations from the mean. What is the value of the standard deviation?

©Ezy Math Tutoring | All Rights Reserved

$$z = \frac{x - \mu}{\sigma}$$

$$1.6 = \frac{33.5 - 17.5}{\sigma}$$

$$\sigma = \frac{33.5 - 17.5}{1.6} = 10$$

5) The data point 41 lies within a set of data having a standard deviation of 6. If the data point is 4 standard deviations from the mean, what is the value of the mean?

$$z = \frac{x - \mu}{\sigma}$$

$$4 = \frac{41 - \mu}{4}$$

$$41 - \mu = 16$$

$$\mu = 25$$

6) If a set of data is normally distributed what percentage of the scores are within 1 standard deviation from the mean?

Approximately 68.2%

7) 95% of people in a group are between 77kg and 103 kg. What is the mean and standard deviation if we assume the data is normally distributed?

Approximately 2 standard deviations contain 95% of normally distributed data

$$103 - 77 = 26$$

26 represents 4 standard deviations (2 above the mean and 2 below)

Therefore $\sigma = 6.5, and\ \mu = 90$

8) A teacher gives a maths test with the pass mark being 25 out of 50. The class scores the following marks:

12, 14, 10, 22, 35, 38, 13, 22, 40, 11, 22, 24, 25, 30, 5, and 18

©Ezy Math Tutoring | All Rights Reserved

The teacher sees that the majority of the class will fail the test, and he decides to standardise the marks. He will only fail a student that is more than one standard deviation below the mean

How many students now pass the test?

Mean of scores $= 341 \div 16 = 21.3125$

Standard deviation 10.10395

Therefore the only students that will fail are those that scored $10, 11, and\ 5$

9) Another teacher is determining the term marks for his class and wants to grade according to the following formula

Standard Deviations from mean	Grade
Score ≥2 s.d.	A
1 s.d. ≤ score < 2 s.d.	B
0 s.d. ≤ score < 1 s.d.	C
-1 s.d. ≤ score < 0 s.d.	D
Score< -1 s.d	E

Grade the following students

NAME	SCORE	GRADE
James	62	C
Mark	38	E
Karen	90	A
Janine	70	C
Carol	65	C
June	68	C
Peter	44	E
Kevin	48	D
Brian	56	D
Alan	66	C
Bree	53	D

$\mu = 60.9, \sigma = 13.6$

©Ezy Math Tutoring | All Rights Reserved

10) Deliveries of sand made by a nursery are advertised as 100 kg. The mean of the deliveries is 100 kg with a standard deviation of 1.2 kg

a) Within what weight range will 95% of the deliveries be?

95% covers 2 standard deviations above and below the mean

Range is therefore from 97.6 to 102.4 kg

b) What percentage of deliveries will be between 100 kg and 101.2 kg?

Approximately 34% of data is contained in one standard deviation above the mean

c) The company offers money back if any of the deliveries are 3 or more standard deviations below the mean. If they made 5000 deliveries in one month, how many of these will have to be refunded?

Approximately 0.1% of normally distributed data is 3 or more standard deviations below the mean.

$0.1\% \times 5000 = 5$ deliveries

(Assume the data is normally distributed)

©Ezy Math Tutoring | All Rights Reserved

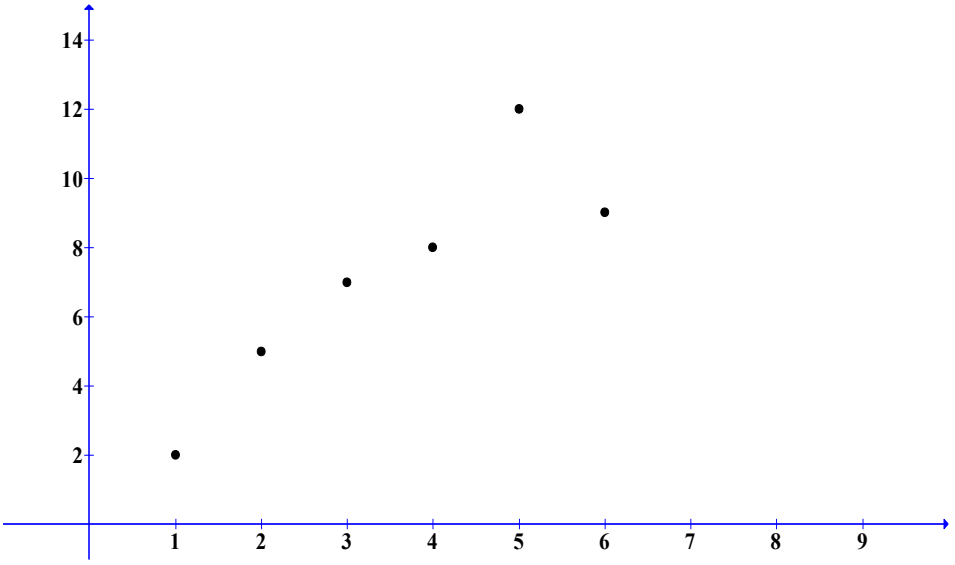
Exercise 6

Correlation

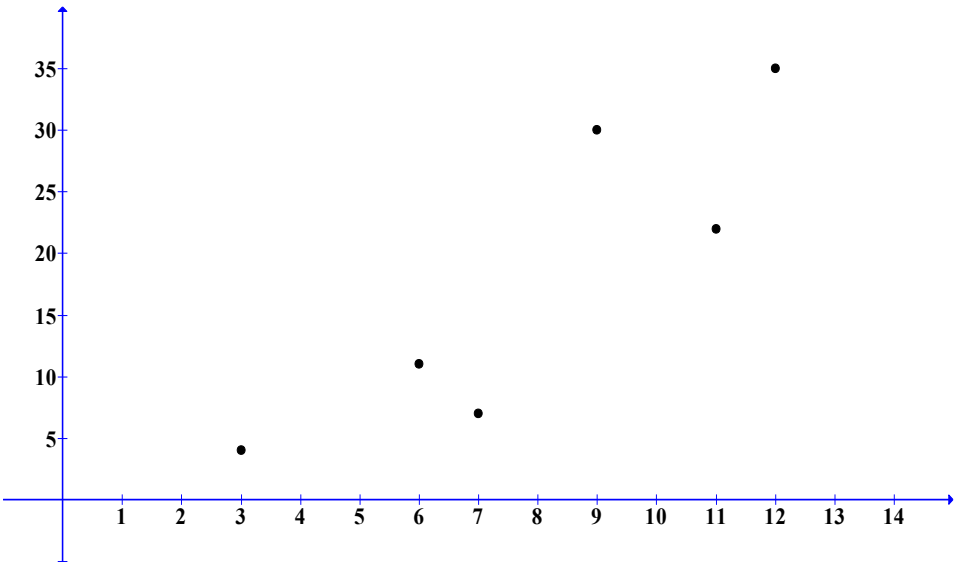
©Ezy Math Tutoring | All Rights Reserved

1) Plot the following sets of ordered pairs on their own scatter plot

a) $(1,2)\ (2,5)\ (3,7)\ (4,8)\ (5,12)\ (6,9)$

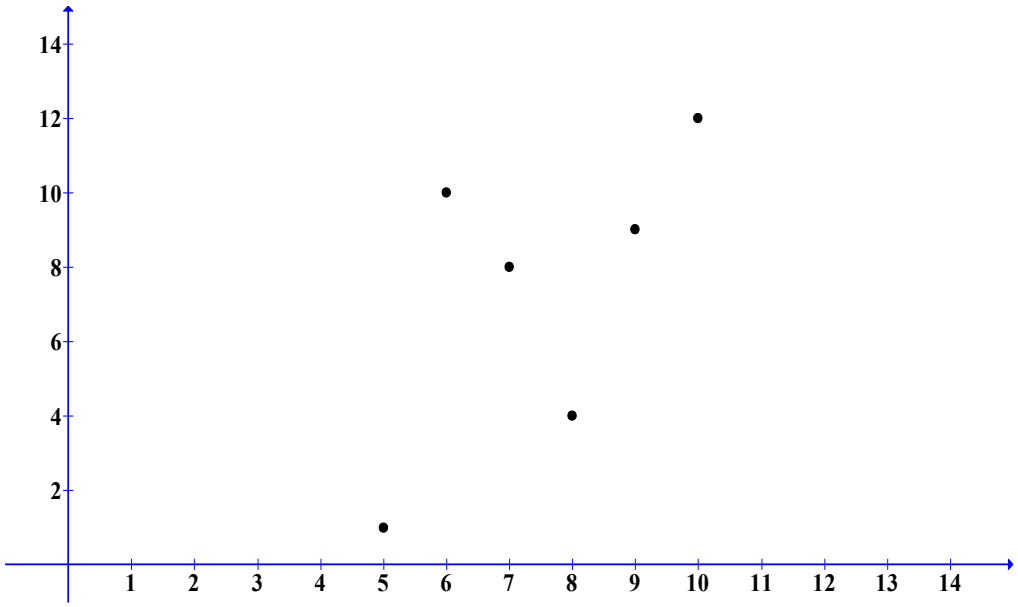


b) $(3,4)\ (6,11)\ (7,7)\ (9,30)\ (11,22)\ (12,35)$

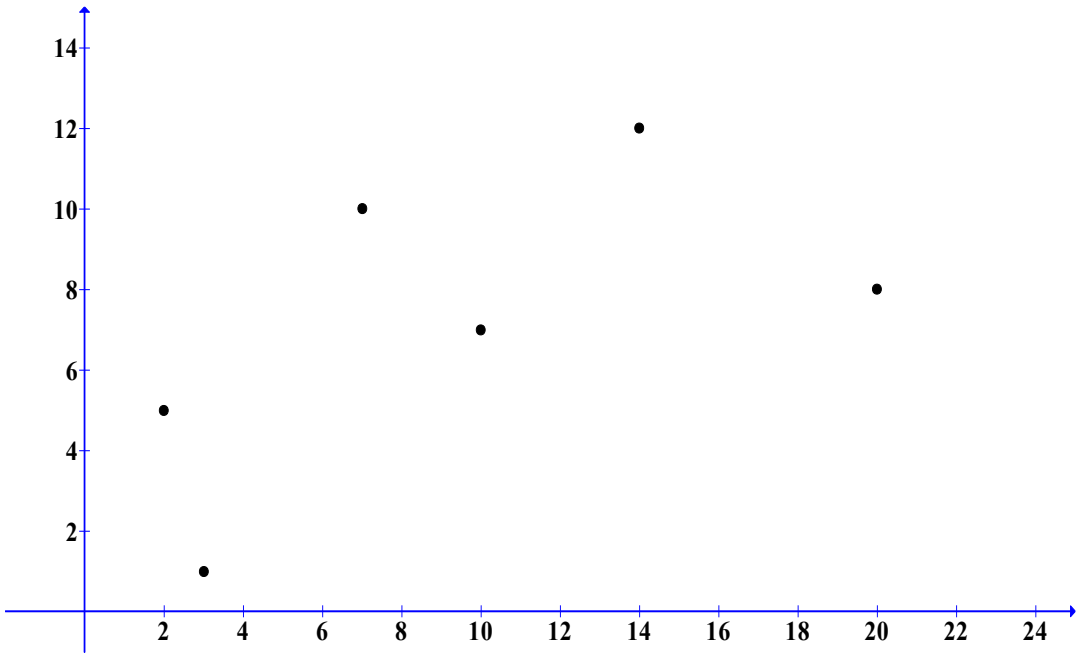


©Ezy Math Tutoring | All Rights Reserved

c) $(10, 12)$ $(9, 9)$ $(8, 4)$ $(7, 8)$ $(6, 10)$ $(5, 1)$

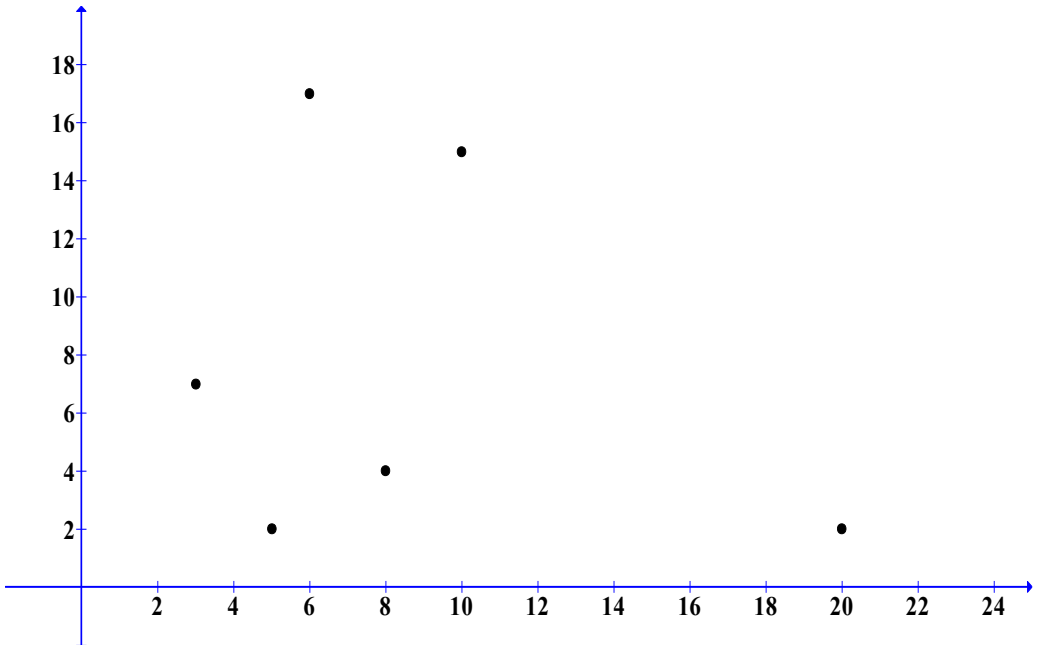


d) $(20, 8)$ $(14,12)$ $(10, 7)$ $(7, 10)$ $(3, 1)$ $(2,5)$

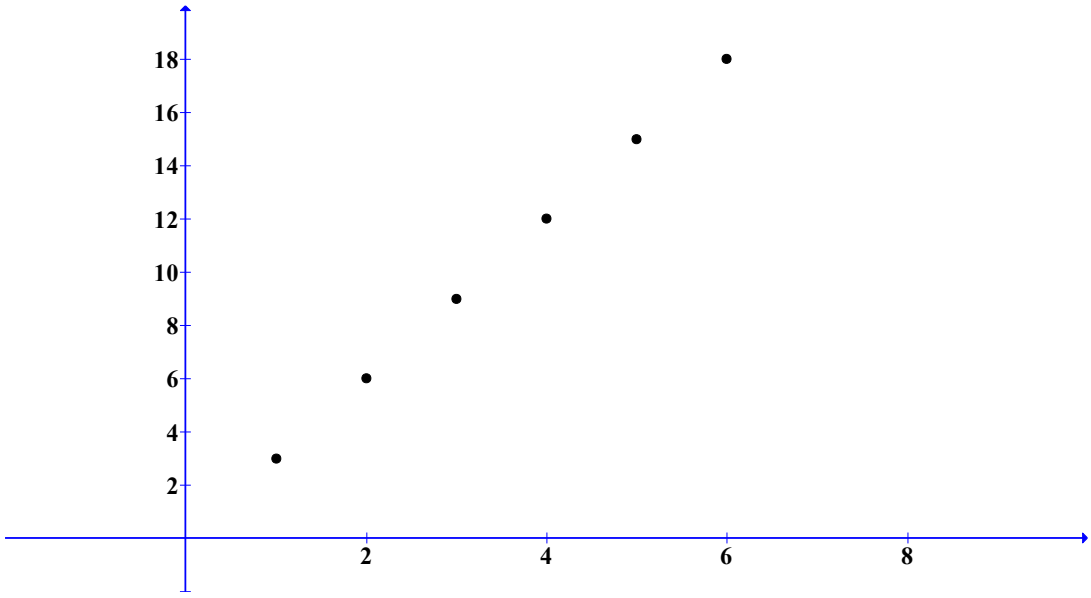


©Ezy Math Tutoring | All Rights Reserved

e) (20, 2) (10,15) (3, 7) (8, 4) (5, 2) (6,17)



f) (4,12) (2,6) (3, 9) (1, 3) (5, 15) (6, 18)



©Ezy Math Tutoring | All Rights Reserved

2) For each set of data points in question 1, describe the relationship between the points as strong/medium/weak and positive/negative. Also indicate if any relationship is perfect or there is no relationship at all.

a) Strong positive

b) Medium positive

c) Weak positive

d) Medium positive

e) No real relationship

f) Perfect positive

3) For any set of data from question 1 for which there is a relationship, draw the line of best fit through the data, and determine the gradient and vertical intercept. Hence determine the equation of the line of best fit

NOTE: Equations may differ due to different fit, but should be consistent with solutions in terms of approximate slope and y intercept

a) $y = 1.6x + 1.5$

b) $y = 3.3x - 8.5$

c) $y = 1.4x - 3$

d) $y = 0.4x + 4$

e) XX

f) $y = 3x$

©Ezy Math Tutoring | All Rights Reserved

4) For each of the equations derived in question 3, predict the y value obtained when substituting the point $(3, y)$ into the equation

a) 6.3

b) 0.8

c) 1.2

d) 5.2

e) XX

f) 9

5) Explain why you could not predict the y value of the point $(40, y)$ in any of the equations above

The value $x = 40$ does not fall into the range of the data set, therefore predictions cannot be accurately made

6) Describe the relation between the two variables of a scatter plot that have the following correlation coefficients

a) $r = 1$

Perfect positive

b) $r = 0.8$

Strong positive

c) $r = -0.1$

Weak negative

d) $r = 0.6$

Medium positive

©Ezy Math Tutoring | All Rights Reserved

e) $r = -0.85$

Strong negative

f) $r = 0.09$

No meaningful correlation

7) When the relationship between the sale of blankets in Canada and the sale of air conditioners in Australia at different times of a year is graphed in a scatter plot, the correlation coefficient for the line of best fit is 0.8. Does this mean that the number of air conditioners bought in Australia affects the number of blankets bought in Canada? Explain your answer

No, the high correlation is due to a third factor that affects both data sets. When it is summer in Australia (when sales of air conditioners are high) it is winter in Canada (which would increase blanket sales). One of the two does not cause the other

8) A scatter plot was produced that showed the relationship between the average life expectancy and the number of television sets per person for a number of countries. The correlation coefficient was very high $(r = 0.92)$. Does this mean that in order to increase life expectancy in third world countries, simply introduce more television sets? Explain your answer

No, the high correlation is due to the fact that countries that have more television sets are more likely to have better technology and industry. These countries would generally be able to provide a better standard of living and health care, hence increasing life expectancy

9) Describe the likely scatter plot between the ages and heights of a randomly selected group of 5000 people. What do you think the value of the correlation coefficient may be, and are there any restrictions on the validity of the correlation coefficient? Explain your answer

There would be a strong positive relationship between the two variables, as people generally grow taller as they get older. However the data would only exhibit this strong correlation for people up to about 25, as growth is usually finished by then. In fact as people get very old they trend to shrink a little, and this would produce a negative correlation for the correctly selected age groups!

©Ezy Math Tutoring | All Rights Reserved



General Mathematics

Measurement

©Ezy Math Tutoring | All Rights Reserved

Exercise 1

Units of Measurement

©Ezy Math Tutoring | All Rights Reserved

1) Convert the following to cm

a) 8 mm

0.8 cm

b) 1.5 m

150 cm

c) 0.3 km

30,000 cm

d) 412 mm

41.2 cm

e) 22.65 m

0.2265 cm

f) 0.025 km

25

2) Convert the following to m^2

a) 4900 cm^2

0.49

b) 0.04 km^2

40,000

c) 320000 mm^2

0.32

©Ezy Math Tutoring | All Rights Reserved

d) 0.005 km^2

5000

e) 22250 cm^2

2.25

3) Brian uses a ruler marked in centimetres to measure the lengths of various lines. What is the percentage error for each of the following measurements?

a) 400 cm

$$\frac{0.5}{400} = 0.125\%$$

b) 12 cm

$$\frac{0.5}{12} = 4.167\%$$

c) 2 m

$$\frac{0.5}{200} = 0.25\%$$

d) 1200 mm

$$\frac{0.5}{120} = 0.4167\%$$

e) 0.3 km

$$\frac{0.5}{30000} = 0.00167\%$$

f) 3000 cm

$$\frac{0.5}{3000} = 0.0167\%$$

©Ezy Math Tutoring | All Rights Reserved

4) Convert the following to metres per minute

a) 3 km per second

3 km per second = 3000 m per second = 180000 m per minute

b) 10000 mm per hour

10000 mm per hour = 10 m per hour = 0.167 m per minute

c) 1500 m per day

1500 m per day = 62.5 m per hour = 1.04 m per minute

d) 20 km per hour

20 km per hour = 20000 m per hour = 333.33 m per minute

e) 525.6 km per year

525.6 km per year = 1.44 km per day = 1440 m per day = 60 m per hour = 1 m per minute

5) The concentration of an additive in a solution is 1:500000. How much additive is present in the following amounts of solution?

a) 1 kg

2 mg

b) 800 g

1.6 mg

c) 10 kg

20 mg

©Ezy Math Tutoring | All Rights Reserved

d) 0.6 kg

1.2 mg

e) 10000 g

20 mg

f) 300 kg

600 mg

6) The concentration of an additive in a solution is 1 mg per 750 ml. How much additive is there in the following volumes?

a) 2 litres

2.67 mg

b) 500 ml

0.67 mg

c) 3 litres

4 mg

d) 20 litres

26.67 mg

e) How much solution is there if it contains 12 g of additive?

9 litres

7) What percentage of the original quantity remains after the following additions and reductions occur?

©Ezy Math Tutoring | All Rights Reserved

a) There is an increase of 10% then a decrease of 10%

$100 \times 110\% = 110$

$110 \times 90\% = 99$

99% of original remains

b) There is a decrease of 10% followed by an increase of 10%

$100 \times 90\% = 90$

$90 \times 110\% = 99$

99 % of original remains

c) There is an increase of 50% followed by a decrease of 50%

$100 \times 150\% = 150$

$150 \times 50\% = 75$

75% of original remains

d) There is an increase of 100% followed by a decrease of 100%

$100 \times 200\% = 200$

$200 - 100\% = 0$

Nothing remains

e) Does the answer change if the decrease occurs before the increase?

No

f) Develop a formula to calculate the above changes in one step, and validate it by checking it against the answer for a 20% decrease followed by a 20% increase.

©Ezy Math Tutoring | All Rights Reserved

$$x\left(1-\frac{y^2}{10000}\right)$$

Where x is the original amount and y is the percentage change

Substituting $x = 100\ and\ y = 20$

Amount remaining $= 100\left(1-\frac{400}{10000}\right) = 100 - 4 = 96$

8) The recommended dosage of a medicine is 5 ml plus an extra 1.5 ml per kg of weight of the patient over 50kg. What dosage should be given to patients with the following weights?

a) 41 kg

5 ml

b) 103 kg

5 ml $+1.5 \times (103 - 50) = 84.5\ ml$

c) 75 kg

5 ml $+1.5 \times (75 - 50) = 42.5\ ml$

d) 30 kg

5 ml

e) If a patient was given 20 ml of the medicine, what was their weight?

5 ml $+1.5 \times (x - 50) = 20\ ml$

Solving gives weight = 60 kg

©Ezy Math Tutoring | All Rights Reserved

9) Two powders (A and B) are to be mixed in the ratio 3:5. How much of powder A must be added to the following quantities of powder B?

a) 1.5 kg

$1.5 \div 5 = 0.3$

$0.3 \times 3 = 0.9\ kg$

b) 600 g

$600 \div 5 = 120$

$120 \times 3 = 360\ g$

c) 10 kg

$10 \div 5 = 2$

$2 \times 3 = 6\ kg$

d) 200 mg

$200 \div 5 = 40$

$40 \times 3 = 120\ g$

e) 1.4 g

$1.4 \div 5 = 0.28$

$0.28 \times 3 = 0.84\ g$

f) 1000 kg

$1000 \div 5 = 200$

$200 \times 3 = 600\ kg$

©Ezy Math Tutoring | All Rights Reserved

10) Solve the following

a) A mixture to make 12 cakes needs 300g of sugar, how much sugar is needed to make 16 cakes?

$$\frac{16}{12} \times 300 = 400$$

b) A car requires 65 litres of fuel to travel 800 km, how much fuel does it need to travel 900 km?

$$\frac{900}{800} \times 65 = 73.125$$

c) A plate of radius 10 cm holds 30 biscuits laid flat. What is the radius of a plate that holds 8 biscuits?

$$\frac{8}{30} \times 10 = 2.67$$

d) 15 cats require a total of 2.25 kg of food per day. How much food is needed for 35 cats in 2 days?

e) In 6 minutes a train travels 25 km. If its speed is constant, how far will it travel in 11 minutes?

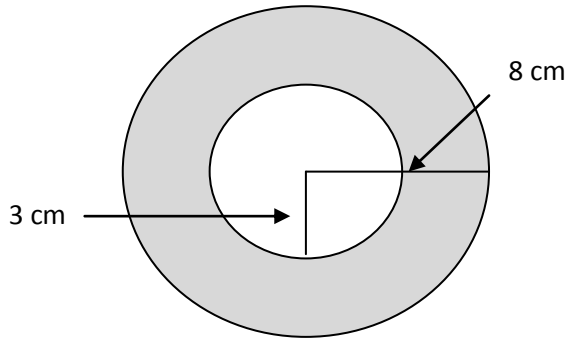
©Ezy Math Tutoring | All Rights Reserved

Exercise 2

Applications of Area & Volume

©Ezy Math Tutoring | All Rights Reserved

1) Calculate the area of the annulus



Area of large circle $= \pi r^2 = \pi \times 8 \times 8 = 64\pi$

Area of white circle $= \pi \times 3 \times 3 = 9\pi$

Area of annulus $= 55\pi\ cm^2$

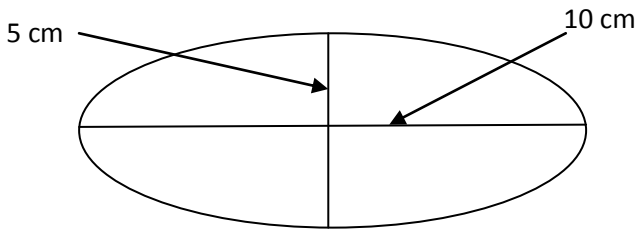
2) If the radius of the larger circle from question 1 is halved, and the radius of the smaller circle is doubled, what is the change in the area of the new annulus formed?

New area of outside circle $= \pi \times 6 \times 6 = 36\pi$

New area of inside circle $= \pi \times 4 \times 4 = 16\pi$

Area of new annulus $= 20\pi\ cm^2$

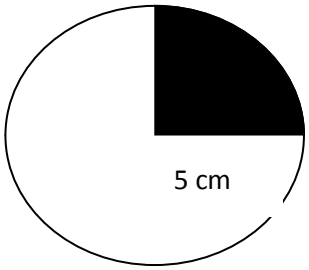
3) Calculate the area of the following figure



Area of ellipse $= \pi \times 5 \times 10 = 50\pi\ cm^2$

©Ezy Math Tutoring | All Rights Reserved

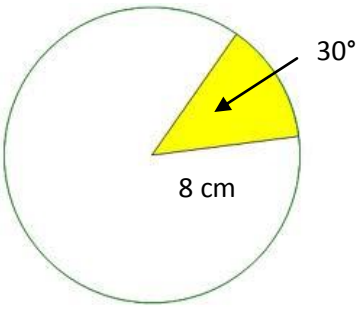
4) Calculate the shaded area



Area of whole circle $= \pi \times 5 \times 5 = 25\pi$

Area of quarter circle $= \frac{25}{4}\pi\ cm^2$

5) Calculate the shaded area

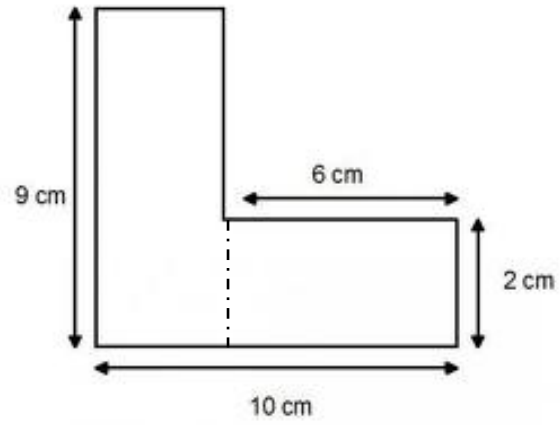


Area of sector $= \frac{30}{360} \times \pi \times 8 \times 8 = \frac{16}{3}\pi\ cm^2$

For questions 6 – 9, calculate the total area of each composite shape

©Ezy Math Tutoring | All Rights Reserved

6)

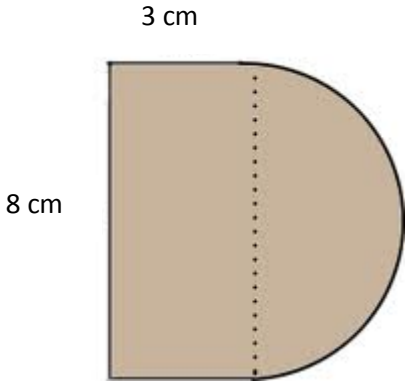


Area of left hand rectangle $= 9 \times 4 = 36$

Area of right hand rectangle $= 6 \times 2 = 12$

Total area $= 48\ cm^2$

7)



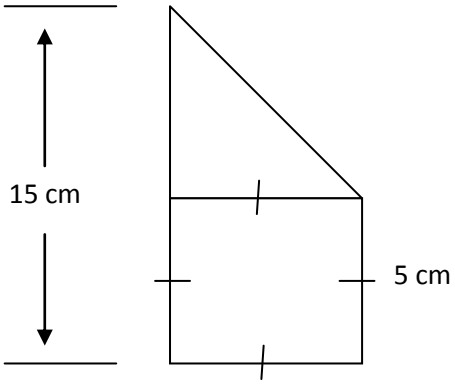
Area of rectangle $= 8 \times 3 = 24$

Area of semicircle $= \frac{1}{2} \times \pi \times 4 \times 4 = 8\pi$

Total area $\cong 49.12\ cm^2$

©Ezy Math Tutoring | All Rights Reserved

8)

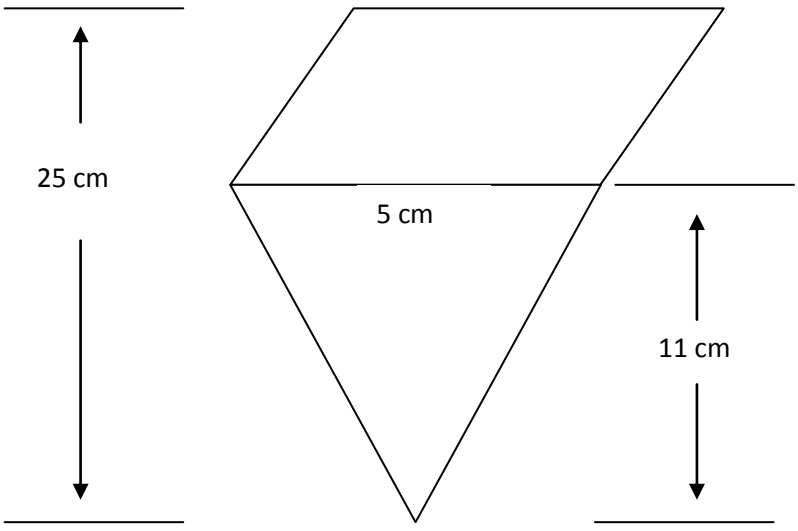


Area of triangle $= \frac{1}{2} \times 15 \times 5 = 37.5$

Area of square $= 5 \times 5 = 25$

Total area $= 62.5\ cm^2$

9)



Area of triangle $= \frac{1}{2} \times 5 \times 11 = 27.5$

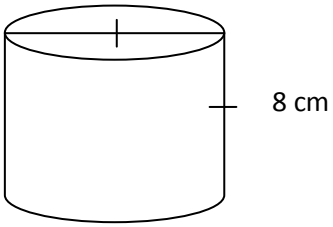
Area of parallelogram $= 5 \times 25 = 125$

Total area $= 152.5\ cm^2$

©Ezy Math Tutoring | All Rights Reserved

10) Calculate the surface area of the following cylinders (parts c and d are open cylinders; they have no top or bottom)

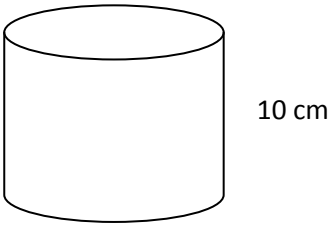
a)



TSA $= 2\pi r^2 + 2\pi rh = (2\pi \times 4 \times 4) + (2\pi \times 4 \times 8)$

$= 16\pi + 64\pi = 80\pi\ cm^2$

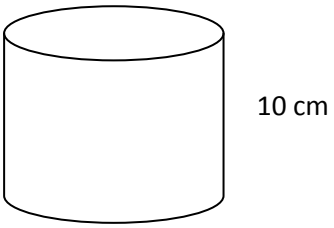
b) $r = 5\ cm$



TSA $= 2\pi r^2 + 2\pi rh = (2\pi \times 5 \times 5) + (2\pi \times 5 \times 10)$

$= 50\pi + 100\pi = 150\pi\ cm^2$

c) $r = 5\ cm$

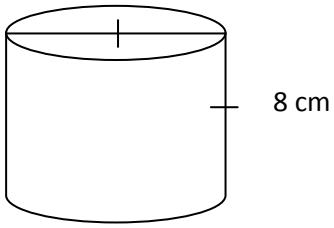


TSA = surface area from part b, subtract areas of ends

$2\pi rh = 100\pi\ cm^2$

©Ezy Math Tutoring | All Rights Reserved

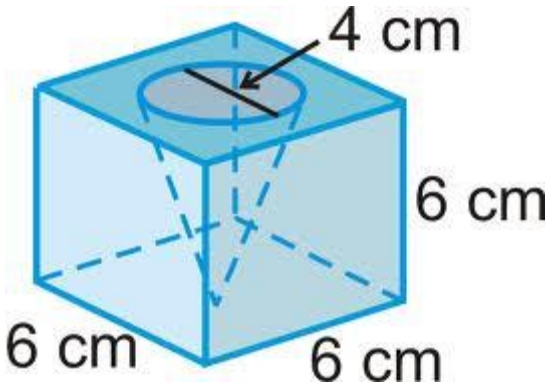
d)



TSA = surface area from part a, subtract area of ends

$= 2\pi rh = 64\pi\ cm^2$

11) What is the total surface area of the following solid, which is a cube with a conic section cut out?



Area of one side of cube $= 6 \times 6 = 36$

TSA of cube $= 6 \times 36 = 216$

Top of cube is missing circular piece of radius 2cm

Missing area $= \pi \times 2 \times 2 = 4\pi$

The conic section has a surface area that has to be added to the cubic surface area

TSA of conic section $= \pi rs$

To calculate s, note it forms a right angled triangle with the height and radius of the cone

$s^2 = 6^2 + 2^2 = 40$

©Ezy Math Tutoring | All Rights Reserved

$s = \sqrt{40}$

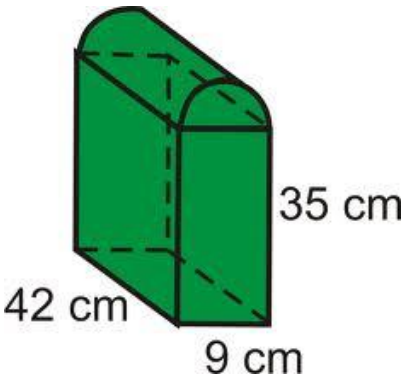
Surface area of conic section $= \pi \times 2 \times \sqrt{40}$

NOTE: We do not include the area of the base of the cone since it is a hole

TSA of shape $= 216 - 4\pi + 2\pi\sqrt{40} \cong 216 - 12.56 + 39.72 \cong 243.16\ cm^2$

12) Calculate the volume of the following solids

a)



Volume = Volume of rectangular prism + Volume of top section (half cylinder)

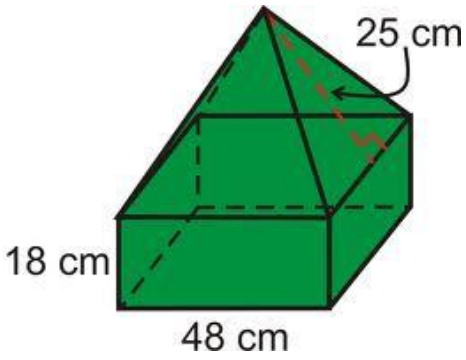
Volume of prism $= 9 \times 42 \times 35 = 13230$

Volume of top section $= \frac{1}{2}\pi r^2 h = \frac{1}{2}\pi \times 4.5^2 \times 42 = 425.25\pi \cong 1335.28$

Total volume $\cong 14565\ cm^3$

©Ezy Math Tutoring | All Rights Reserved

b)



Volume = Volume of rectangular prism + volume of pyramid

Volume of rectangular prism $= 18 \times 48 \times 48 = 41472$

Height of pyramid forms a right angled triangle with slant length and half the length of the base

$25^2 = 24^2 + h^2$

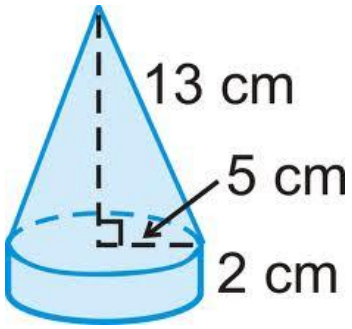
$h^2 = 625 - 576 = 49$

$h = 7$

Volume of pyramid $= \frac{1}{3} \times 48 \times 48 \times 7 = 5376$

Total volume $= 46848\ cm^3$

c)



©Ezy Math Tutoring | All Rights Reserved

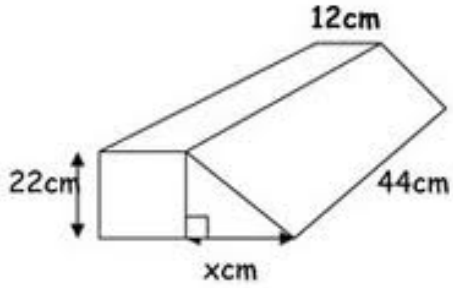
Total volume = Volume of cylinder + volume of cone

Volume of cylinder $= \pi \times 5 \times 5 \times 2 = 50\pi$

Volume of cone $= \frac{1}{3} \times \pi \times 5 \times 5 \times 13 = 108.33\pi$

Total volume $\cong 158.33\pi \cong 497.16\ cm^3$

13) The volume of the solid below is 16456 cm^3. What is the value of x?



Total volume = Volume of rectangular prism + volume of triangular prism

Volume of rectangular prism $22 \times 12 \times 44 = 11616$

Volume of triangular prism $= 16456 - 11616 = 4840$

$\frac{1}{2}x \times 22 \times 44 = 4840$

$484x = 4840$

$x = 10\ cm$

14) Calculate the surface area of a sphere with the following radii

a) 4 cm

SA of sphere $= 4\pi r^2 = 4 \times \pi \times 4 \times 4 = 64\pi\ cm^2$

©Ezy Math Tutoring | All Rights Reserved

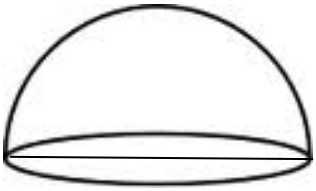
b) 6 cm

SA of sphere $= 4\pi r^2 = 4 \times \pi \times 6 \times 6 = 144\pi\ cm^2$

c) 10 cm

SA of sphere $= 4\pi r^2 = 4 \times \pi \times 10 \times 10 = 400\pi\ cm^2$

15) Calculate the total surface area of the shape below



12 cm

SA of sphere $= 4\pi r^2 = 4 \times \pi \times 6 \times 6 = 144\pi\ cm^2$

SA of hemisphere $= \frac{1}{2} \times 144\pi = 72\pi$

Area of circular base $= \pi r^2 = \pi \times 6 \times 6 = 36\pi$

TSA $= 108\pi\ cm^2$

©Ezy Math Tutoring | All Rights Reserved

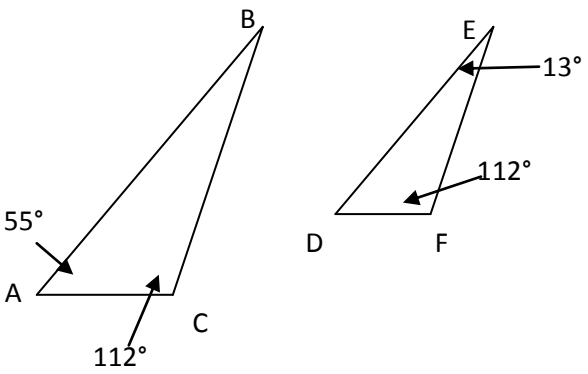
Exercise 3

Similarity

©Ezy Math Tutoring | All Rights Reserved

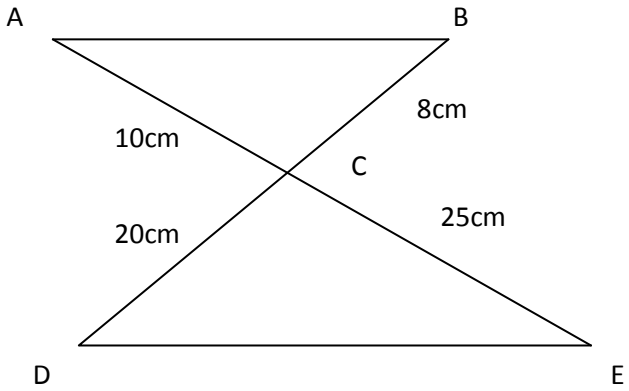
1) Determine if each pair of triangles is similar. If so, state the similarity conditions met

a)



AAA

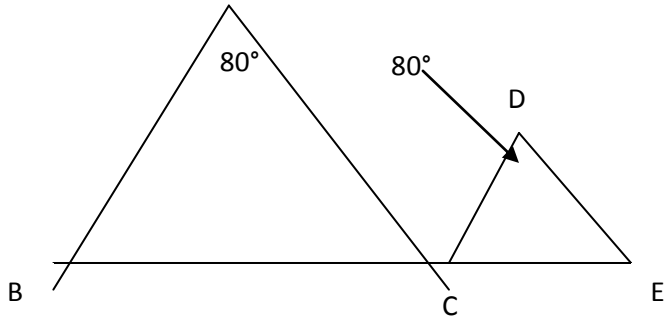
b)



SAS

A

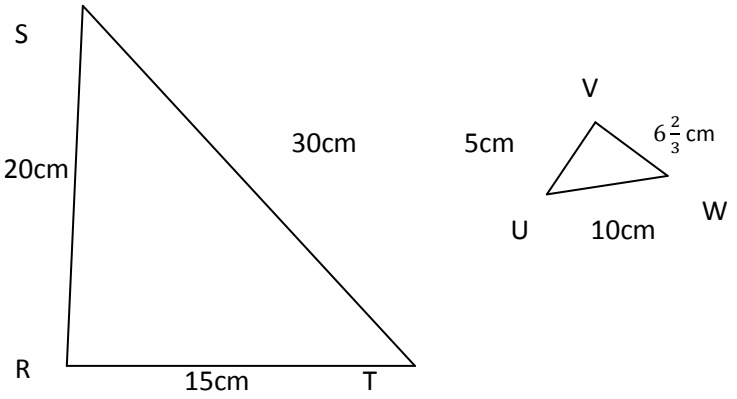
c) AB || DC



AAA

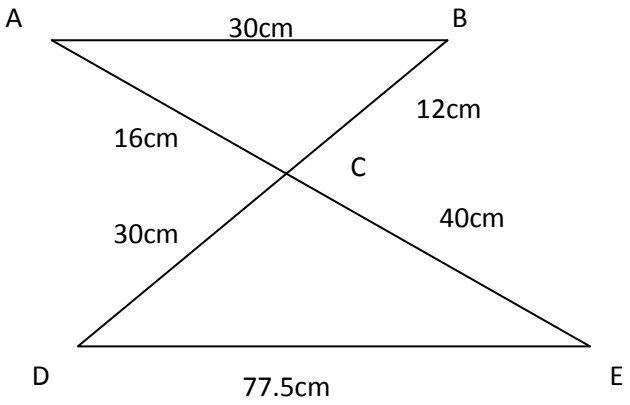
©Ezy Math Tutoring | All Rights Reserved

d)



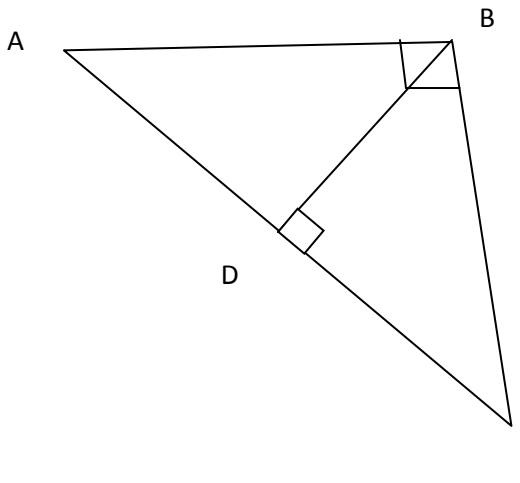
SSS

e)



Not similar

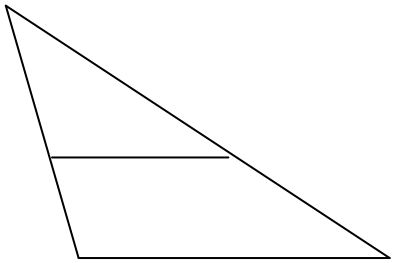
f)



AAA

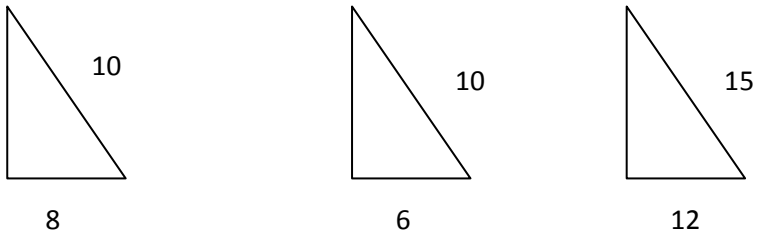
©Ezy Math Tutoring | All Rights Reserved

2) What additional information is needed to show that the two triangles are similar by AAA?



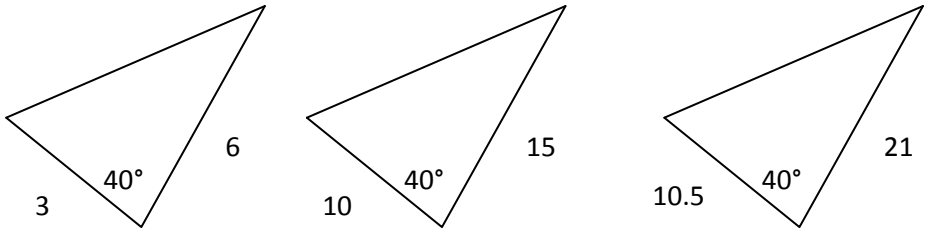
The two bases have to be parallel

3) Of the following three right-angled triangles, which two are similar and why?



A and C; SSS

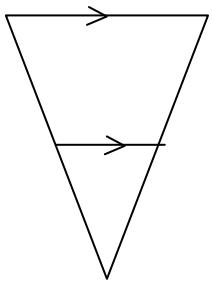
4) Of the following three triangles, which are similar and why?



A and C; SAS

©Ezy Math Tutoring | All Rights Reserved

5) Prove that the two triangles in the diagram are similar



Since the bases are parallel the sets of angles formed by them are equal, and the triangles share a common angle

6) Prove that if two angles of a triangle are equal then the sides opposite those angles are equal

7) A tower casts a shadow of 40 metres, whilst a 4 metre pole nearby casts a shadow of 32 metres. How tall is the tower?

$$\frac{40}{32} = \frac{x}{4}$$

$$x = 5\ metres$$

8) A pole casts a 4 metre shadow, whilst a man standing near the pole casts a shadow of 0.5 metres. If the man is 2 metres tall, how tall is the pole?

$$\frac{x}{4} = \frac{2}{0.5}$$

$$x = 16\ metres$$

9) A ladder of length 1.2 metres reaches 4 metres up a wall when placed on a safe angle on the ground. How long should a ladder be if it needs to reach 10 metres up the wall, and be placed on the same safe angle?

$$\frac{x}{10} = \frac{1.2}{4}$$

$$x = 3\ metres$$

©Ezy Math Tutoring | All Rights Reserved

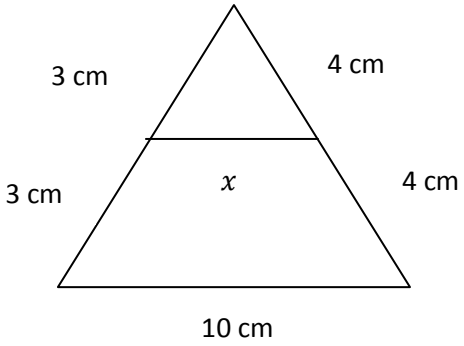
10) A man stands 2.5 metres away from a camera lens, and the film is 1.25 centimetres from the lens (the film is behind the lens). If the man is 2 metres tall how tall is his image on the film?

$2.5\ m = 250\ cm$

$\frac{x}{2} = \frac{1.25}{251.25}$

$x = 1\ cm$

11) What is the value of x in the following diagram?



$\frac{x}{10} = \frac{3}{6} = \frac{4}{8}$

$x = 5\ cm$

©Ezy Math Tutoring | All Rights Reserved

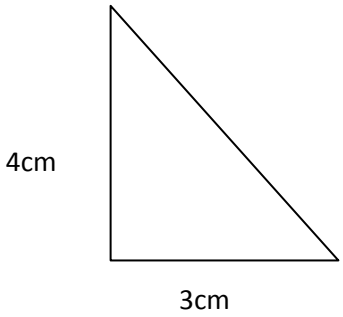
Exercise 4

Right Angled Triangles

©Ezy Math Tutoring | All Rights Reserved

1) Calculate the length of the hypotenuse in the following triangles

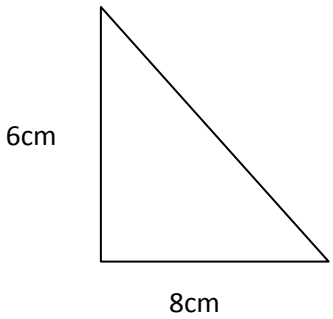
a)



$x^2 = 3^2 + 4^2$

$x = 5\ cm$

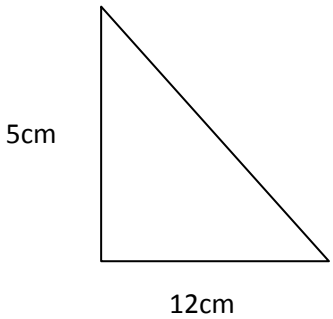
b)



$x^2 = 6 + 8^2$

$x = 10\ cm$

c)

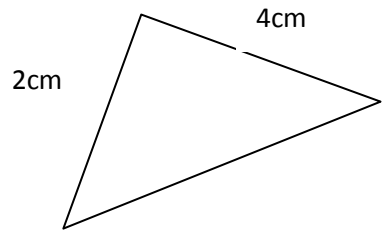


$x^2 = 5 + 12^2$

$x = 13\ cm$

©Ezy Math Tutoring | All Rights Reserved

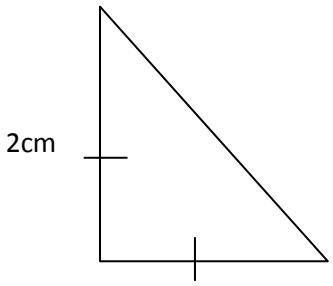
d)



$x^2 = 2^2 + 4^2$

$x = \sqrt{20}\ cm$

e)



$x^2 = 2^2 + 2^2$

$x = \sqrt{8}\ cm$

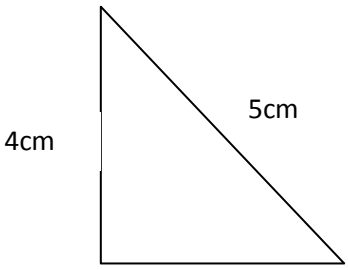
2) Explain why an equilateral triangle cannot be right-angled

The hypotenuse must be larger than the other two sides

©Ezy Math Tutoring | All Rights Reserved

3) Calculate the missing side length in the following triangles

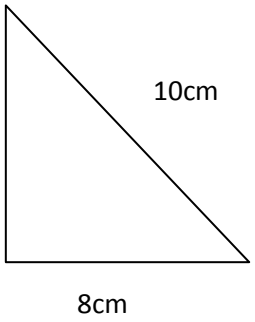
a)



$x^2 = 5^2 - 4^2$

$x = 3\ cm$

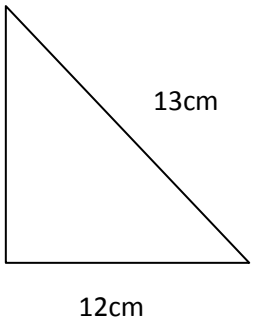
b)



$x^2 = 10^2 - 8^2$

$x = 6\ cm$

c)

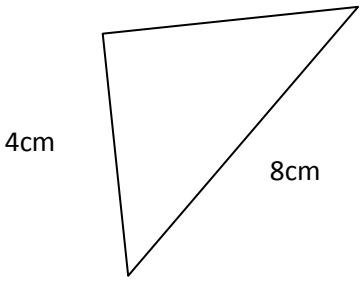


$x^2 = 13^2 - 12^2$

$x = 5\ cm$

©Ezy Math Tutoring | All Rights Reserved

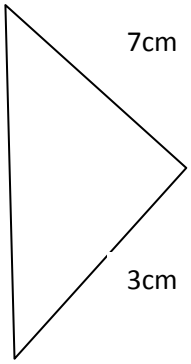
d)



$x^2 = 8^2 - 4^2$

$x = \sqrt{48}\ cm$

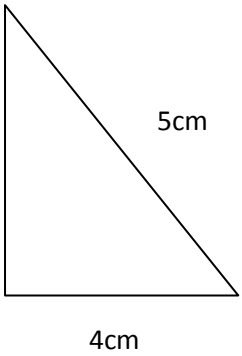
e)



$x^2 = 7^2 - 3^2$

$x = \sqrt{40}\ cm$

4) What is the area of the following triangle? (Use Pythagoras' to find required length)



From previous, missing length is 3 cm; $Area = \frac{1}{2} \times 4 \times 3 = 6cm^2$

©Ezy Math Tutoring | All Rights Reserved

5) The equal sides of an isosceles right-angled triangle measure 8cm. What is the length of the third side?

$$x^2 = 8^2 + 8^2$$

$$x = \sqrt{128}\ cm$$

6) A man stands at the base of a cliff which is 120 metres high. He sees a friend 100 metres away along the beach. What is the shortest distance from his friend to the top of the cliff?

$$x^2 = 120^2 + 100^2$$

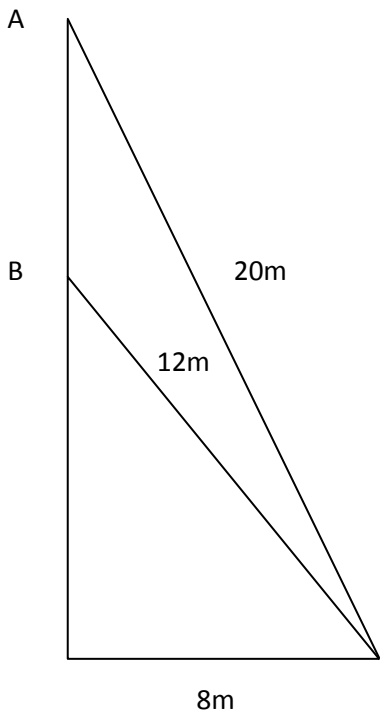
$$x \cong 156.2\ cm$$

7) A steel cable runs from the top of a building to a point on the street below which is 80 metres away from the bottom of the building. If the building is 40 metres high, how long is the steel cable?

$$x^2 = 40^2 + 80^2$$

$$x = 89.4\ m$$

8) What is the distance from point A to point B?



©Ezy Math Tutoring | All Rights Reserved

Distance from ground to A

$$x^2 = 20^2 - 8^2$$

$$x = 18.33\ m$$

Distance from ground to B

$$x^2 = 12^2 - 8^2$$

$$x = 8.94\ m$$

Distance from A to B = 9.39 m

9) A right angled triangle has an area of 20 cm^2. If its height is 4cm, what is the length of its hypotenuse?

$$Area = \frac{1}{2} \times base \times 4 = 20$$

$$Base = 10\ cm$$

$$x^2 = 10^2 + 4^2$$

$$x = 10.77\ cm$$

10) What is the length of a diagonal of a square of side length 5cm?

$$x^2 = 5^2 + 5^2$$

$$x = \sqrt{50}\ cm$$

11) A man is laying a slab for a shed. The shed is to be 6m wide and 8m long. To check if he has the corners as exactly right angles, what should the slab measure from corner to corner?

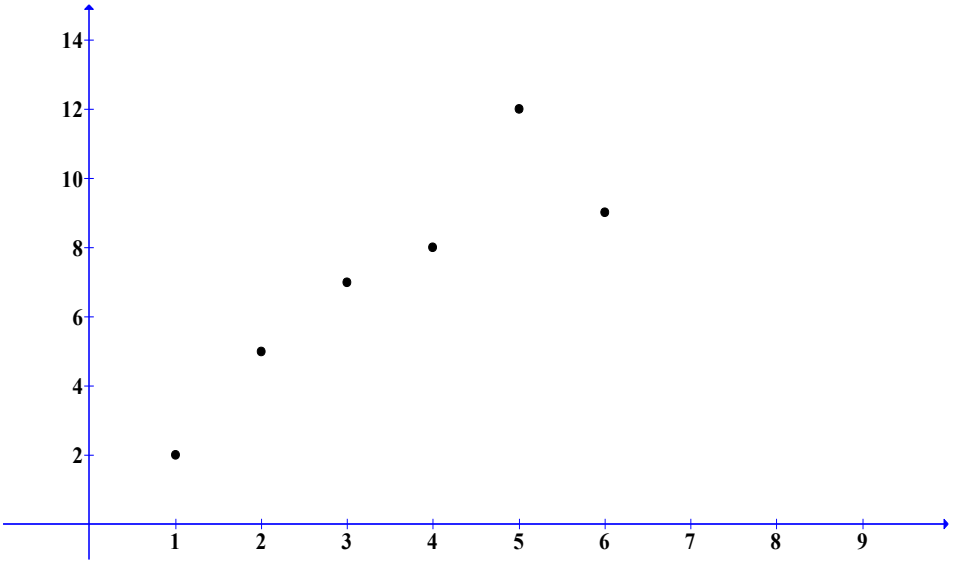
$$x^2 = 6^2 + 8^2$$

$$x = 10\ m$$

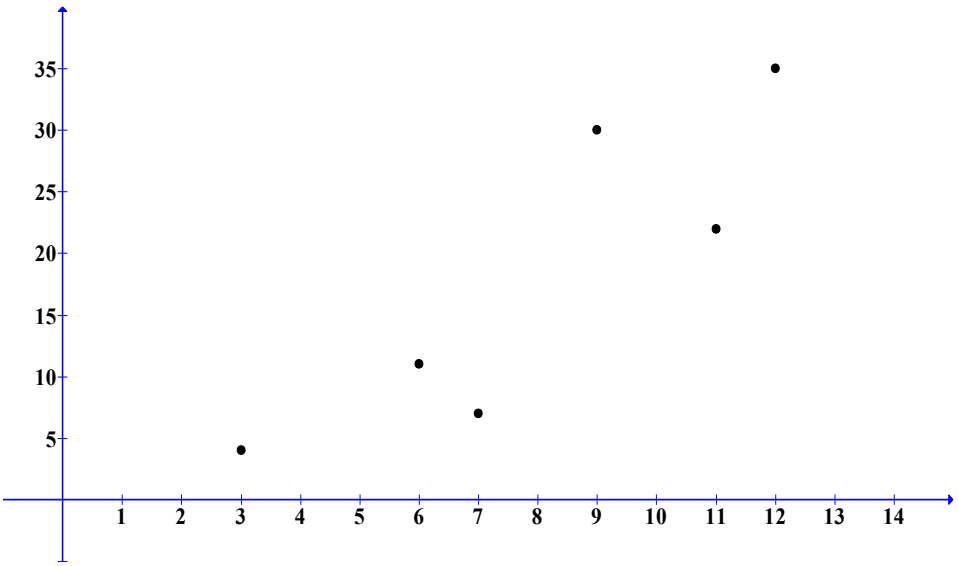
©Ezy Math Tutoring | All Rights Reserved

1) Plot the following sets of ordered pairs on their own scatter plot

a) $(1,2)\ (2,5)\ (3,7)\ (4,8)\ (5,12)\ (6,9)$



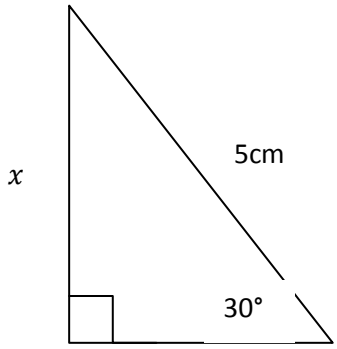
b) $(3,4)\ (6,11)\ (7,7)\ (9,30)\ (11,22)\ (12,35)$



©Ezy Math Tutoring | All Rights Reserved

14) Calculate the length of x in each of the diagrams below

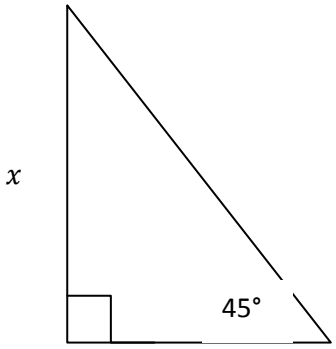
a)



$$\sin 30° = \frac{x}{5}$$

$$x = 2.5\ cm$$

b)

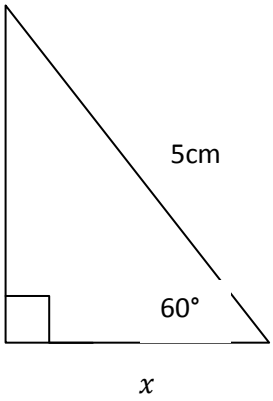


$$\tan 45° = \frac{x}{7}$$

$$x = 7\ cm$$

©Ezy Math Tutoring | All Rights Reserved

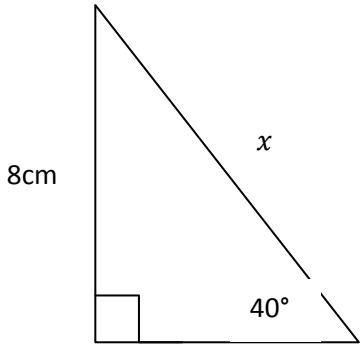
c)



$$\cos 60° = \frac{x}{5}$$

$$x = 2.5\ cm$$

d)



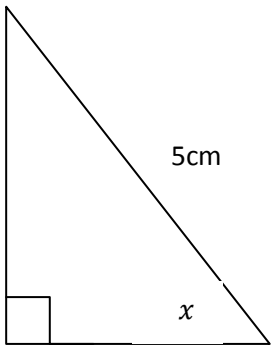
$$\sin 40° = \frac{8}{x}$$

$$x = 12.36\ cm$$

©Ezy Math Tutoring | All Rights Reserved

15) Calculate the size of angle x in the diagrams below, correct to the nearest degree.

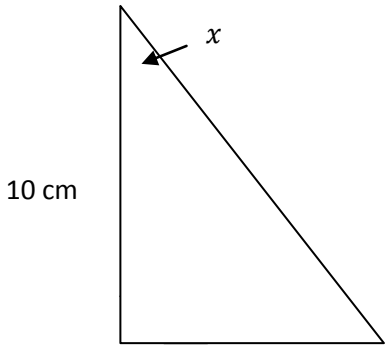
a)



$$\cos x = \frac{3}{5}$$

$$x = 53°$$

b)

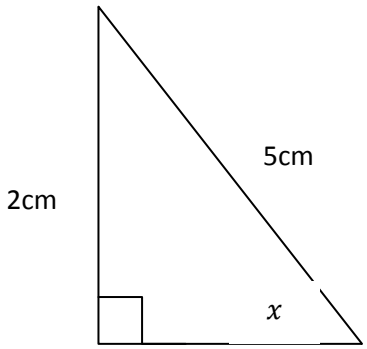


$$\tan x = \frac{6}{10}$$

$$x = 31°$$

©Ezy Math Tutoring | All Rights Reserved

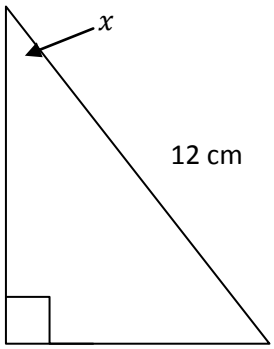
c)



$$\sin x = \frac{2}{5}$$

$$x = 24°$$

d)



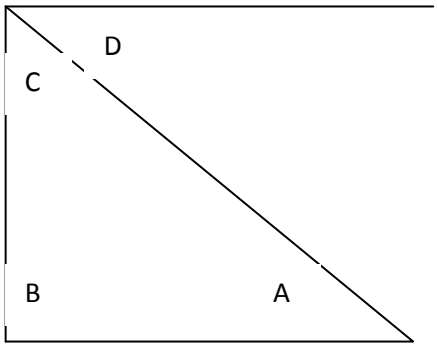
6 cm

$$\sin x = \frac{6}{12}$$

$$x = 30°$$

©Ezy Math Tutoring | All Rights Reserved

16) Identify the angles of elevation and depression in the diagram below



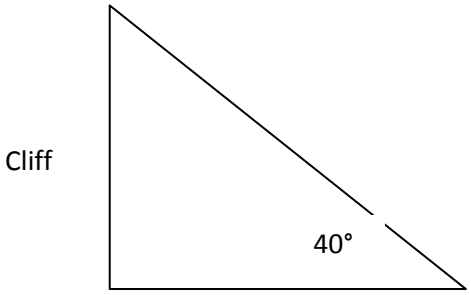
Elevation: A

Depression: D

Complete the statement: The angle of elevation is ..... the angle of depression

Equal to

17) A man standing 100 metres away from the base of a cliff measures the angle of elevation to the top of the cliff to be 40 degrees. How high is the cliff?

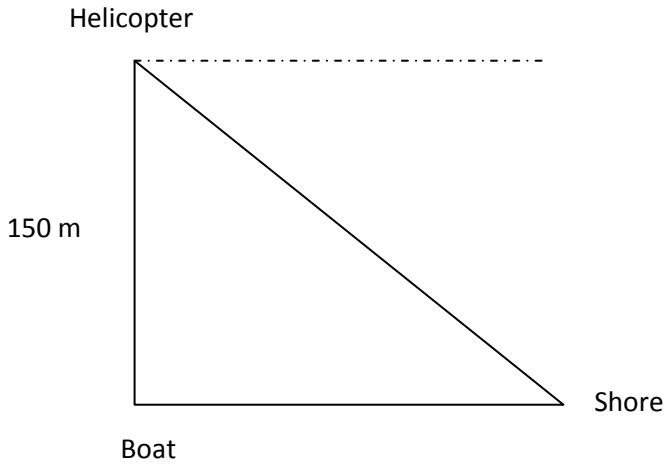


$$\tan 40 = \frac{x}{100}$$

$$x = 83.9\ m$$

©Ezy Math Tutoring | All Rights Reserved

18) A helicopter is hovering 150 metres above a boat in the ocean. From the helicopter, the angle of depression to the shore is measured to be 25 degrees. How far out to sea is the boat? (You need to fill in angle of depression on diagram)



$$\tan 25 = \frac{150}{x}$$

$$x = 321.7\ m$$

19) A ramp is built to allow wheelchair access to a lift. If the angle of elevation to the lift is 2 degrees, and the bottom of the lift is 50 cm above the ground how long is the ramp?

$$\sin 2 = \frac{50}{x}$$

$$x = 14.32\ m$$

20) The angle of elevation to the top of a tree is 15 degrees. If the tree is 10 metres tall how far away from the base of the tree is the observer?

$$\tan 15 = \frac{10}{x}$$

$$x = 37.3\ m$$

21) From the top of a tower a man sees his friend on the ground at an angle of depression of 30 degrees. If his friend is 80 metres from the base of the tower how tall is the tower?

©Ezy Math Tutoring | All Rights Reserved

$$\tan 30 = \frac{x}{80}$$

$$x = 46.2\ m$$

©Ezy Math Tutoring | All Rights Reserved

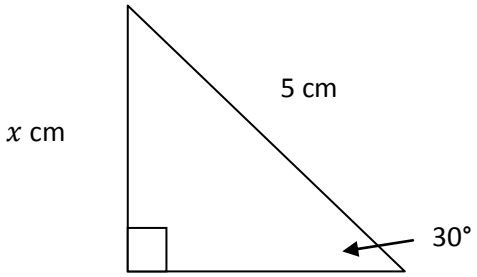
Exercise 5

Further Applications of Trigonometry

©Ezy Math Tutoring | All Rights Reserved

1) Calculate the value of x in the following diagrams

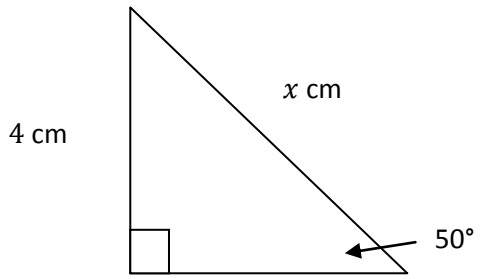
a)



$$\sin 30° = \frac{x}{5}$$

$$x = 5 \sin 30° = 2.5\ cm$$

b)

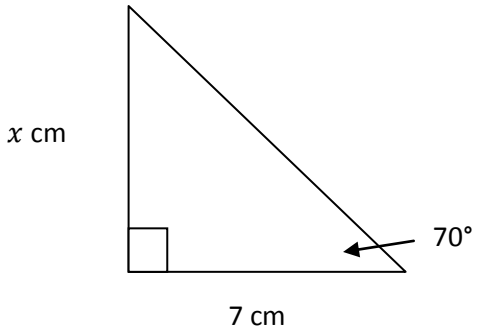


$$\sin 50° = \frac{4}{x}$$

$$x = \frac{4}{\sin 50°} \cong 5.22\ cm$$

©Ezy Math Tutoring | All Rights Reserved

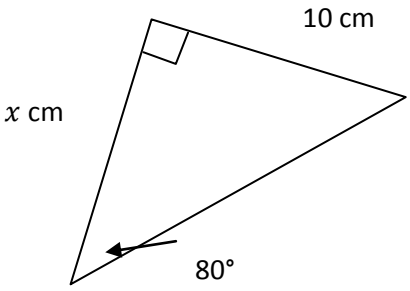
c)



$$\tan 70° = \frac{x}{7}$$

$$x = 7 \tan 70° \cong 19.23\ cm$$

d)

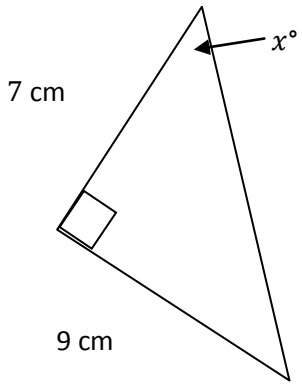


$$\tan 80° = \frac{10}{x}$$

$$x = \frac{10}{\tan 80°} \cong 1.76\ cm$$

©Ezy Math Tutoring | All Rights Reserved

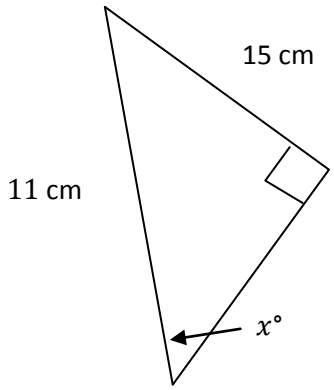
e)



$$\tan x = \frac{9}{7}$$

$$x \cong 52°$$

f)



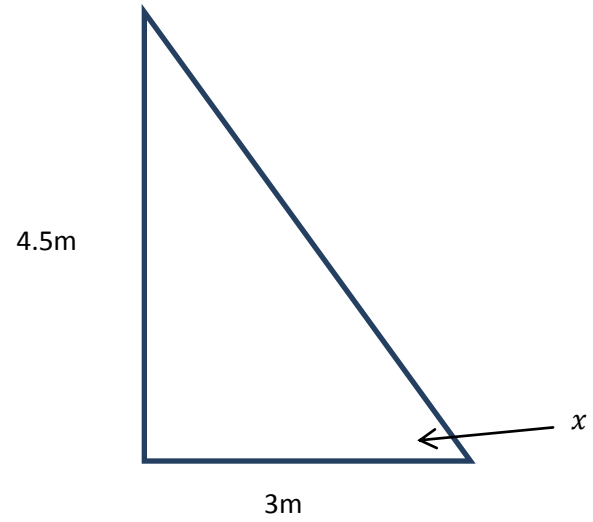
$$\sin x = \frac{15}{11}$$

There is no solution as sine cannot have a value greater than 1

Also the hypotenuse must be the longest side of a valid right angled triangle

©Ezy Math Tutoring | All Rights Reserved

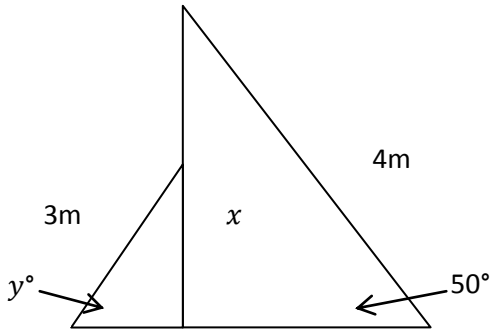
2) The foot of a ladder is 3 metres away from the base of a wall. If the ladder reaches 4.5 metres up the wall, what angle doe the foot of the ladder make with the ground?



$\tan x = \frac{4.5}{3}$

$x \cong 56.3°$

3) Two sails sit back to back on a yacht. The first sail reaches half way up the second



The longest part of the second sail is 4 metres, and it makes an angle of 50 degrees to the deck. If the longest part of the first sail is 3 metres, what angle does it make with the deck?

$\sin 50° = \frac{x}{4}$

$x = 4 \sin 50° \cong 3.06\ m$

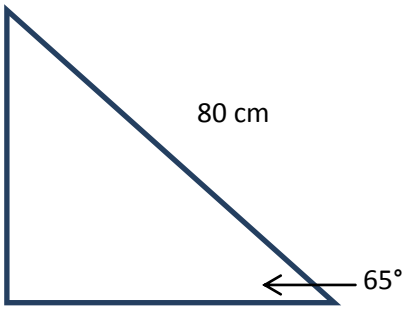
©Ezy Math Tutoring | All Rights Reserved

Half of this length (length of back of smaller sail) = 1.53 m

$$\sin y = \frac{1.53}{3}$$

$$y \cong 30.7°$$

4) A piece of carpet is in the shape of a right angled triangle. The longest side is 80 cm, and it makes an angle of 65 degrees with the next side. What is the area of the piece of carpet?



$$\text{Area} = \frac{1}{2} \times base \times height$$

$$\cos 65° = \frac{base}{80}$$

$$\text{Base} = 80 \cos 65° \cong 33.8\ cm$$

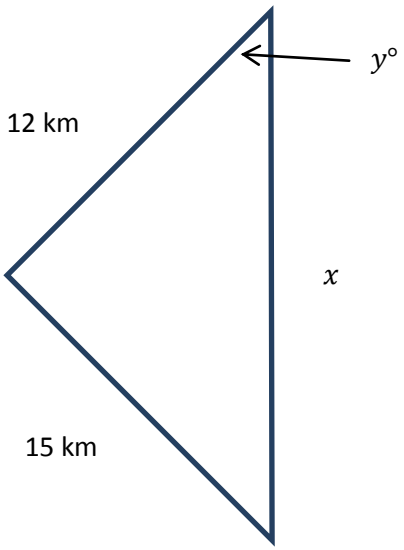
$$\sin 65° = \frac{height}{80}$$

$$\text{Height} = 80 \sin 65° \cong 72.5\ cm$$

$$\text{Area} = \frac{1}{2} \times 33.8 \times 72.5 = 1225.25\ cm^2$$

©Ezy Math Tutoring | All Rights Reserved

5) Tom walks at an average speed of 4 km per hour in a north east direction. Ben walks at 5 km per hour, starting from the same point but in a south east direction. After 3 hours what is the shortest distance between them, and what is the angle from Tom to Ben?



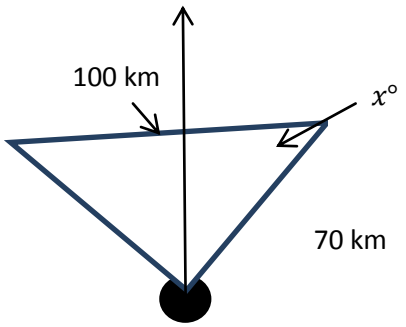
Using Pythagoras, $x^2 = 12^2 + 15^2 = 369$

$x \cong 19.2\ km$

$\tan y = \frac{15}{12}$

$y \cong 51.3°$

6) A ship is on a bearing of 040 from a lighthouse, and a marker buoy is on a bearing of 310 from the same lighthouse. If the ship and the buoy are 100 km apart and the ship is 70 km from the lighthouse, what is the angle of the buoy from the ship?



©Ezy Math Tutoring | All Rights Reserved

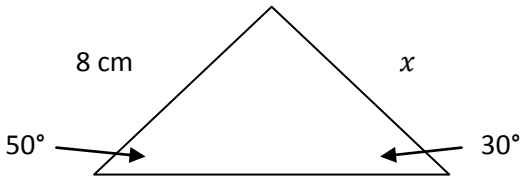
Angle from ship to buoy is $x°$

$$\cos x = \frac{70}{100}$$

$$x \cong 45.6°$$

7) Calculate the value of x in the following diagrams

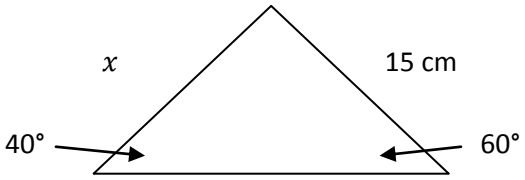
a)



$$\frac{x}{\sin 50°} = \frac{8}{\sin 30°}$$

$$x = \frac{8 \sin 50°}{\sin 30°} \cong 12.3\ cm$$

b)

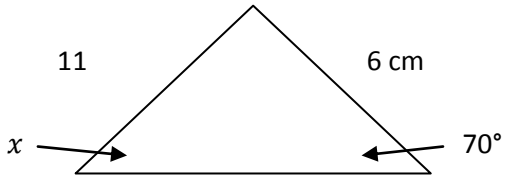


$$\frac{x}{\sin 60°} = \frac{15}{\sin 40°}$$

$$x = \frac{15 \sin 60°}{\sin 40°} \cong 20.2\ cm$$

©Ezy Math Tutoring | All Rights Reserved

c)

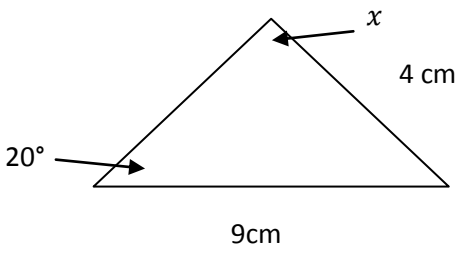


$$\frac{6}{\sin x} = \frac{11}{\sin 70°}$$

$$\sin x = \frac{6 \sin 70°}{11}$$

$$x \cong 30.9°$$

d)

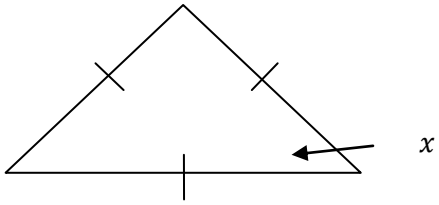


$$\frac{9}{\sin x} = \frac{4}{\sin 20°}$$

$$\sin x = \frac{9 \sin 20°}{4}$$

$$x \cong 50.4°$$

e)

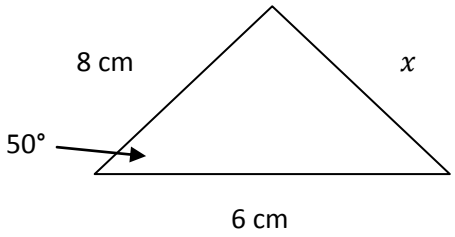


Equilateral triangle, therefore $x = 60°$

©Ezy Math Tutoring | All Rights Reserved

8) Calculate the value of x in the following diagrams

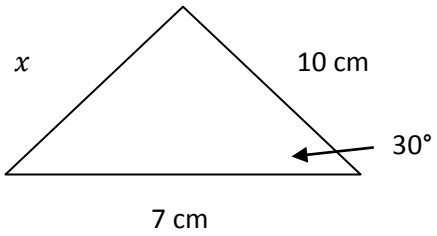
a)



$x^2 = 8^2 + 6^2 - (2 \times 8 \times 6 \times \cos 50°) \cong 100 - 61.7$

$x \cong 6.2\ cm$

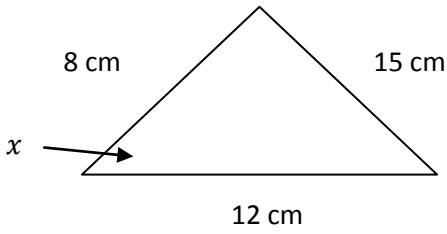
b)



$x^2 = 10^2 + 7^2 - (2 \times 10 \times 7 \times \cos 30°) \cong 149 - 121.2$

$x \cong 5.3\ cm$

c)

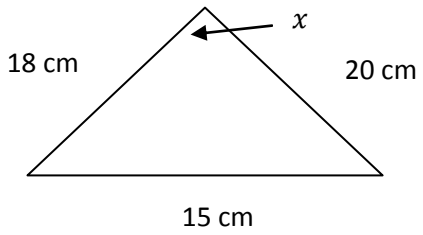


$$\cos x = \frac{8^2 + 12^2 - 15^2}{2 \times 8 \times 12}$$

$x \cong 95°$

©Ezy Math Tutoring | All Rights Reserved

d)



$$\cos x = \frac{18^2 + 20^2 - 15^2}{2 \times 18 \times 20}$$

$$x \cong 46.1°$$

9) Calculate the area of each of the triangles in question 8

a) Area $= \frac{1}{2} \times 8 \times 6 \times \sin 50° \cong 18.4\ cm^2$

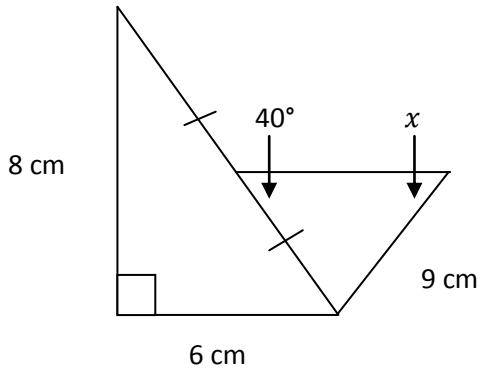
b) Area $= \frac{1}{2} \times 10 \times 7 \times \sin 30° = 17.5\ cm^2$

c) Area $= \frac{1}{2} \times 8 \times 12 \times \sin 95° \cong 47.8\ cm^2$

d) Area $= \frac{1}{2} \times 18 \times 20 \times \sin 46.1° \cong 130\ cm^2$

10) Calculate the value of x in the following

a)



By Pythagoras, length of hypotenuse of right angled triangle is 10 cm

©Ezy Math Tutoring | All Rights Reserved

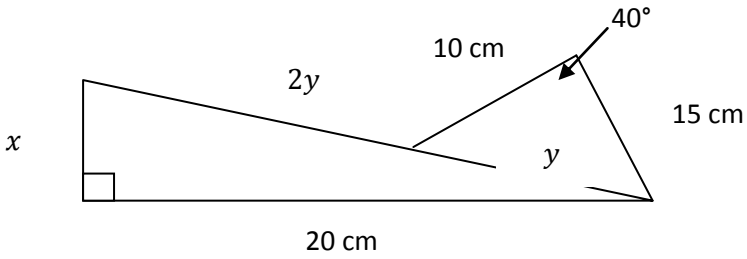
Therefore length of two equal parts is 5 cm

By sine rule, $\frac{\sin x}{5} = \frac{\sin 40°}{9}$

$\sin x = \frac{5 \sin 40°}{9}$

$x \cong 21°$

b)



By cosine rule, $y^2 = 10^2 + 15^2 - (2 \times 10 \times 15 \times \cos 40°)$

$y \cong 9.76\ cm$

$2y = 19.52\ cm$

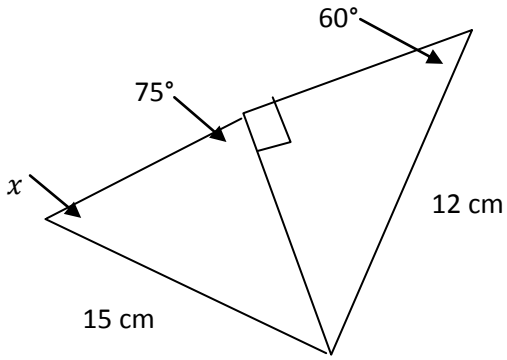
Length of hypotenuse $= 29.28\ cm$

By Pythagoras, $x^2 + 20^2 = 29.28^2$

$x \cong 21.4\ cm$

©Ezy Math Tutoring | All Rights Reserved

c)



In non-right angled triangle, let y be length opposite angle x

In right angled triangle, $\sin 60° = \frac{y}{12}$

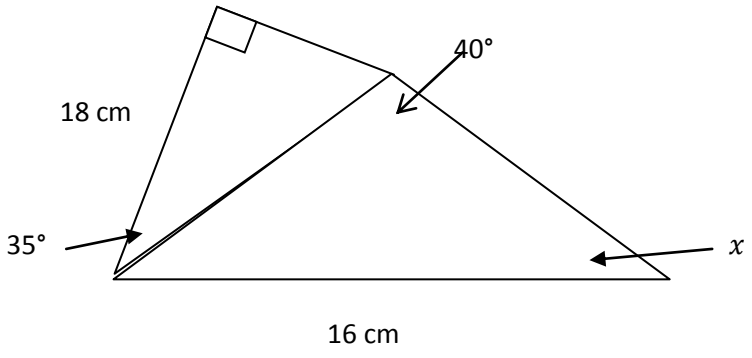
$y = 12 \sin 60° \cong 10.4\ cm$

By sine rule, $\frac{10.4}{\sin x} = \frac{15}{\sin 75°}$

$\sin x = \frac{10.4 \sin 75°}{15}$

$x \cong 42°$

d)



Let y be length of side opposite angle x

From right angled triangle

$\cos 35° = \frac{18}{y}$

©Ezy Math Tutoring | All Rights Reserved

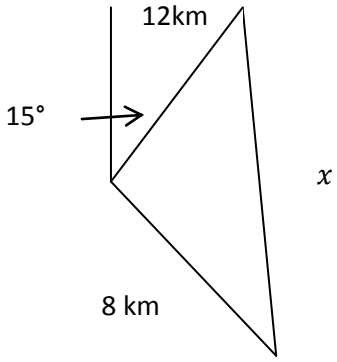
$$y = \frac{18}{\cos 35°} \cong 22$$

By sine rule, $\frac{\sin x}{22} = \frac{\sin 40}{16}$

$$\sin x = \frac{22 \sin 40°}{16}$$

$$x \cong 62°$$

11) Thomas walks on a bearing of 15 degrees for 12 km, and Karl walks on a bearing of 125 degrees for 8 km. What is the shortest distance them after their walks?

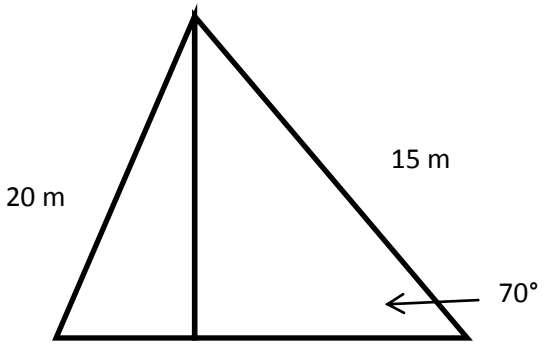


Angle between the men is $125 - 15 = 110°$

By cosine rule, $x^2 = 12^2 + 8^2 - (2 \times 12 \times 8 \cos 110°)$

$x \cong 15.7\ km$

12) Two wire ropes are attached to a tower; one on each side. The first rope makes an angle of 70° with the ground and is 15 metres long. If the second rope is 20 metres long, what angle does it make with the ground?



©Ezy Math Tutoring | All Rights Reserved

Let x be height of tower to attachment point

$$\sin 70° = \frac{x}{15}$$

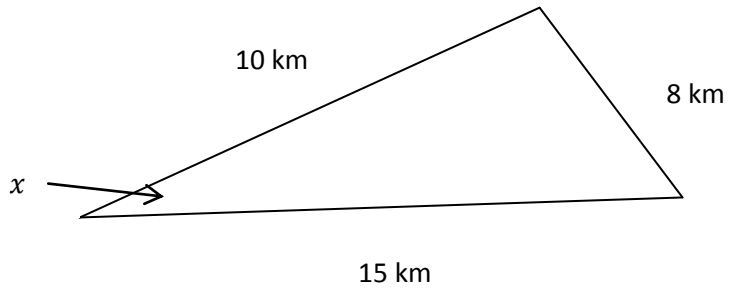
$$x = 15 \sin 70° \cong 14.1\ m$$

Let y be missing angle

$$\sin y = \frac{14.1}{20}$$

$$y \cong 44.8°$$

13) Three legs of a yacht race form a triangular course. The first leg is 10 km, and sails at some angle to the east of north, the second is 8 km, and the third leg is 15 km. The start and finish points are the same. What angle is the first marker from the start point?



$$\cos x = \frac{10^2 + 15^2 - 8^2}{2 \times 10 \times 15} = \frac{261}{300}$$

$$x \cong 29.5°$$

©Ezy Math Tutoring | All Rights Reserved

Exercise 6

Spherical Geometry

©Ezy Math Tutoring | All Rights Reserved

1) Complete the following table

Angle subtended by arc	Radius of circle	Arc length
90°	10 cm	5π cm
40°	25 cm	$\frac{50}{9}\pi$ cm
70°	$\frac{1440}{7\pi}$ cm	80 cm
125°	$\frac{108}{5\pi}$ cm	15 cm
$\frac{540}{\pi} \cong 172°$	30 cm	90 cm
$\frac{90}{\pi} \cong 29°$	90 cm	45 cm

2) State whether the following are true or false

a) All lines of latitude form great circles

False: lines of longitude are great circles as is the equator

b) Any two points on the same longitude form part of a great circle

True

c) Any two points on a sphere are parts of a circle

True

d) There is only one circle that can pass through 3 points on a sphere

True

e) The equator is a great circle

True

3) Find the latitude and longitude of the following cities to the nearest degree

a) Adelaide

$35° S$

b) Barcelona

$41° N$

c) Cairo

$30° N$

d) Jakarta

$6° S$

©Ezy Math Tutoring | All Rights Reserved

e) Lima

$12° S$

f) Mexico City

$19° N$

g) Osaka

$34° N$

h) Rome

$42° N$

i) Warsaw

$52° N$

4) Convert the following to nautical miles

a) 1.852 km

$1\ NM$

b) 18.52 km

$10\ NM$

c) 312 km

$168.5\ NM$

d) 74.3 km

$40.1\ NM$

e) 1000 km

$540\ NM$

5) Convert the following to km

a) 1 nautical mile

1.85 km

b) 5 nautical miles

9.26 km

c) 0.1 nautical miles

0.185 km

d) 6.6 nautical miles

12.21 km

7) Calculate the shortest distances (in nautical miles and kilometres) between the following pairs of points (Assume Earth is a perfect sphere with a radius of 6400 km)

Points are same longitude, hence are part of a great circle, and we need to only consider changes in latitude

a) 26°N 40°W and 50°N 40°W

Change in latitude $= 24°$

L= $2\pi \times 6400 \times \frac{24}{360}$ ≅2680km

©Ezy Math Tutoring | All Rights Reserved

$\cong 1447.2\ NM$

b) 10°N 30°E and 40°N 30°E

Change in latitude $= 30°$

L= $2\pi \times 6400 \times \frac{30}{360}$ ≅3350km

$= 1809\ NM$

c) 45°N 25°W and 32°S 25°W

Change in latitude $= 77°$

L= $2\pi \times 6400 \times \frac{77}{360}$ ≅8597km

$\cong 4642\ NM$

d) 9°N 75°W and 43°S 75°W

Change in latitude $= 52°$

L= $2\pi \times 6400 \times \frac{52}{360}$ ≅5805km

$\cong 3135\ NM$

8) Calculate the time differences between the following cities using their longitudes (ignore daylight saving)

15 degrees equals one hour
4 degrees converts to 1 minute

a) Athens and Adelaide

24E to 139E=115°

Converts to $(7 \times 15) + (10)$=7 hours 2.5 minutes

b) London and New York

0E to 74W=74°

Converts to $(4 \times 15) + (14)$=4 hours 3.5 minutes

c) Moscow and Anchorage,

38E to 150W=188°

Shortest distance covers
360 – 188 = 172°

Converts to $(11 \times 15) + (7)$=11 hours 1.75 minutes

d) Sydney and Nairobi

151E to 37E=114°

Converts to $(7 \times 15) + (9)$=7 hours 2.25 minutes

e) Bogota and Cairo

74W to 31E=105°

Converts to (7×15)
=7 hours

f) Tehran and Beijing

52E to 116E=68°

Converts to $(4 \times 15) + (4)$=4 hours 1 minute

©Ezy Math Tutoring | All Rights Reserved

9) How much time would one gain or lose by flying between the following pairs of cities, given the flight time?

a) Cairo to Moscow takes 3 and a quarter hours

Time difference is +2 hours

One would lose 5 and a quarter hours

b) London to New York takes 6 and a quarter hours

Time difference is -5 hours

One would lose an hour and three quarters

c) Melbourne to Perth takes 3 hours

Time difference is +3 hours

One would neither gain nor lose time

d) Paris to Tokyo takes 11 hours

Time difference is +8 hours

One would lose 19 hours

e) Istanbul to New Delhi takes 5 hours

Time difference is +3 hours

One would lose 8 hours

©Ezy Math Tutoring | All Rights Reserved

Year 7 Mathematics

Probability

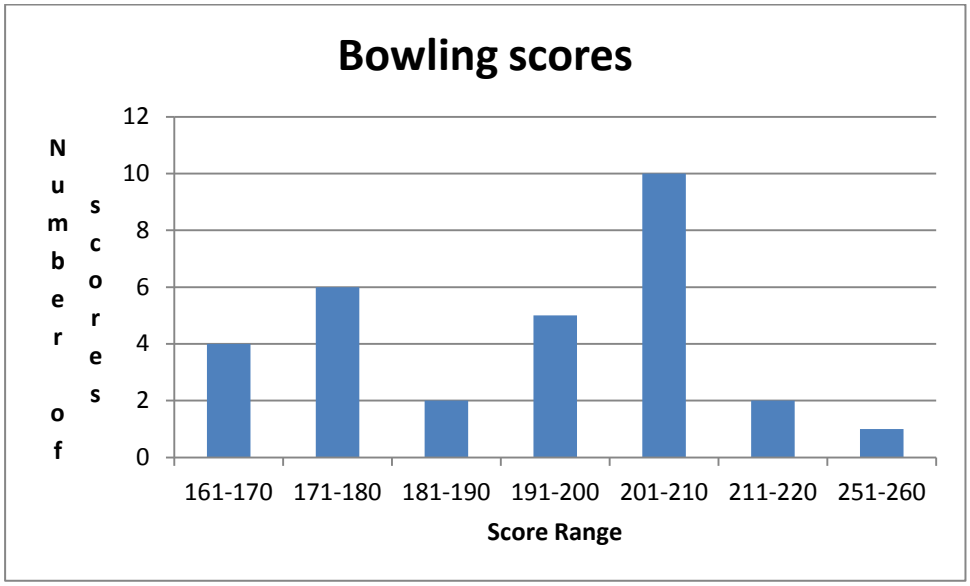
©Ezy Math Tutoring | All Rights Reserved

Exercise 1

Simple Probability

©Ezy Math Tutoring | All Rights Reserved

1) Peter plays ten pin bowling; his last 30 scores have been graphed in a frequency chart, shown here



Basing you answers on the chart data

a) Is Peter more likely to score 205 or 185 when he next bowls?

205

b) Is he more or less likely to score over 200 when he next bowls?

Under 200

c) What would be his probability of scoring over 250 when next he bowls?

$\frac{1}{30}$

d) What would be his probability of scoring between 201 and 210 when next he bowls?

$\frac{10}{30}$

e) Discuss a major drawback with using this chart to predict the probabilities of future scores

©Ezy Math Tutoring | All Rights Reserved

The scores are only for 30 games, and do not indicate any factors that may have affected Peter's performance in any of the games. There is no indication of the time between games, or how long until his next game

2) Craig rolled a pair of dice 360 times and recorded the sum of the two each time. He summarized his results in the table below

SUM of TWO DICE	Frequency
2	8
3	21
4	30
5	42
6	49
7	62
8	51
9	41
10	28
11	21
12	7

Based on his table:

a) What total is most likely to be rolled by two dice?

7

©Ezy Math Tutoring | All Rights Reserved

b) What is the most likely double?

Double 4 (8)

c) What total is least likely to be rolled by two dice

12

d) Is he more likely to roll a sum of 10 or a sum of 6 with two dice?

6

e) Is this data more reliable than that of Q1? Give two reasons to support your answer

It is to a point since there are (assuming) no external factors affecting the roll of the dice. The sample is however still quite small

3) What is the theoretical probability of each of the following?

a) A head being thrown when a coin is tossed

$\frac{1}{2}$

b) A blue sock being taken from a draw containing 3 blue and 5 red socks

$\frac{3}{8}$

c) The number 2 being rolled on a dice

$\frac{1}{6}$

d) An even number being rolled on a dice

$\frac{3}{6} = \frac{1}{2}$

©Ezy Math Tutoring | All Rights Reserved

4) A card is drawn from a standard pack of 52 cards. What is the probability of the card being:

a) A black card

$$\frac{26}{52} = \frac{1}{2}$$

b) A club

$$\frac{13}{52} = \frac{1}{4}$$

c) An ace

$$\frac{4}{52} = \frac{1}{13}$$

d) A black 2

$$\frac{2}{52} = \frac{1}{26}$$

e) A picture card

$$\frac{8}{52} = \frac{2}{13}$$

f) The 2 of diamonds

$$\frac{1}{52}$$

5) A man throws two coins into the air

a) List the possible combinations, and from this table:

HH, TH, HT, TT

©Ezy Math Tutoring | All Rights Reserved

b) What is the probability of throwing two heads?

$\frac{1}{4}$

c) What is the probability of throwing a head and a tail?

$\frac{2}{4} = \frac{1}{2}$

d) If the first coin lands on a head, is the second coin more likely or less likely to be a head?

It makes no difference

6) A coin is tossed and a dice is rolled

a) List the possible combinations of the coin and dice, and from this table:

H1, H2, H3, H4, H5, H6, T1, T2, T3, T4, T5, T6

b) What is the probability of throwing a six and a head?

$\frac{1}{12}$

c) What is the probability of throwing an odd number and a tail?

$\frac{3}{12} = \frac{1}{4}$

d) What is the probability of throwing a number higher than 4 and a head?

$\frac{2}{12} = \frac{1}{6}$

e) What is the probability of throwing a head and a 2 or a head and a 4?

$\frac{2}{12} = \frac{1}{6}$

©Ezy Math Tutoring | All Rights Reserved

7) A card is drawn from a normal pack. It is not replaced and a second card is drawn.

a) If the first card is red, what is the probability that the second card is also red?

$\frac{12}{51}$

b) If the first card is red, what is the probability that the second card is black?

$\frac{13}{51}$

c) If the first card is an ace, what is the probability that the second card is also an ace?

$\frac{3}{51} = \frac{1}{17}$

d) If the first card is the jack of clubs, what is the probability that the second card is the jack of clubs?

Zero

8) A set of cards consists of 10 red cards, numbered 1 to 10 and 10 black cards numbered 1 to 10

a) What is the probability of pulling a 10 at random?

$\frac{2}{20} = \frac{1}{10}$

b) What is the probability of pulling a black card at random?

$\frac{10}{20} = \frac{1}{2}$

c) What is the probability of pulling a red 2 at random?

$\frac{1}{20}$

©Ezy Math Tutoring | All Rights Reserved

d) What is the probability of pulling a red 2 on the second draw if the first card is a black 2, and it is not replaced?

$\frac{1}{19}$

e) What is the probability of pulling an 8 on the second draw if the first card is an 8, and it is not replaced?

$\frac{1}{19}$

9) Consider the word ANATOMICALLY

a) What is the probability that a randomly chosen letter from this word will be an L?

$\frac{2}{12} = \frac{1}{6}$

b) What is the probability that a randomly chosen letter from this word will be an A?

$\frac{3}{12} = \frac{1}{4}$

c) What is the probability that a randomly chosen letter from this word will not be a vowel

$\frac{7}{12}$

d) What is the probability that a randomly chosen letter from this word will be a Z?

Zero

10) What is the probability that a digit chosen randomly from all digits (0- 9) is:

a) A prime number?

©Ezy Math Tutoring | All Rights Reserved

$$\frac{4}{10} = \frac{2}{5}$$

b) An even number?

$$\frac{4}{10} = \frac{2}{5}$$

c) Not 7?

$$\frac{9}{10}$$

d) Greater than 4?

$$\frac{5}{10} = \frac{1}{2}$$

e) Less than 10?

1 (certain)

©Ezy Math Tutoring | All Rights Reserved

Exercise 2

Multi-stage Events

©Ezy Math Tutoring | All Rights Reserved

1) Construct a tree diagram that shows the possible outcomes of tossing a coin 3 times. List the sample space

Sample space is:

HHH
HHT
HTH
HTT
TTT
TTH
THT
THH

2) Construct a tree diagram that shows the possible outcomes of rolling a four sided dice (numbered 1 to 4) twice. List the sample space

Sample space is:

1, 1
1, 2
1, 3
1, 4
2, 1
2, 2
2, 3
2, 4
3, 1
3, 2
3, 3
3, 4
4, 1
4, 2
4, 3
4, 4

3) Peter has 3 green, 2 white and 4 black shirts in a draw. If he takes 3 out without replacing them construct a tree diagram that shows all possible outcomes and list the sample space

GGG
GGW
GGB
GWG
GWW
GWB
GBG
GBW
GBB
WGG
WGW
WGB
WWG
WWB
WBG
WBW
WBB
BGG
BGW
BGB
BWG
BWW
BWB
BBG
BBW
BBB

4) Repeat question 3, but assume Peter replace the shirt each time he pulls one out

As above, but WWW is now possible

©Ezy Math Tutoring | All Rights Reserved

5) For each of the above questions, relate the number of choices available for EACH event to the number of outcomes in the sample space

The number of outcomes in the sample space equals the number of outcomes for each event raised to the power of the number of events

For example, there are 2 possible outcomes for tossing a coin. If this is repeated three times, the sample space has $2^3 = 8$ combinations

Similarly for question 2, $4^2 = 16$

For question 4, $3^3 = 27$

For question 3, the rule cannot be applied, since in one scenario, there is no chance of pulling a white shirt from the draw on the third try (since there are only 2 and they are not replaced). In effect, the number of possible outcomes changes for the third pull

6) A man wants to visit three different towns; Alpha, Beta, and Gamma. If he can visit them in any order, but can only visit each town once per trip, how many different trips are possible? (List the possible trips)

A then B then G
A then G then B
B then A then G
B then G then A
G then A then B
G then B then A

There are 6 possible ways to make the trip

7) In how many different ways can four separate coloured cards be arranged on a table?

Let the four colours be A, B, C and D

Then if A is first, the cards can be arranged

ABCD
ABDC
ACBD
ACDB
ADBC
ADCB

There are six combinations if A is put first

Similarly, there are 6 combinations for each of the other three cards being placed first

There are therefore 24 combinations of these cards

©Ezy Math Tutoring | All Rights Reserved

8) From your answers to questions 6 and 7, establish a rule for determining the number of arrangements of any number of different objects. Use your rule to calculate the number of ways a man could read 5 books given that they can be read once only, but in any order

If there are d objects, then there are d choices for the first object (3 towns, 4 cards). Once the first choice has been made, there is one less choice for the second (2 towns, 3 cards), etc.

Therefore for x objects, there are

$$(d) \times (d-1) \times (d-2) \times \ldots \times 1$$

combinations

For 5 books, there are $5 \times 4 \times 3 \times 2 \times 1 = 120$ combinations

9) From a group of 4 people one is to wear a blue badge, and another a red badge. How many different combinations of people could wear the badges? (List the possibilities)

Let the people be A, B, C and D

Listing the person to wear the blue badge first, then the person to wear the red badge gives

AB, AC, AD, BA, BC, BD, CA, CB, CD, DA, DB, DC

This gives 12 possible combinations

10) From a list of 5 books, John and Alex can choose one each. How many different combinations of books can they choose? (Note they cannot choose the same book as the other)

Let the books be titled A, B, C, D and E

If two are chosen, they could be

AB, AC, AD, AE, BA, BC, BD, BE, CA, CB, CD, CE, DA, DB, DC, DE, EA, EB, EC, ED

11) From your answers to questions 9 and 10, determine a rule for calculating how many different combinations of selections can be made from a list. Use your rule to determine how many groups of President, Secretary and Treasurer can be made from a committee of 5 people.

For x items, there are x choices for the first selection, then $(x-1)$ for the second selection etc.

So for 3 specific titles from 5 people, there are $5 \times 4 \times 3 = 60$ possible combinations

©Ezy Math Tutoring | All Rights Reserved

12) From a group of 4 people 2 are to be selected. How many different combinations are there?

Let the people be A, B, C and D

Possible combinations are

A and B
A and C
A and D
B and C
B and D
C and D

This gives 6 possible pairings

Note that since order of selection does not matter here, AB and BA represent the same choice and therefore cannot be listed twice

13) From a group of 5 pizza toppings, a customer can choose two. How many different pizzas can be made?

Let toppings be A, B, C, D and E

Possible combinations are

ABC
ABD
ABE
ACD
ACE
ADE
BCD
BCE
BDE
CDE

This gives 10 possible combinations

Note again that in this case ABC is the same as

ACB, BAC, BCA, CBA, CAB

And therefore only one possibility is listed.

(There is no difference between a ham and pineapple pizza and a pineapple and ham pizza!)

14) From your answers to questions 12 and 13, determine a rule for selecting a certain quantity from a group. How does this differ from your answer to question 11? Use your result to calculate the number of groups of 3 people that can be chosen from a larger group of 8

The answers differ because the order of selection does not matter, hence the possible combinations must be reduced to allow for non-doubling

Let P be the number of combinations as per question 8.

Let Q be the number of combinations of selections to be made

Let R be the number of non-selections that can be made

©Ezy Math Tutoring | All Rights Reserved

Then number $= \frac{P}{Q \times R}$

To select 3 items from a group of 8

Number
$= \frac{8 \times 7 \times 6 \times 5 \times 4 \times 3 \times 2 \times 1}{(5 \times 4 \times 3 \times 2 \times 1) \times (3 \times 2 \times 1)} = 56$

15) Use tree diagrams to calculate the following probabilities

a) A coin is tossed three times and lands on heads each time

$\frac{1}{8}$

b) A four sided dice is rolled three times and the numbers 2, 4, and 1 are rolled (in any order)

$\frac{6}{64} = \frac{3}{32}$

c) A four sided dice is rolled and the numbers 2, 4, and 1 are rolled, in that order

$\frac{1}{64}$

16) A man pulls 3 shirts from a draw that initially contains 3 green, 2 white and 4 black shirts. If he does not replace the shirts what is the probability of drawing

a) One of each colour?

Probability of GWB
$= \frac{3}{9} \times \frac{2}{8} \times \frac{4}{7} = \frac{24}{504}$

All combinations of three colours have the same probability

There are 6 ways to get three different coloured shirts $(3 \times 2 \times 1)$

Total probability
$= 6 \times \frac{24}{504} = \frac{2}{7}$

b) 2 black and 2 green?

There are only 3 shirts being taken from draw so impossible to get a total of 4!

c) All white?

There are only 2 white shirts to begin with; since they are not replaced it is impossible to get 3 whites

d) All black?

$\frac{4}{9} \times \frac{3}{8} \times \frac{2}{7} = \frac{1}{21}$

17) Repeat question 16, but assume that the man replaces the shirts each time he pulls one out

©Ezy Math Tutoring | All Rights Reserved

a) Probability of GWB
$= \frac{3}{9} \times \frac{2}{9} \times \frac{4}{9} = \frac{8}{243}$

Again there are 6 ways to get the three colours

Probability $= 6 \times \frac{8}{243} = \frac{16}{81}$

b) Same as previous question

c) Probability of all white
$= \frac{2}{9} \times \frac{2}{9} \times \frac{2}{9} = \frac{8}{729}$

d) Probability of all black
$= \frac{4}{9} \times \frac{4}{9} \times \frac{4}{9} = \frac{64}{729}$

©Ezy Math Tutoring | All Rights Reserved

Exercise 3

Applications of Probability

©Ezy Math Tutoring | All Rights Reserved

1) Tom tosses a coin four times in a row

a) How many different outcomes are there in each toss?

2 (heads or tails)

b) What is the probability of throwing a tail on any toss?

$\frac{1}{2}$

c) How many different outcomes are there for the four tosses?

16

d) Of these outcomes, how many times should Tom expect to throw four tails in a row?

Once

e) Raise the probability of throwing a tail on one toss to the power of the number of tosses.

$\frac{1}{16}$

f) What do you notice about your answers to parts d and e?

The probability of throwing 4 tails in a row is equal to the probability of throwing one tail raised to the power of the number of throws

g) Tom actually took 25 trials to throw four tails in a row. Does this mean the calculations are wrong? Explain your answer

No. Theoretical probability is not necessarily in actual experiment, especially for such a small number of throws. For very large numbers of trials, theoretical and experimental numbers should converge

2)

a) What is the probability of drawing a diamond from a standard pack of 52 cards?

$\frac{1}{4}$

b) What is the theoretical probability of drawing 5 diamonds in a row? (Assume the card is replaced each time)

$\left(\frac{1}{4}\right)^5 = \frac{1}{1024}$

©Ezy Math Tutoring | All Rights Reserved

c) Do you expect that an experiment would produce the exact result calculated in part b? Explain

No. As per previous question, experimental and theoretical probabilities do not coincide exactly, especially for a small number of trials

3) Tim buys a ticket in a raffle which has three prizes. First receives \$300, second gets \$150 and third prize is \$50. If there are 1000 tickets at \$1 each, what is the financial expectation of Tim's ticket?

$$E = \frac{(300+150+50)}{1000\times\$1} = 0.50$$

4) Glen always bets \$5 on red at the roulette table. If the ball lands on red, Glen gets \$10 back. If the ball lands on black, Glen loses his \$5. If there are equal quantities of red and black numbers (18 of each). What is Glen's financial expectation?

$$E = \left(\frac{1}{2}\right) \times \$5 - \left(\frac{1}{2}\right) \times \$5 = 0$$

In the long run Glen should break even

5) In reality there are also two green numbers on the wheel (0 and 00). If the ball lands on either of these, Glen (and every other player) loses. What is the new financial expectation?

$$E = \left(\frac{18}{38}\right) \times \$5 - \left(\frac{18}{38}\right) \times \$5 - \left(\frac{2}{38}\right) \times \$5 = -\$\frac{10}{38} \cong 26\ cents$$

Glen will lose an average of 26 cents per spin in the long term

6) Colin plays a game where there is a 30% chance of winning \$4, a 20% chance of winning \$10 and a 50% chance of losing \$10. Each game costs 50 cents to play. What is his financial expectation?

$$E\left(\frac{3}{10} \times \$4\right) + \left(\frac{2}{10} \times \$10\right) - \left(\frac{5}{10} \times \$10\right) - 0.50 = -\$2.30$$

7) A group of 1000 people were asked whether smoking should be banned in restaurants, totally, only allowed in designated areas, or allowed anywhere in the restaurant. The results of the survey are shown in the following table

	Smokers	**Non-smokers**	**Total**
Banned	25	600	625

©Ezy Math Tutoring | All Rights Reserved

Special areas	75	100	175
Allowed	150	50	200
	250	750	1000

a) What is the probability that a person chosen at random wants smoking banned?

$$\frac{625}{1000} = \frac{5}{8}$$

b) What is the probability that a smoker wants smoking banned?

$$\frac{25}{250} = \frac{1}{10}$$

c) What is the probability that a person who wants smoking to be allowed in special areas is a non-smoker?

$$\frac{100}{175} = \frac{4}{7}$$

d) What is the probability that a person who wants smoking banned is a non-smoker?

$$\frac{600}{625} = \frac{24}{25}$$

e) The surveyors claimed that the survey proves the majority of the population wants smoking banned in restaurants. How would you respond to this claim?

The sample is small, only 1000 people surveyed

There were more non-smokers surveyed than smokers which would naturally bias the results toward having smoking banned

8) One thousand people take a lie detector test. Of 800 people known to be telling the

©Ezy Math Tutoring | All Rights Reserved

truth the lie detector indicates that 23 are lying. Of 200 people known to be lying, the lie detector indicates that 156 are lying. Present this information in a two-way table

	Accurate	Not accurate	Total
Lying	156	44	200
Truthful	777	23	800
Total	933	67	1000

9) A proposed test for a medical condition was trialled on 1000 volunteers, some who had the condition and some who did not. The trial was taken to determine how accurate the test was. The results are summarized in the table

	Accurate	Not accurate	Total
With condition	195	5	200
Without condition	730	70	800
Total	925	75	1000

a) Why were only 200 people with the condition included in the trial of 1000 people?

There are more people in the community who do not have the condition, therefore the survey attempted to reflect a fair sample of the population

b) What was the overall correct diagnosis percentage?

92.5%

c) What is the probability that a person with the condition is properly diagnosed?

©Ezy Math Tutoring | All Rights Reserved

$$\frac{195}{200} = \frac{39}{40}$$

d) What is the probability that a person who did not have the condition was incorrectly diagnosed (that is told they had the condition)?

$$\frac{70}{800} = \frac{7}{80}$$

e) What is the probability that a person who was diagnosed incorrectly did not have the condition?

$$\frac{70}{75} = \frac{14}{15}$$

f) Comment on the overall effectiveness of the test

Although the overall accuracy of the test was high (92.5%), there is concern that the majority of wrong diagnoses were given to people who did not have the condition; that is they would have been told they did have it. This may have caused trauma, cost time and money to have further tests and/or unnecessary treatment.

2.5% of the people who had the condition were diagnosed as clear

©Ezy Math Tutoring | All Rights Reserved

General Mathematics

Algebraic Modelling

©Ezy Math Tutoring | All Rights Reserved

Exercise 1

Algebraic Skills & Techniques

©Ezy Math Tutoring | All Rights Reserved

1) Substitute the value $x = 2$ into each of the following linear equations, and hence evaluate the equation

a) $x + 2$

4

b) $x + 3$

5

c) $x - 5$

-3

d) $2x + 1$

5

e) $3x - 4$

2

f) $6x + 5$

17

g) $2x - 10$

-6

h) $3x - 6$

0

i) $\frac{1}{2}x + 8$

9

j) $\frac{1}{4}x - 4$

$-\frac{7}{2}$

k) $\frac{7}{2}x - 7$

0

l) $0x + 23456$

23456

2) Substitute the value $x = 3$ into each of the following quadratic equations, and hence evaluate the equation

a) x^2

9

b) $x^2 + 3$

12

c) $x^2 - 2$

7

d) $2x^2 + 2$

20

e) $3x^2 - 1$

26

©Ezy Math Tutoring | All Rights Reserved

f) $5x^2 + 2$

47

g) $2x^2 - 20$

-2

h) $4x^2 - 30$

6

i) $\frac{1}{2}x^2 + 1$

$\frac{11}{2}$

j) $\frac{1}{3}x^2 - 3$

0

k) $\frac{2}{3}x^2 + 5$

7

l) $0x^2 + 21232$

21232

3) Substitute the value $x = 2$ into each of the following cubic equations, and hence evaluate the equation

a) x^3

8

b) $x^3 + 1$

9

c) $x^3 - 2$

6

d) $2x^3$

16

e) $3x^3 - 24$

0

f) $\frac{1}{2}x^3 - 6$

-2

g) $0x^3 + 123432$

123432

4) Simplify the following expressions

a) $4x - 2x$

$2x$

b) $3x + x$

$4x$

c) $3x + y - x$

$2x + y$

©Ezy Math Tutoring | All Rights Reserved

d) $4x + 2y - x + y$

$3x + 3y$

e) $3x - 2y + 3y - 4x$

$y - x$

f) $2x^2 - 3x + x^2 - 2x$

$3x^2 - 5x$

g) $x - x^2 + y - x^2 - y$

$x - 2x^2$

h) $2x^2 - y + y^2 - y - 2x^2$

$y^2 - 2y$

i) $-y + 3x - 2x^2 - 3x + y$

$-2x^2$

j) $(2x + y) - (y - 2x)$

$4x$

k) $(y - 3x) + (3x^2 - y)$

$3x^2 - 3x$

5) Multiply the following, expressing your answer in index form

a) $2x \times y$

$2xy$

b) $3x \times x$

$3x^2$

c) $3x^2 \times x^2$

$3x^4$

d) $y^2 \times x \times y$

xy^3

e) $x \times y \times z \times y^2$

xy^3z

f) $a \times b^2 \times a^2 \times b$

a^3b^3

6) Simplify the following

a) $\frac{3x^2}{x}$

$3x$

b) $\frac{4xy}{2y}$

$2x$

c) $\frac{8x^3y^2}{2xy}$

$4x^2y$

d) $\frac{15xy^3z}{5xyz}$

$3y^2$

©Ezy Math Tutoring | All Rights Reserved

e) $\frac{x^4y}{x^2y^2}$

$\frac{x^2}{y}$

f) $\frac{x \times x^2 \times z^2 \times z^2}{x^3 \times z^4}$

1

7) Make t the subject of the following equations

a) $y = 3t - 1$

$3t = y + 1$

$t = \frac{y+1}{3}$

b) $y = \frac{1}{2}t + 2$

$\frac{1}{2}t = y - 2$

$t = 2y - 4$

c) $2y = 3t - 5$

$3t = 2y + 5$

$t = \frac{2y+5}{3}$

d) $2y - 3 = 2t + 4$

$2t = 2y - 7$

$t = y - \frac{7}{2}$

e) $y = t^2$

$t = \sqrt{y}$

f) $y = \frac{1}{2}t^2$

$t^2 = 2y$

$t = \sqrt{2y}$

g) $y = x^2 - t^2$

$t^2 = x^2 - y$

$t = \sqrt{x^2 - y}$

h) $y = at^2 + c$

$at^2 = y - c$

$t^2 = \frac{y-c}{a}$

$t = \sqrt{\frac{y-c}{a}}$

8) Solve for y by substituting the value given into the equation

a) $y = 3x - 2$when $x = 3$

$y = 7$

b) $y = 7x - 7$ when $x = 1$

$y = 0$

©Ezy Math Tutoring | All Rights Reserved

c) $y = \frac{1}{2}x^2 - 4$ when $x = 4$

$y = 4$

d) $y = \sqrt{x} - 3$ when $x = 16$

$y = 1 \; or \; y = -7$

e) $y = 2x^2 + 3x - 1$ when $x = 3$

$y = 26$

f) $y = 3(x^2 - 100)$ when $x = 20$

$y = 900$

g) $y = \frac{100}{x} + x^2$ when $x = 15$

$y = 231\frac{2}{3}$

h) $x = 3y^2$ when $x = 27$

$3y^2 = 27$

$y^2 = 9$

$y = \pm 3$

i) $x = \frac{1}{2}y^2 - 2$ when $x = 70$

$\frac{1}{2}y^2 = 72$

$y^2 = 144$

$y = \pm 12$

j) $x = \frac{2y-4}{3}$ when $x = 6$

$\frac{2y - 4}{3} = 6$

$2y - 4 = 18$

$y = 11$

9) Express the following in scientific notation

a) 0.0356

3.56×10^{-2}

b) 21223.19

2.122319×10^4

c) 409.754

4.09754×10^2

d) 0.00787

7.87×10^{-3}

e) 19003

1.9003×10^4

f) 32.856

3.2856×10^1

g) 0.00342

3.42×10^{-3}

©Ezy Math Tutoring | All Rights Reserved

h) 499.005

4.99005×10^2

10) Use guess check and improve to calculate the value of x in the following

a) $3^x = 12$

$3^2 = 9$

$3^3 = 27$

$x \cong 2.2$

b) $2^x = 14$

$2^3 = 8$

$2^4 = 16$

$x \cong 3.8$

c) $2^{x-1} = 9$

$2^3 = 8$

$2^4 = 16$

$x - 1 \cong 3.2$

$x \cong 4.2$

d) $0.5^{2x} = 0.25$

$0.5^2 = 0.25$

$2x = 2$
$x = 1$

e) $10^{4x} = 100$

$10^2 = 100$

$4x = 2$

$x = \frac{1}{2}$

f) $1^x = 700$

One to any power equal one, therefore this equation has no solution

g) $34^x = 1$

Any number to the power zero equals 1, therefore $x = 0$

11) A tree loses 20% of the leaves it has each day. After how many days will it have 10% of its original number of leaves left?

Let original number of leaves be N

Day	% of leaves left
1	0.8N
2	0.8 x 0.8 = 0.64N
3	0.64 x 0.8 = 0.512N
4	0.512 x 0.8 = 0.4096N
5	0.4096 x 0.8 = 0.32768N
6	0.32768 x 0.8 = 0.262144N
7	0.262144 x 0.8 = 0.2097152N
8	0.2097152 x 0.8 = 0.16777216N
9	0.16777216 x 0.8 = 0.134217728N
10	0.134217728 x 0.8 = 0.107374

©Ezy Math Tutoring | All Rights Reserved

At the end of the eleventh day the tree will have 0.107374 x 0.8 = 0.086N of its original leaves left

The amount of leaves on the tree will drop below 10% on the eleventh day

12) A balloon is blown up so its size increases by 25% each minute. It bursts after 8 minutes. How much bigger than its original size was it when it burst?

Let original size be R

Minutes	Times original size
1	1.25R
2	1.25 x 1.25 = 1.5625R
3	1.5625 x 1.25 =1.953125R
4	1.952125 x 1.25 = 2.44149625R
5	2.44149625 x 1.25 = 3.05R
6	3.05 x 1.25 = 3.815R
7	3.815 x 1.25 = 4.7684R
8	4.7684 x 1.25 = 5.96R

The balloon will be approximately 6 times bigger at time of bursting

13) The total resistance of two resistors placed in parallel in an electrical circuit is given by the formula

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2}$$

Where R is the total resistance in the circuit, and R_1 and R_2 are the values of the two resistors

If the value of R_1 is fixed at 10 ohms, draw up a table of values for R when R_2 is 5, 10, 15, ...50 ohms

Example of one calculation

Let $R_2 = 50$

$$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} = \frac{1}{10} + \frac{1}{50} = \frac{6}{50}$$

$$R = \frac{50}{6} \cong 8.333$$

Complete table

$\boldsymbol{R_2}$	$\boldsymbol{R}$
5	3.333
10	5
15	6
20	6.6667
25	7.143
30	7.5
35	7.778
40	8
45	8.1818
50	8.333

©Ezy Math Tutoring | All Rights Reserved

Exercise 2

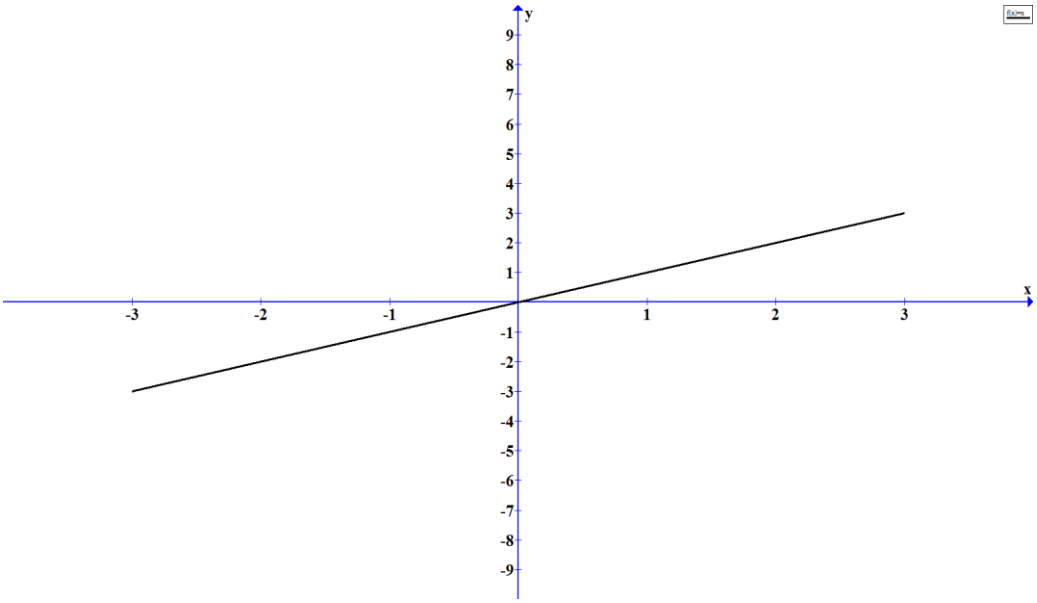
Modelling Linear Relationships

©Ezy Math Tutoring | All Rights Reserved

1) For each of the linear functions, draw a table of values for $-3 \leq x \leq 3$ and sketch the graph of the function from your table

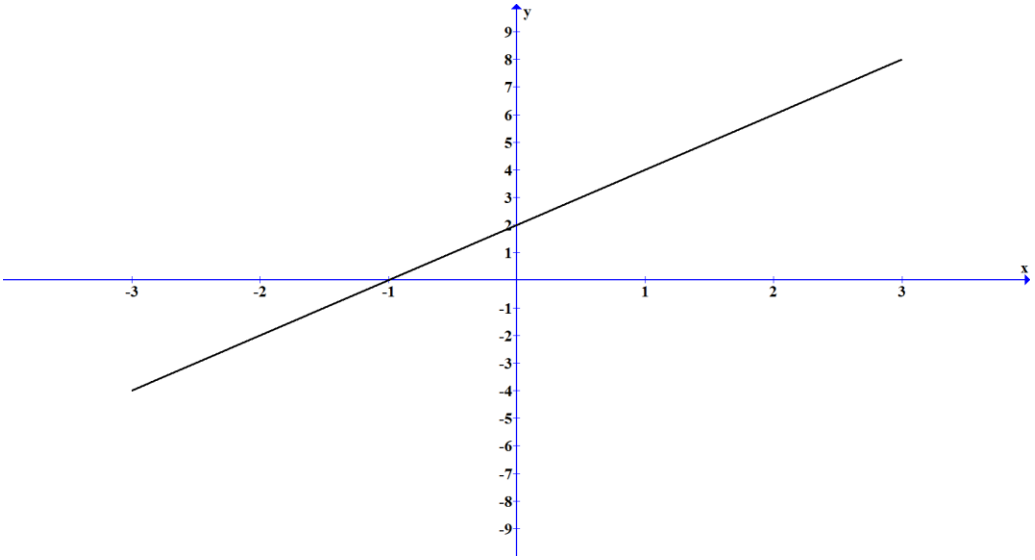
a) $y = x$

x	-3	-2	-1	0	1	2	3
y	-3	-2	-1	0	1	2	3



b) $y = 2x + 2$

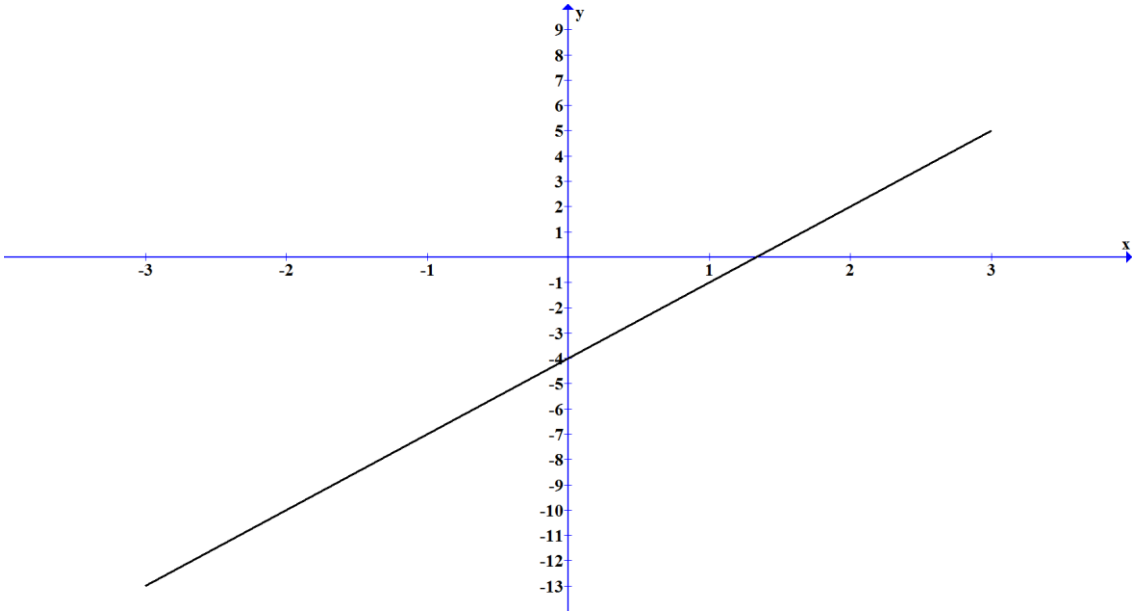
x	-3	-2	-1	0	1	2	3
y	-4	-2	0	2	4	6	8



©Ezy Math Tutoring | All Rights Reserved

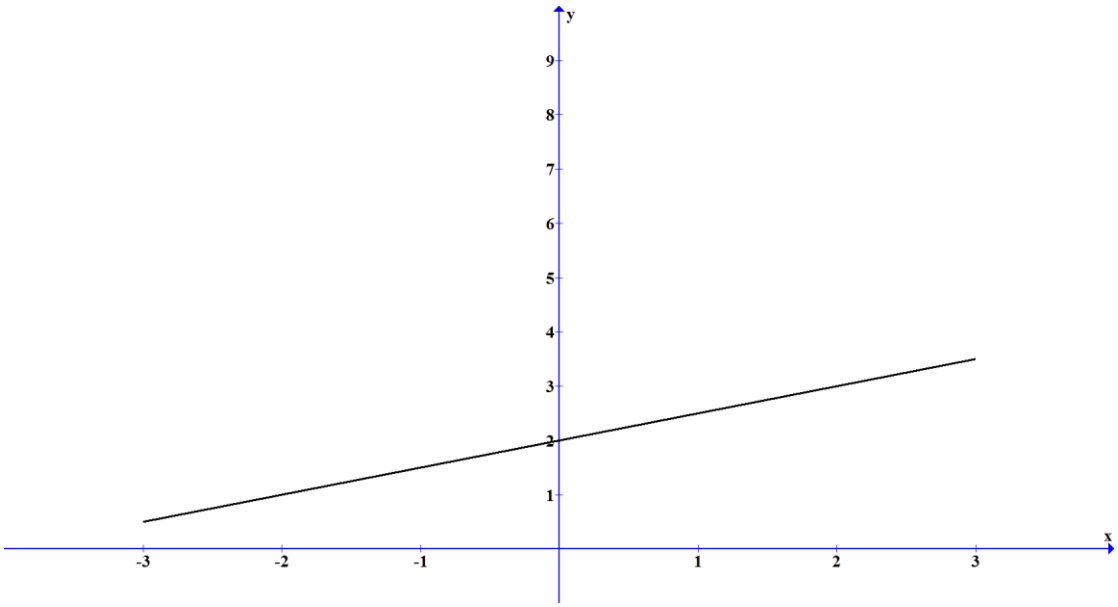
c) $y = 3x - 4$

x	-3	-2	-1	0	1	2	3
y	-13	-10	-7	-4	-1	2	5



d) $y = \frac{1}{2}x + 2$

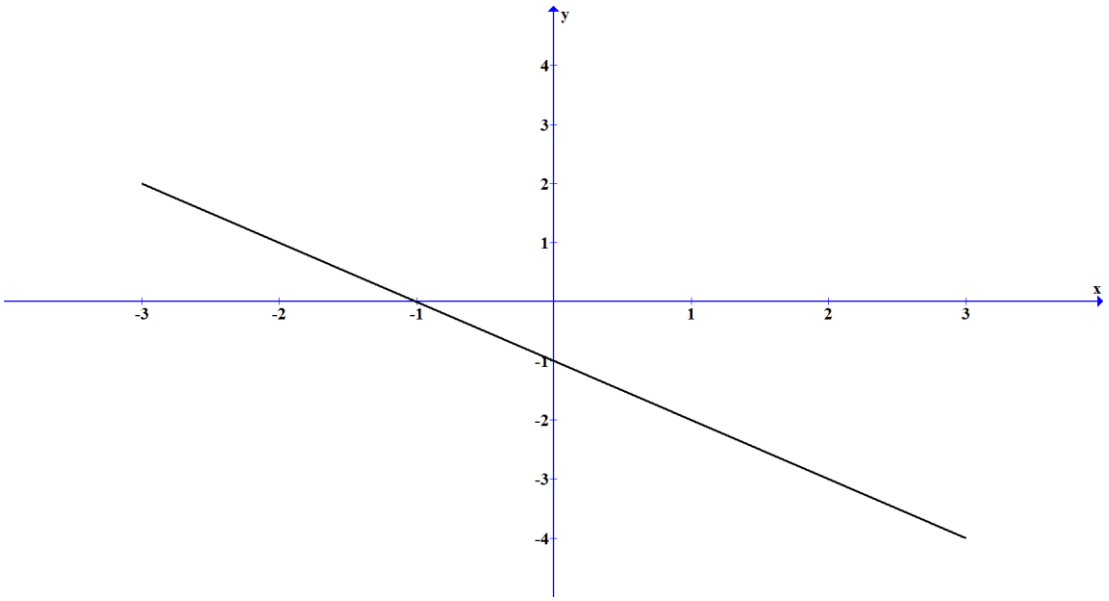
x	-3	-2	-1	0	1	2	3
y	0.5	1	1.5	2	2.5	3	3.5



©Ezy Math Tutoring | All Rights Reserved

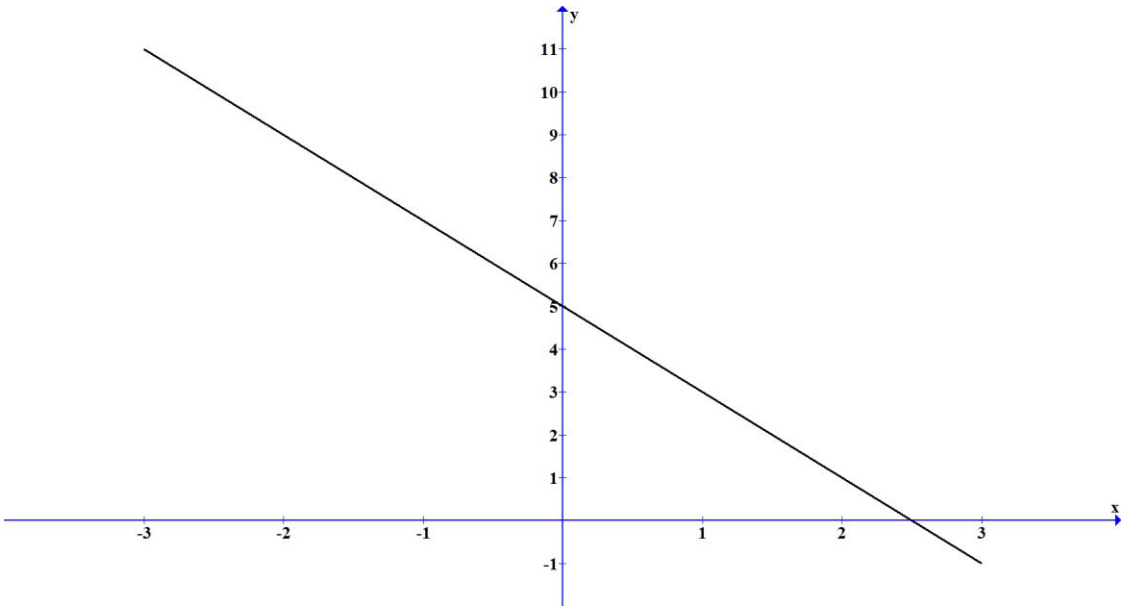
e) $y = -x - 1$

x	-3	-2	-1	0	1	2	3
y	2	1	0	-1	-2	-3	-4



f) $y = -2x + 5$

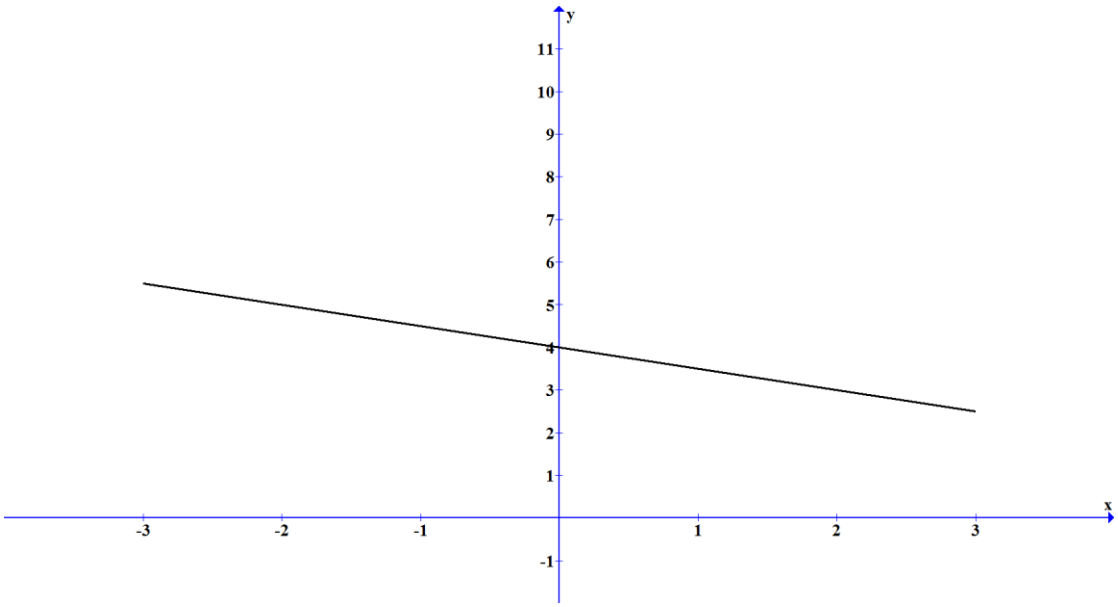
x	-3	-2	-1	0	1	2	3
y	11	9	7	5	3	1	-1



©Ezy Math Tutoring | All Rights Reserved

g) $y = -\frac{1}{2}x + 4$

x	-3	-2	-1	0	1	2	3
y	5.5	5	4.5	4	3.5	3	2.5



2) From your answers to question 1, what is the relationship between the value of the constant in a linear equation, and the graph of the equation?

The graph crosses the y axis at the value of the constant

3) From your answers to question 1, what is the effect of changing the sign of the coefficient of x?

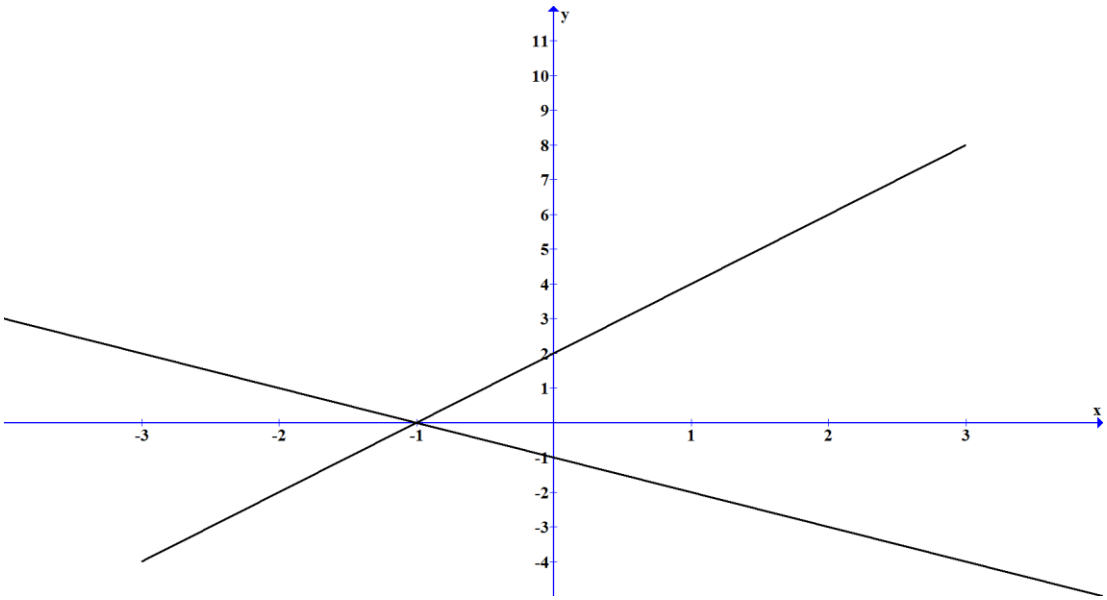
The line slopes upwards for a positive coefficient and down for a negative value

4) Choose two pairs of graphs from question 1 and determine their point(s) of intersection

Choose any two

Example $y = -x - 1 \; and \; y = 2x + 2$

©Ezy Math Tutoring | All Rights Reserved



Lines intersect at the point $x = -1, y = 0$

5) A The instructions for cooking a roast state that it should be cooked for thirty minutes plus 40 minutes for every kg the meat weighs

a) For how long should a roast that weighs 1.5 kg be cooked for?

b) Construct a table of values that relate the weight of the meat to its cooking time

c) Graph the values

d) Determine the gradient of the line produced. How does this value relate to the quantities in the problem?

e) Relate the y intercept to the quantities in the problem

f) Is the graph valid for all weights; that is can the graph be extended indefinitely? Explain your answer

6) A plumber charges a call out fee of $25 plus $20 per hour for his work. If he works for part of the hour he only charges for that part. For example, for 15 minutes work he will charge $5 (plus his call out fee)

a) How much will he charge for 2 hours work?

©Ezy Math Tutoring | All Rights Reserved

b) How much will he charge for 3.5 hours work

c) Construct a table of values that relate the time taken for a job to the total charge

d) Graph the values

e) Determine the gradient of the line produced. How does this value relate to the quantities in the problem

f) Relate the y intercept to the quantities in the problem

g) Is the graph valid for all weights; that is can the graph be extended indefinitely? Explain your answer

7) Another plumber charges a $25 call out fee and $20 per hour for his work. Differently to the previous plumber he charges $20 even if he only works for part of an hour. For example, for 15 minutes work he will charge $20 (plus his call out fee)

a) How much will he charge for 2 hours work?

b) How much will he charge for 3.5 hours work

c) Construct a table of values that relate the time taken for a job to the total charge

d) Graph the values

e) How does the graph differ from that in question 6?

8) To convert from Celsius to Fahrenheit temperature the following formula is used

$$F = \frac{9}{5}C + 32$$

a) Construct a table of values for $C = 0 \; to \; 40$ in steps of 5 degrees

b) Graph the relationship

©Ezy Math Tutoring | All Rights Reserved

c) Determine the gradient of the line produced. How does this value relate to the quantities in the equation?

d) Relate the y intercept to the quantities in the equation

e) Use the graph to extrapolate the value of 42 degrees Celsius in Fahrenheit

f) Use the graph to determine how many degrees Celsius equals 23 degrees Fahrenheit

g) Is the graph valid for all values of C? Explain

9) One Australian dollar currently buys 56.5 Indian rupees

a) Construct a table of values for 0 to 30 Australian dollars in steps of 5 dollars

b) Graph the relationship

c) Determine the gradient of the line produced. How does this value relate to the quantities in the equation?

d) Relate the y intercept to the quantities in the equation

e) How many rupees does 40 Australian dollars buy?

f) How many Australian dollars does 1695 rupees buy?

10) A bath has 200 litres of water in it. The plug is pulled and water flows from it at the rate of 4 litres per second.

a) Construct a table of values that relate the volume of water in the bath to the time since the plug was pulled

b) Graph the relationship

c) From your graph how long until the bath is empty?

©Ezy Math Tutoring | All Rights Reserved

d) Determine the gradient of the line produced. How does this value relate to the quantities in the problem?

e) Relate the y intercept to the quantities in the problem

f) Is the graph valid for all values of t? Explain

©Ezy Math Tutoring | All Rights Reserved

Exercise 3

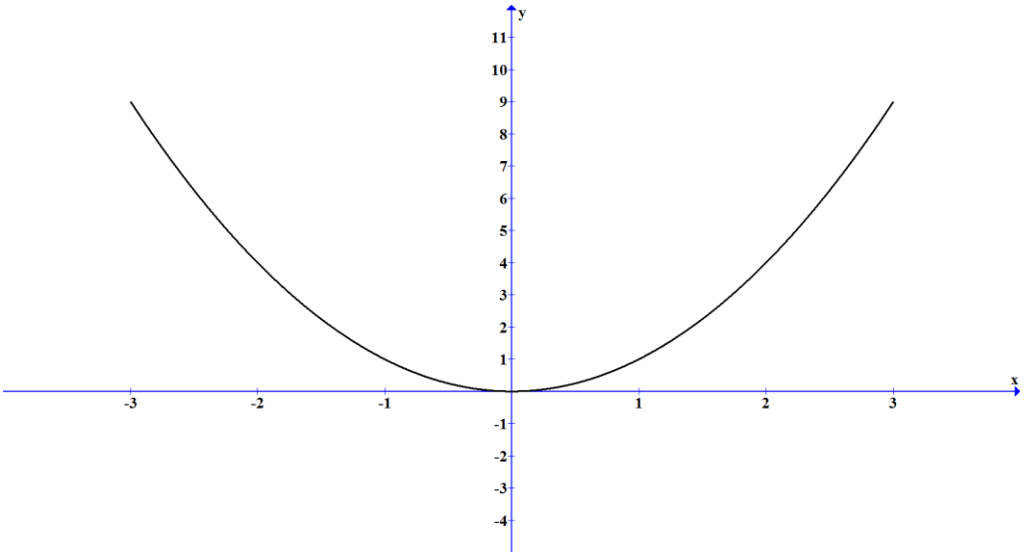
Modelling Non-Linear Relationships

©Ezy Math Tutoring | All Rights Reserved

1) For each of the following equations, generate a table of values for $-3 \leq x \leq 3$ and sketch the graph of the function from your table

a) $y = x^2$

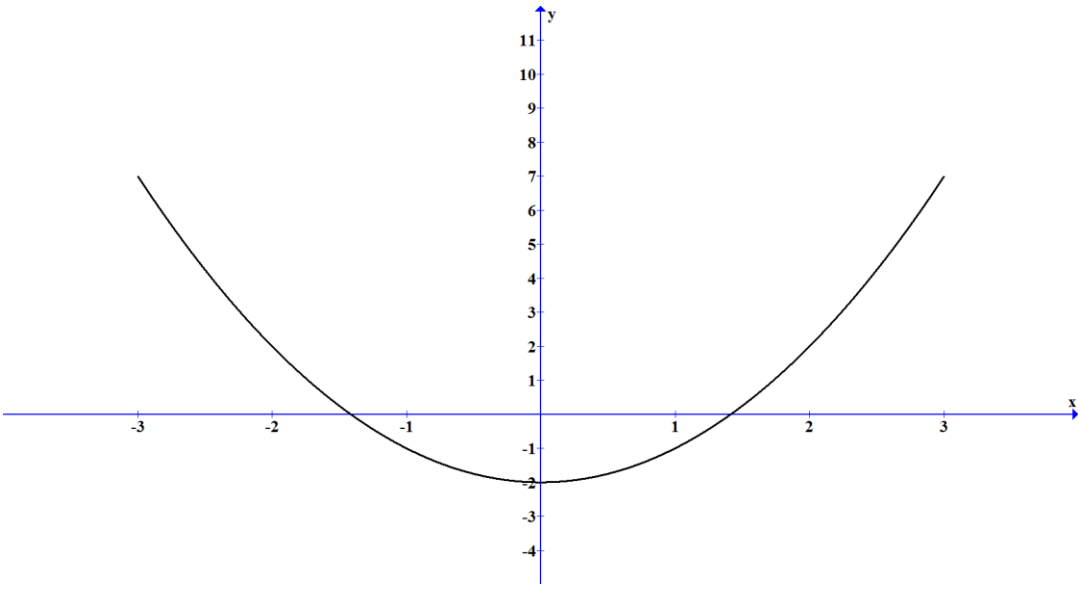
x	-3	-2	-1	0	1	2	3
y	9	4	1	0	1	4	9



b) $y = x^2 - 2$

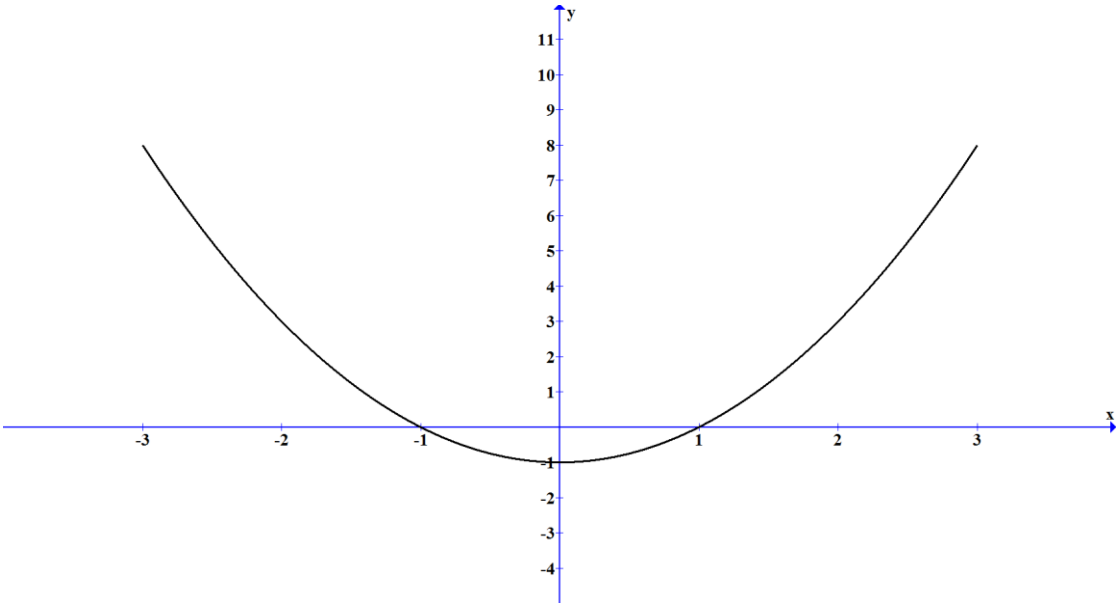
x	-3	-2	-1	0	1	2	3
y	7	2	-1	-2	-1	2	7

©Ezy Math Tutoring | All Rights Reserved



c) $y = x^2 - 1$

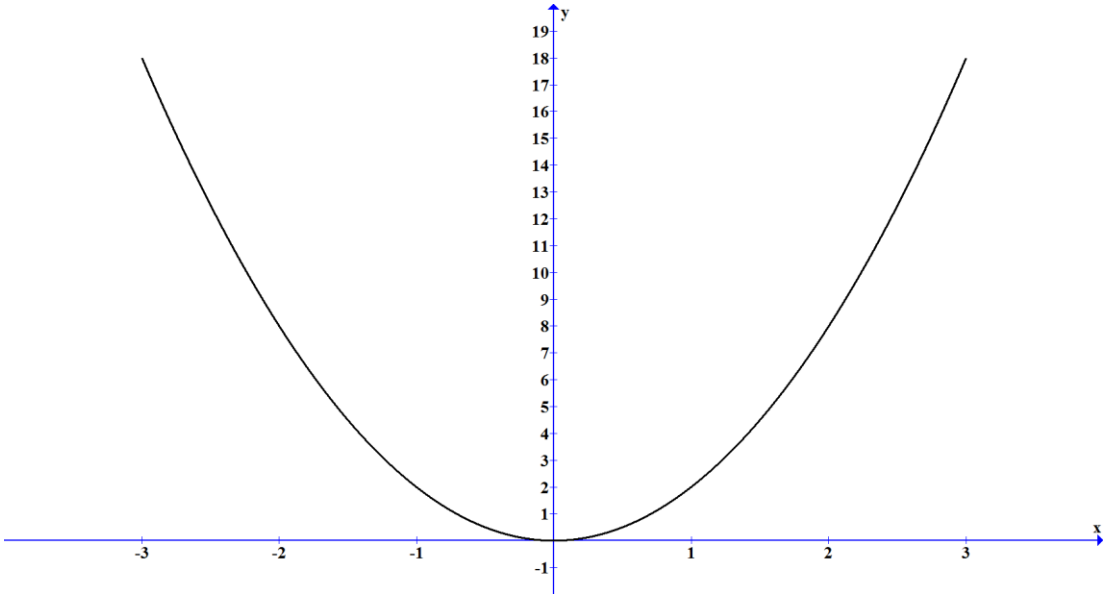
x	-3	-2	-1	0	1	2	3
y	8	3	0	-1	0	3	8



d) $y = 2x^2$

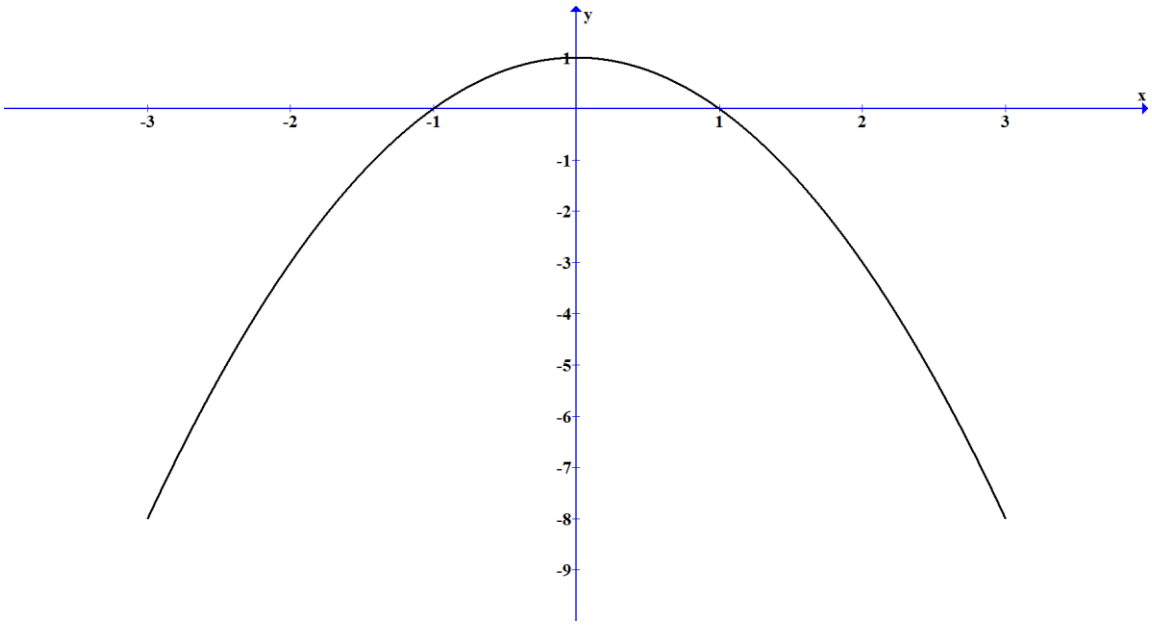
x	-3	-2	-1	0	1	2	3
y	18	8	2	0	2	8	18

©Ezy Math Tutoring | All Rights Reserved



e) $y = -x^2 + 1$

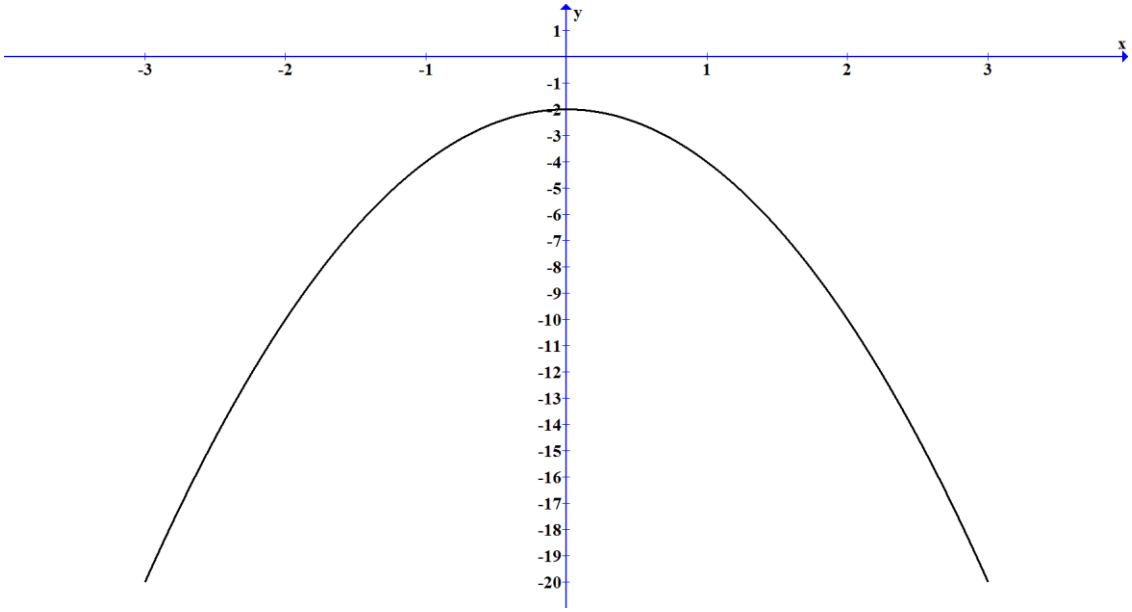
x	-3	-2	-1	0	1	2	3
y	-8	-3	0	1	0	-3	-8



f) $y = -2x^2 - 2$

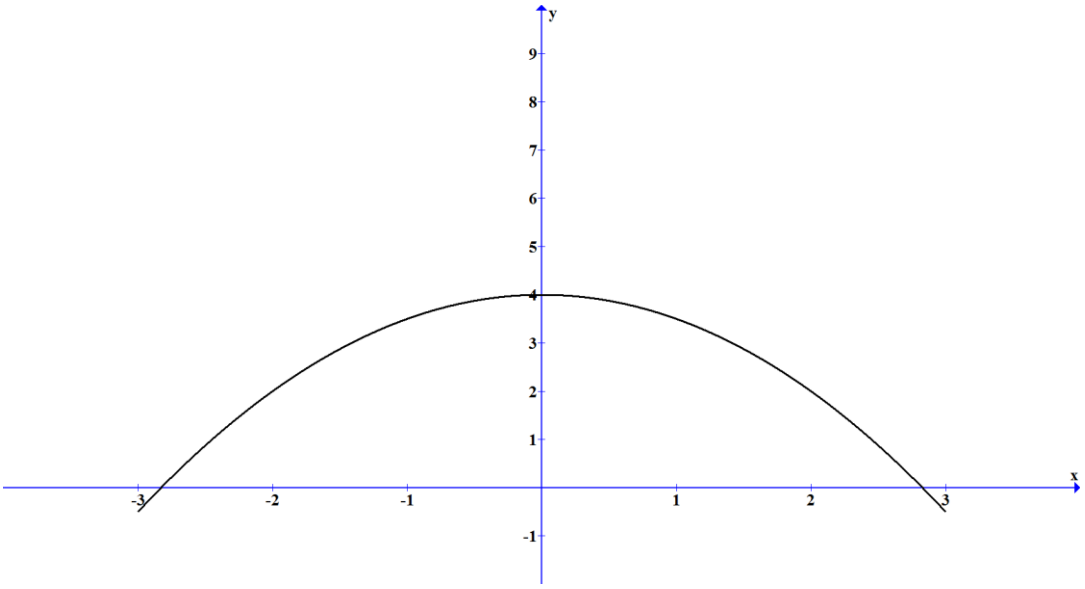
x	-3	-2	-1	0	1	2	3
y	-20	-10	-4	-2	-4	-10	-20

©Ezy Math Tutoring | All Rights Reserved



g) $y = -\frac{1}{2}x^2 + 4$

x	-3	-2	-1	0	1	2	3
y	-0.5	2	3.5	4	3.5	2	-0.5



2) From your answers to question 5, what is the effect of changing the sign and value of the coefficient of x in a quadratic equation?

A larger value makes the graph wider, while a negative value inverts the graph

©Ezy Math Tutoring | All Rights Reserved

3) From your answers to question 5, what is the relationship between the value of the constant in a quadratic equation and the graph of the equation?

The curve intersects the y axis at the value of the constant

4) Using your graphs, find the co-ordinates of the maximum or minimum values of each function in question 5

a) 0

b) -2

c) -1

d) 0

e) 1

f) -2

g) 4

5) Make a table of values for each pair of equations

a) $(x-1)^2+3, \quad x^2-2x+4$

x	-3	-2	-1	0	1	2	3
$(x-1)^2+3$	19	12	7	4	3	4	7
x^2-2x+4	19	12	7	4	3	4	7

b) $(x+2)^2-1, \quad x^2+4x+3$

x	-3	-2	-1	0	1	2	3
$(x+2)^2-1$	0	-1	0	3	8	15	24
x^2+4x+3	0	-1	0	3	8	15	24

c) $(x-2)^2+2, \quad x^2-4x+6$

x	-3	-2	-1	0	1	2	3
$(x-2)^2+2$	27	18	11	6	3	2	3

©Ezy Math Tutoring | All Rights Reserved

$x^2 - 4x + 6$	27	18	11	6	3	2	3

d) $(x+1)^2 + 1, \quad x^2 + 2x + 2$

$\boldsymbol{x}$	-3	-2	-1	0	1	2	3
$(x+1)^2 + 1$	5	2	1	2	5	10	17
$x^2 + 2x + 2$	5	2	1	2	5	10	17

e) $(x-3)^2 - 6, \quad x^2 - 6x + 3$

$\boldsymbol{x}$	-3	-2	-1	0	1	2	3
$(x-3)^2 - 6$	30	19	10	3	-2	-5	-6
$x^2 - 6x + 3$	30	19	10	3	-2	-5	-6

6) What do you notice about the table of values for each pair of equations in question 9, and hence their graphs?

They are the same

7) What can you say about each pair of equations in question 9?

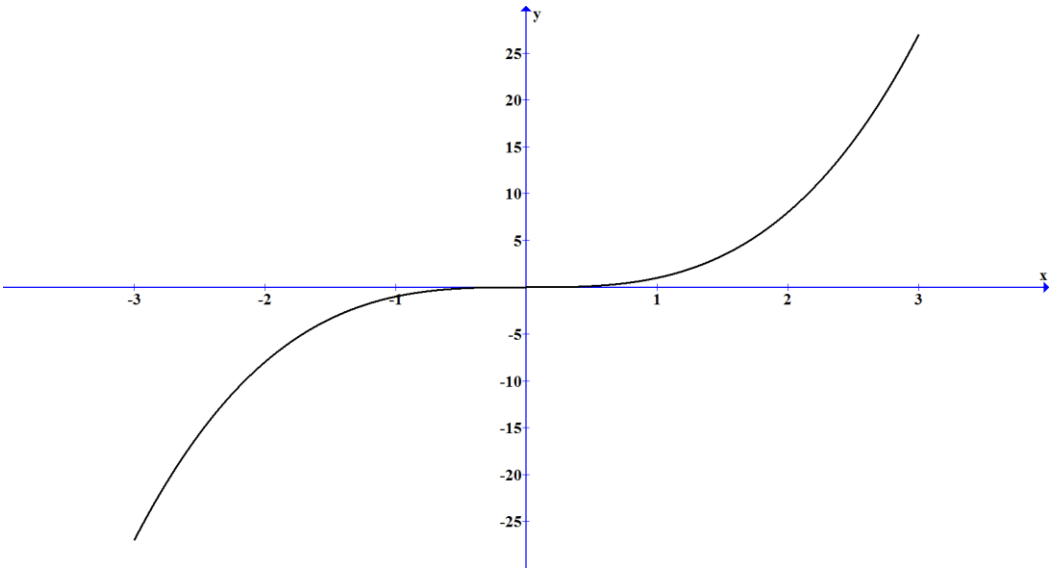
Each pair of equations must be equivalent

8) For each equation, generate a table of values and graph the equation, choosing an appropriate range

a) $y = x^3$

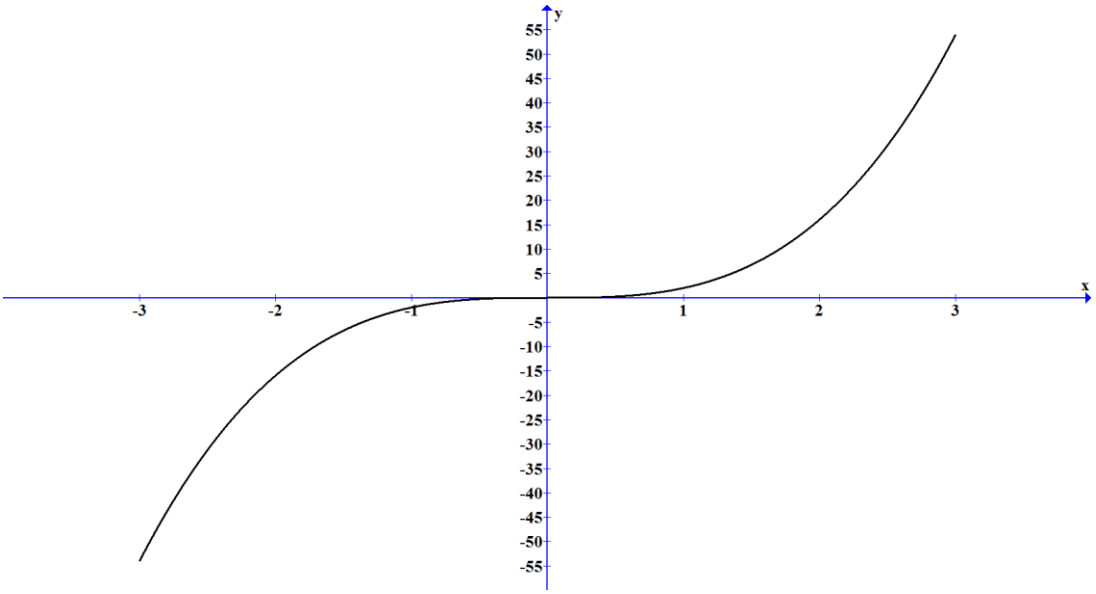
$\boldsymbol{x}$	-3	-2	-1	0	1	2	3
$\boldsymbol{y}$	-27	-8	-1	0	1	8	27

©Ezy Math Tutoring | All Rights Reserved



b) $y = 2x^3$

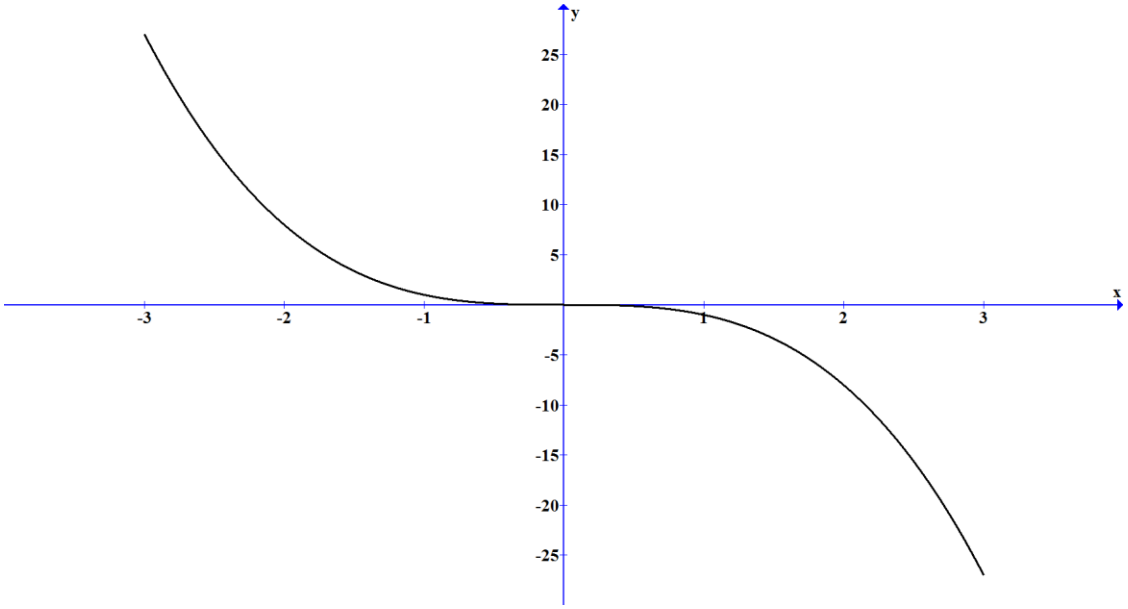
x	-3	-2	-1	0	1	2	3
y	-54	-16	-2	0	2	16	54



c) $y = -x^3$

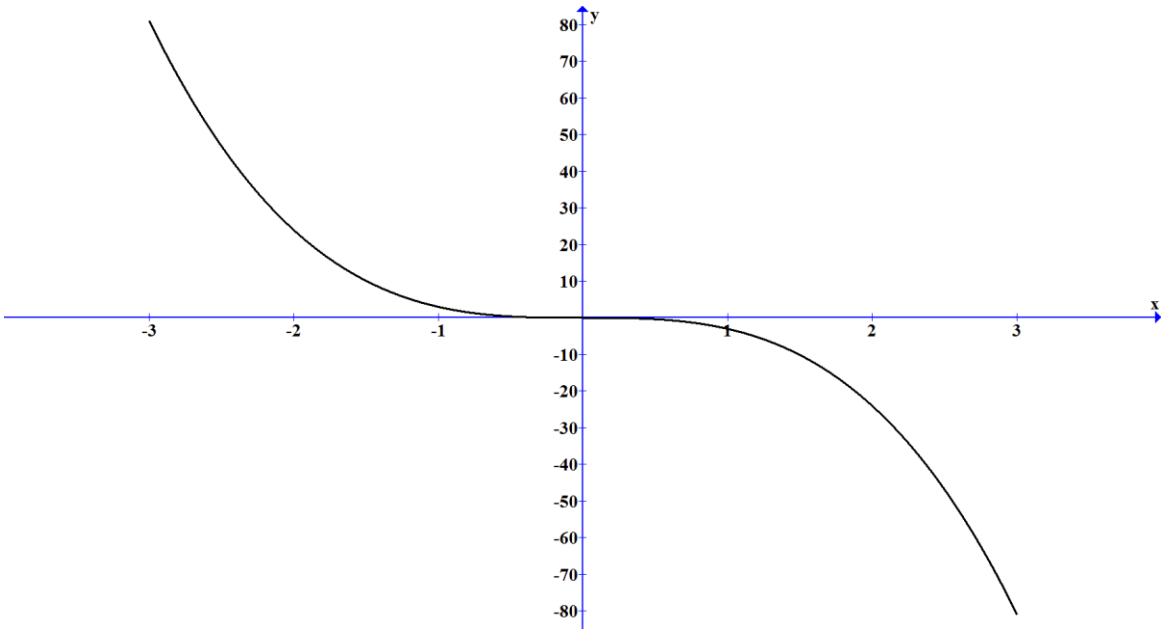
x	-3	-2	-1	0	1	2	3
y	27	8	1	0	-1	-8	-27

©Ezy Math Tutoring | All Rights Reserved



d) $y = -3x^3$

x	-3	-2	-1	0	1	2	3
y	81	24	3	0	-3	-24	-81

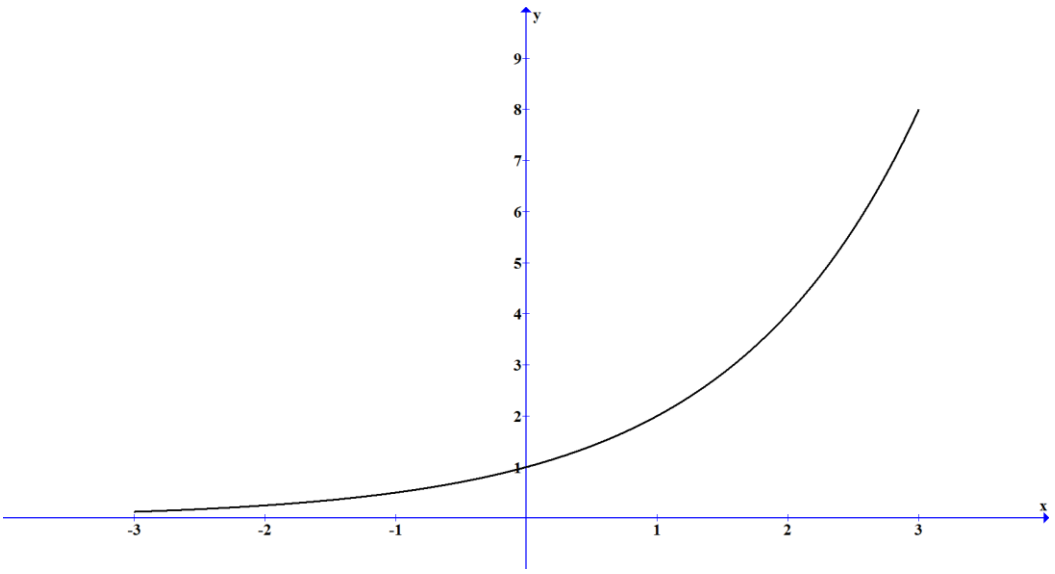


9) For each equation, generate a table of values and graph the equation, choosing an appropriate range

a) $y = 2^x$

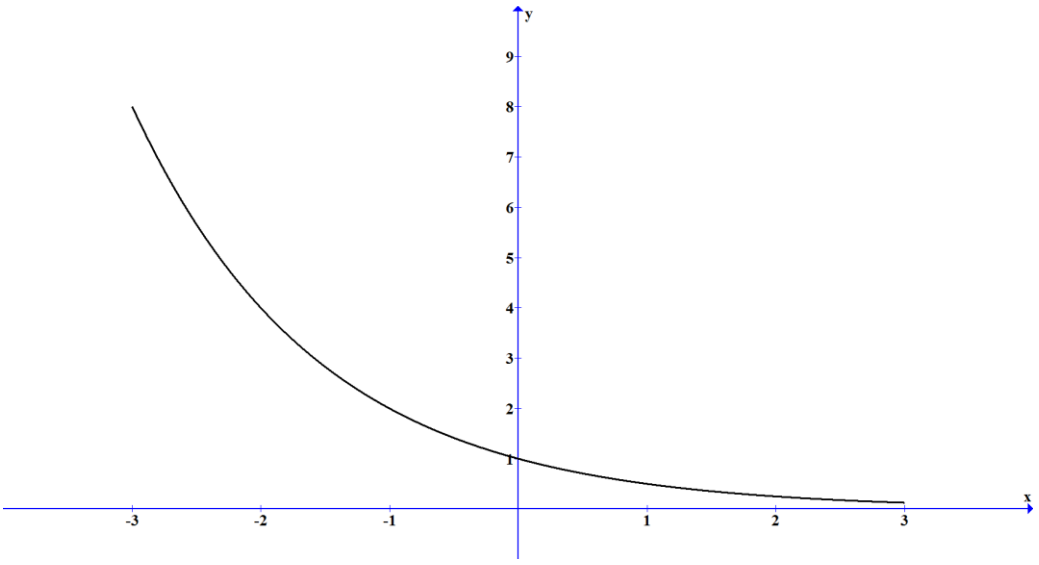
©Ezy Math Tutoring | All Rights Reserved

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{8}$	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4	8



b) $y = \left(\frac{1}{2}\right)^x$

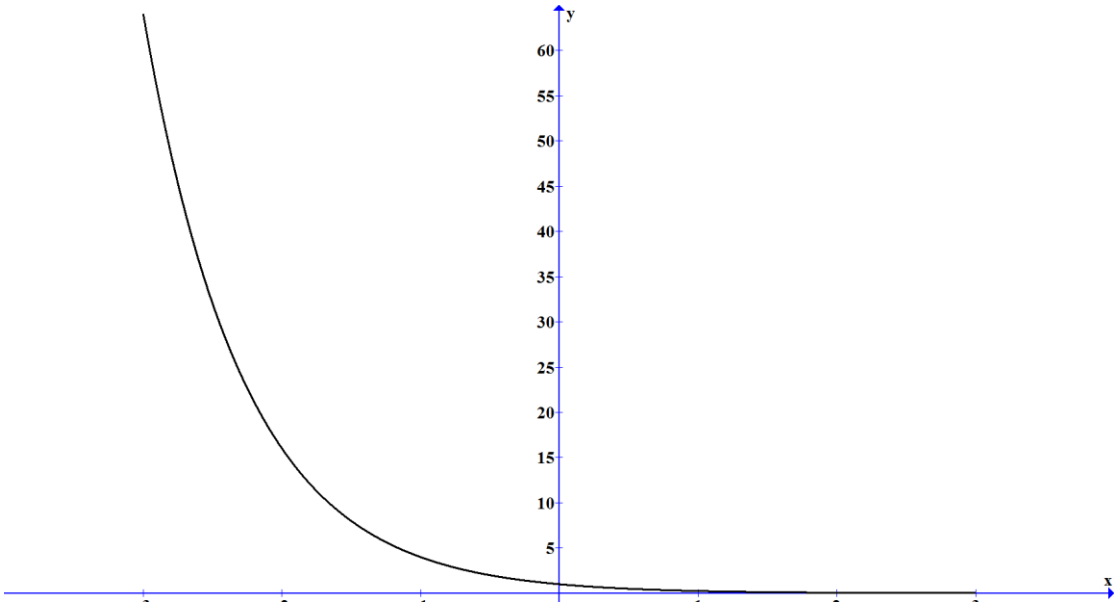
x	-3	-2	-1	0	1	2	3
y	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$



c) $y = \left(\frac{1}{4}\right)^x$

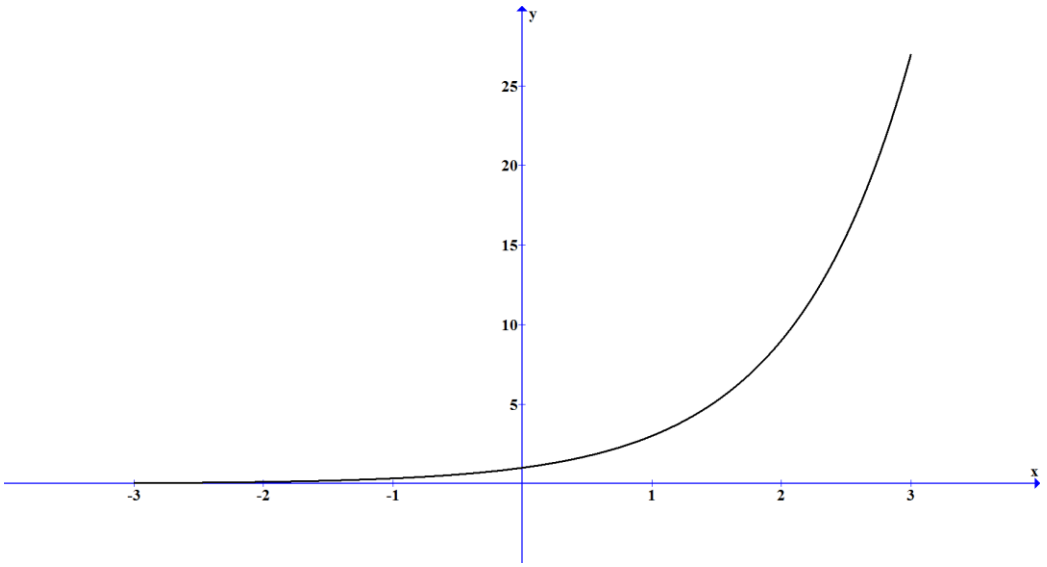
x	-3	-2	-1	0	1	2	3
y	64	16	4	1	$\frac{1}{4}$	$\frac{1}{16}$	$\frac{1}{64}$

©Ezy Math Tutoring | All Rights Reserved



d) $y = 3^x$

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{27}$	$\frac{1}{9}$	$\frac{1}{3}$	1	3	9	27



10) How is the graph of the equations in question 13 different for $a > 1$ or $a < 1$

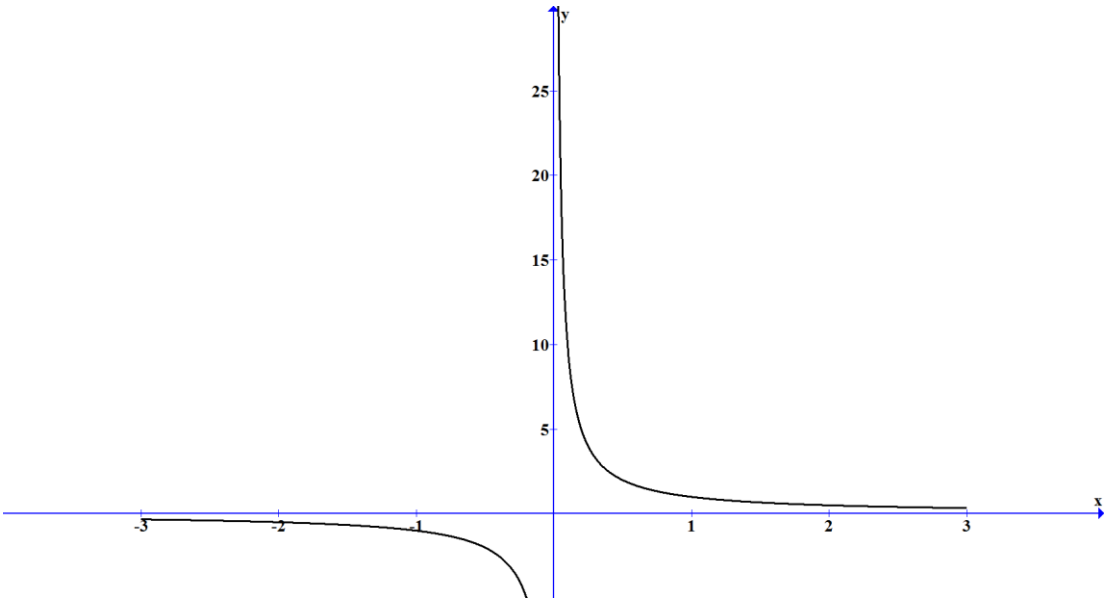
For $a > 1$, the graph increases, for $a < 1$, the graph decreases

©Ezy Math Tutoring | All Rights Reserved

11) For each equation, generate a table of values and graph the equation, choosing an appropriate range

a) $y = \frac{1}{x}$

x	-3	-2	-1	0	1	2	3
y	$-\frac{1}{3}$	$-\frac{1}{2}$	-1	Not defined	1	$\frac{1}{2}$	$\frac{1}{3}$

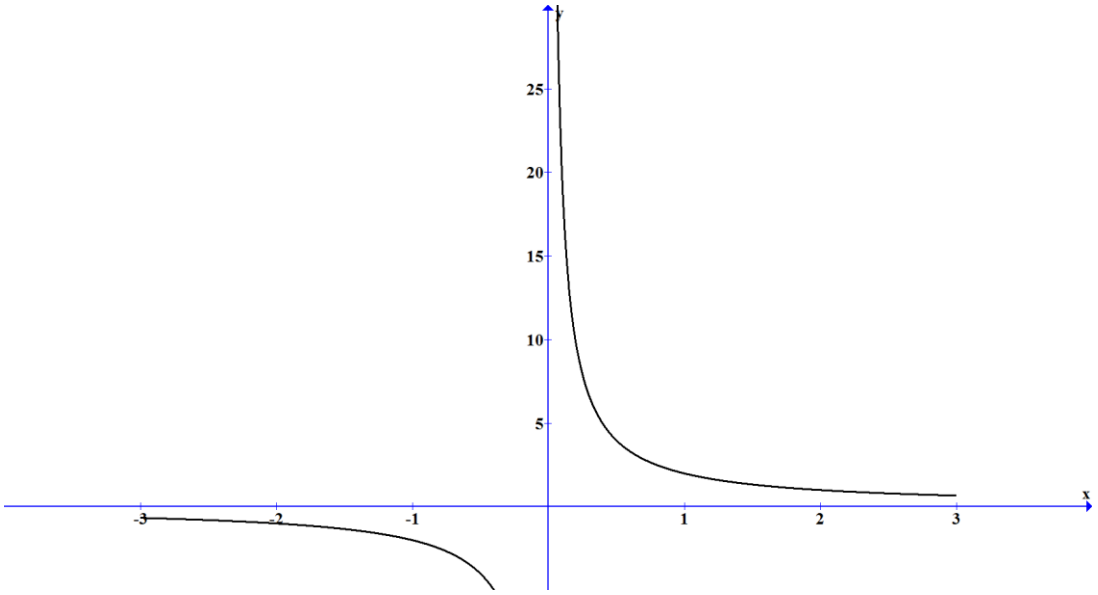


b) $y = \frac{2}{x}$

x	-3	-2	-1	0	1	2	3

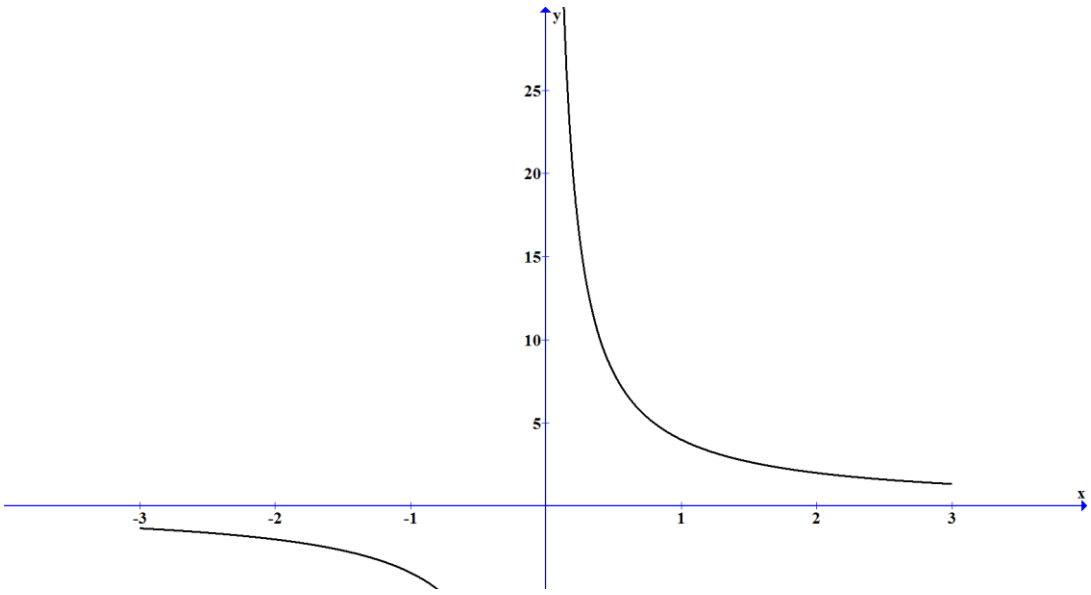
©Ezy Math Tutoring | All Rights Reserved

y	$-\frac{2}{3}$	-1	-2	Not defined	2	1	$\frac{2}{3}$



c) $y = \frac{4}{x}$

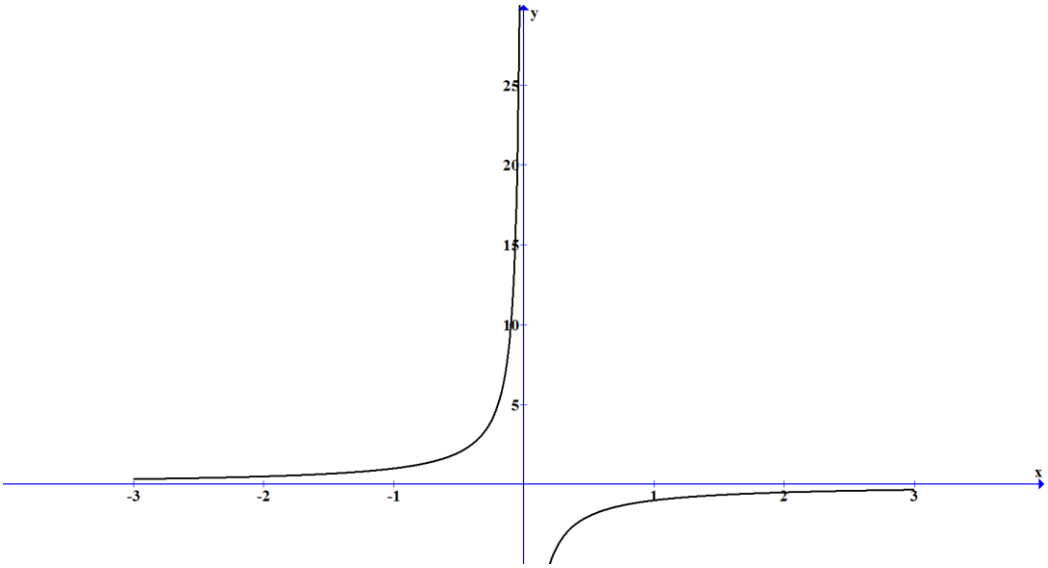
x	-3	-2	-1	0	1	2	3
y	$-\frac{4}{3}$	-2	-4	Not defined	4	2	$\frac{4}{3}$



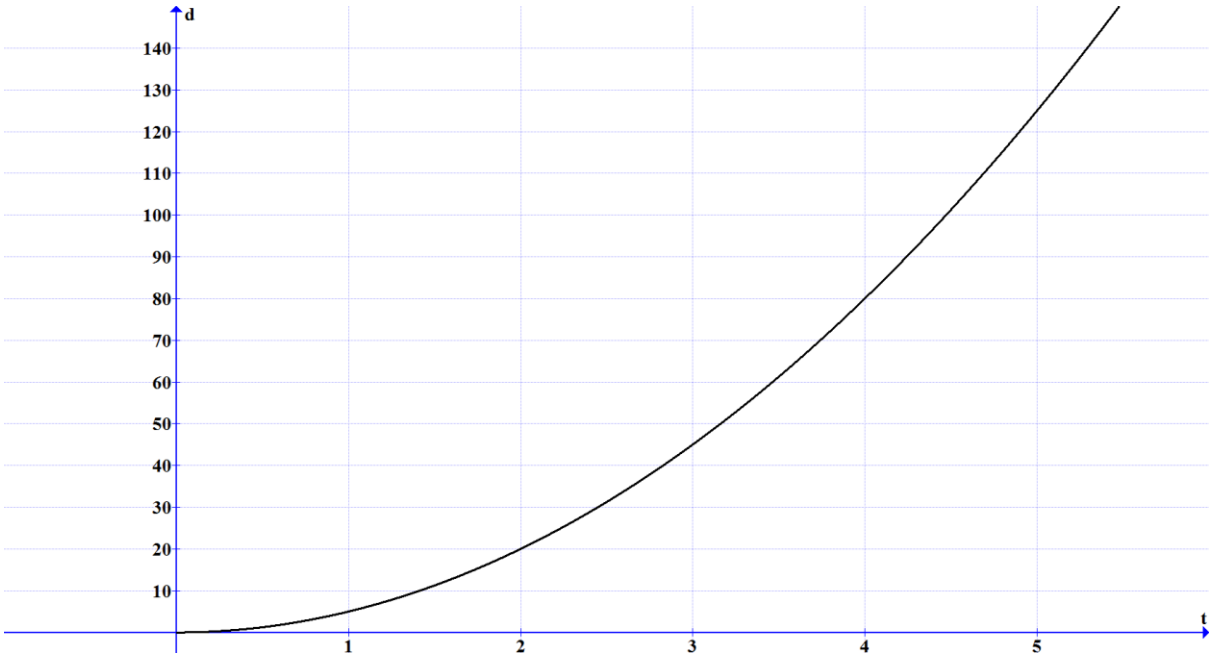
©Ezy Math Tutoring | All Rights Reserved

d) $y = \frac{-1}{x}$

x	-3	-2	-1	0	1	2	3
y	$\frac{1}{3}$	$\frac{2}{3}$	1	Not defined	-1	$-\frac{1}{2}$	$-\frac{1}{3}$



12) The distance an object falls due to gravity on Earth can be approximated by the equation $d = 5t^2$, where d is the distance in metres, and t is the number of seconds. Graph this equation, and use it or a table of values to determine



©Ezy Math Tutoring | All Rights Reserved

a) How far an object falls in 5 seconds

125 metres

b) The time an object has been falling if it gas travelled 80 metres

4 seconds

13) On the moon gravity is weaker, so whilst the equation from question 16 still generally applies, the coefficient is different. After 2 seconds on the moon an object has fallen 3.2 metres.

a) Calculate the new coefficient and hence the equation describing the distance a body falls in t seconds on the moon

$d = at^2$

$3.2 = a(2^2)$

$a = 0.8$

$d = 0.8t^2$

b) How far has a body on the moon fallen after 10 seconds?

When $t = 10, d = 0.8 \times 100 = 80\ metres$

c) A body falls 28.8 metres on the moon. How long has it been falling for?

When $d = 28.8,\ 0.8t^2 = 28.8$

$t^2 = 14.4$

$t \cong 3.8\ seconds$

©Ezy Math Tutoring | All Rights Reserved

14) An ant is removing small rocks from a pile. The number of rocks left in the pile can be approximated by the equation $N = \frac{N_0}{t+1}$ where N is the number of rocks remaining, t is the time in minutes, and N_0 is the number of rocks initially in the pile. After 3 minutes there were 25 rocks in the pile

a) How many rocks were in the pile initially?

When $t = 3, N = 25$

$$25 = \frac{N_0}{4}$$

$$N_0 = 100$$

b) How many rocks had the ant removed after 1 minute?

When $t = 1, N = \frac{100}{1+1} = 50$

There were 50 rocks left so the ant had removed 50

c) How many rocks will remain after 9 minutes?

When $t = 9, N = \frac{100}{9+1} = 10$

d) Explain why this equation can only be considered as an approximation. (Hint look at large values of t)

Some values give meaningless answers, for example, after 2 minutes there are 33.3 rocks left according to the equation

As t gets larger and larger the value of t+1 will eventually be greater than 100, which results in less than 1 rock left for any time greater than 100 minutes

15) Water flows from a large hose at the rate of 16 litres per minute. At this rate it takes 22 hours to fill a small pond. If the flow rate reduces to 4 litres per minute, it takes 88 hours to fill the pond

©Ezy Math Tutoring | All Rights Reserved

a) Calculate the proportionality constant for this situation, and hence produce the equation relating the flow rate to the time taken to fill the pond

Note flow rate must be changed to hourly

$16\ litres\ per\ minute\ \times 60 \times 22 = 4\ litres\ per\ minute\ \times 60 \times 88 = 21120$

$Rate\ \times time = 21120$

$R = \frac{21120}{t}, or\ t = \frac{21120}{R}$

b) How many litres does the pond hold?

21120 litres

c) How long would it take to fill the pond if the flow rate was changed to 32 litres per minute?

$t = \frac{21120}{R}$

When $R = 32, R\ \times 60 = 1920,\ t = 11\ hours$

d) If it took 44 hours to fill the pond, what was the flow rate?

$44 \times R \times 60 = 21120$

$R = \frac{21120}{2640} = 8\ litres\ per\ minute$

16) John deposits \$10000 into a bank account that pays $x\%$ interest compounded annually. He deposits no other funds, and after 3 years his balance is \$12597.12.

a) Calculate the interest rate, and hence write the equation that relates John's balance after t years

$10000 \times \left(1 + \frac{x}{100}\right)^3 = 12597.12$

©Ezy Math Tutoring | All Rights Reserved

$$\left(1+\frac{x}{100}\right)^3 = 1.259712$$

$$1+\frac{x}{100} = 1.08$$

$$\frac{x}{100} = 0.08$$

$$x = 8\%$$

Balance $= 10000 \times (1.08)^t$

b) What will John's balance be after 10 years?

When $t = 10, Balance = 10000 \times 1.08^{10} = \$21589.25$

©Ezy Math Tutoring | All Rights Reserved

©Ezy Math Tutoring | All Rights Reserved